

Introduction to IP (1)

□ 介紹

□ 乘法器

✓ 創建 **Symbol** 之方式

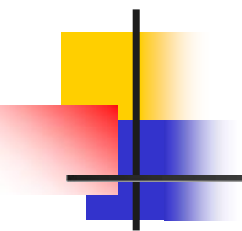
✓ 以 **Verilog** 之方式

✓ 模擬

□ 除法器使用

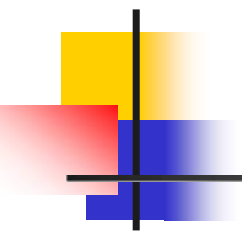
✓ 創建 **Symbol** 之方式

✓ 模擬



Introduction to IP (2)

- 隨著半導體技術的不斷進步，系統單晶片 (System On a Chip, SOC) 也越來越成熟，所以矽智產 (Silicon Intellectual property, IP) 已越來越被廣泛應用。
- **IP:** 通常在一系統中，被分成多個區塊或多個功能模組，因為某些功能常被使用，所以經過定義、驗證過之模組，具有特定功能。之後再次使用到時，可直接拿來使用不用重新設計。



Introduction to IP (1)

☐ 介紹

☐ 乘法器

- ✓ 創建 **Symbol** 之方式

- ✓ 以 **Verilog** 之方式

- ✓ 模擬

☐ 除法器使用

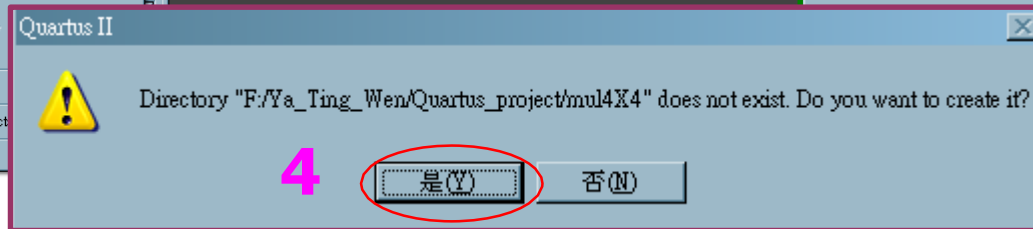
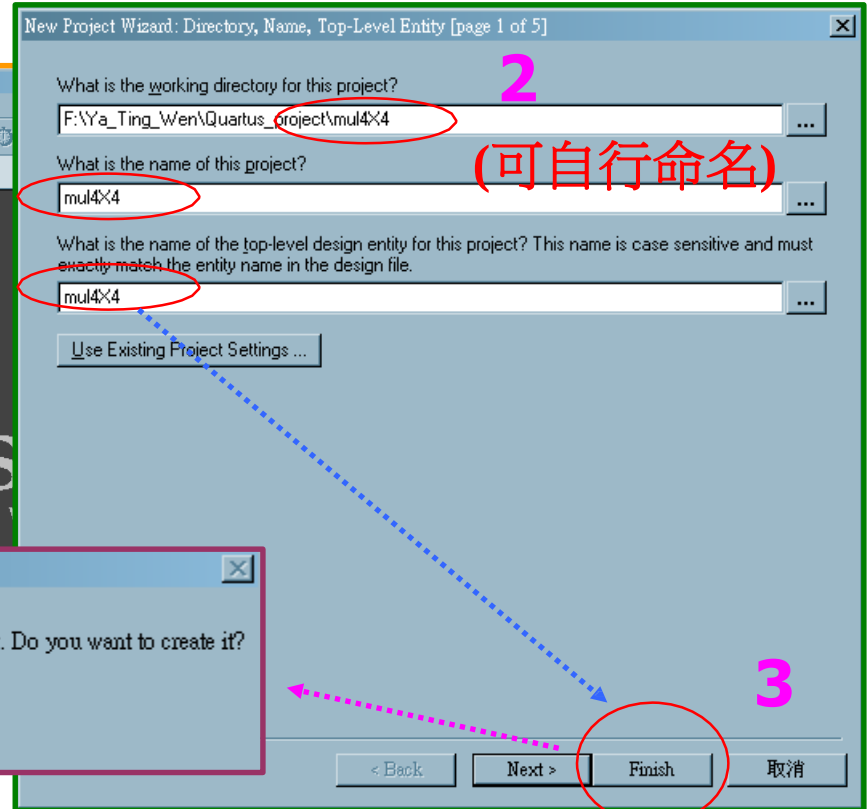
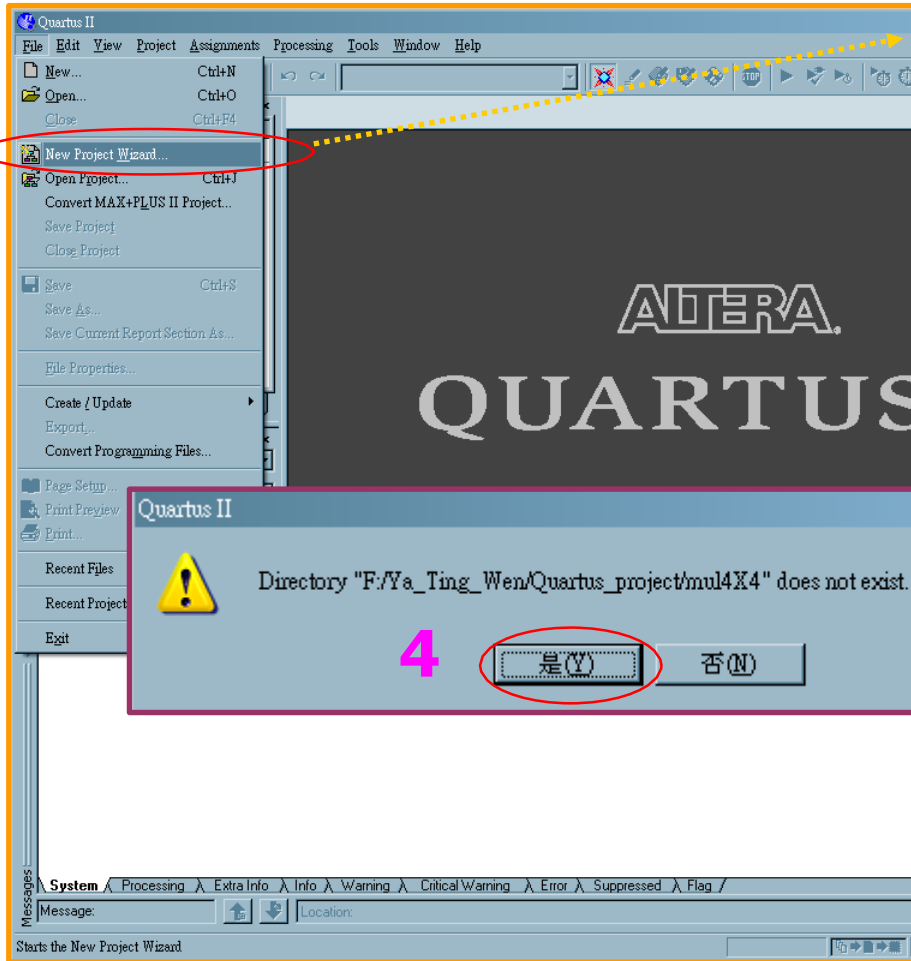
- ✓ 創建 **Symbol** 之方式

- ✓ 模擬

創建 Symbol 之方式_Step 1

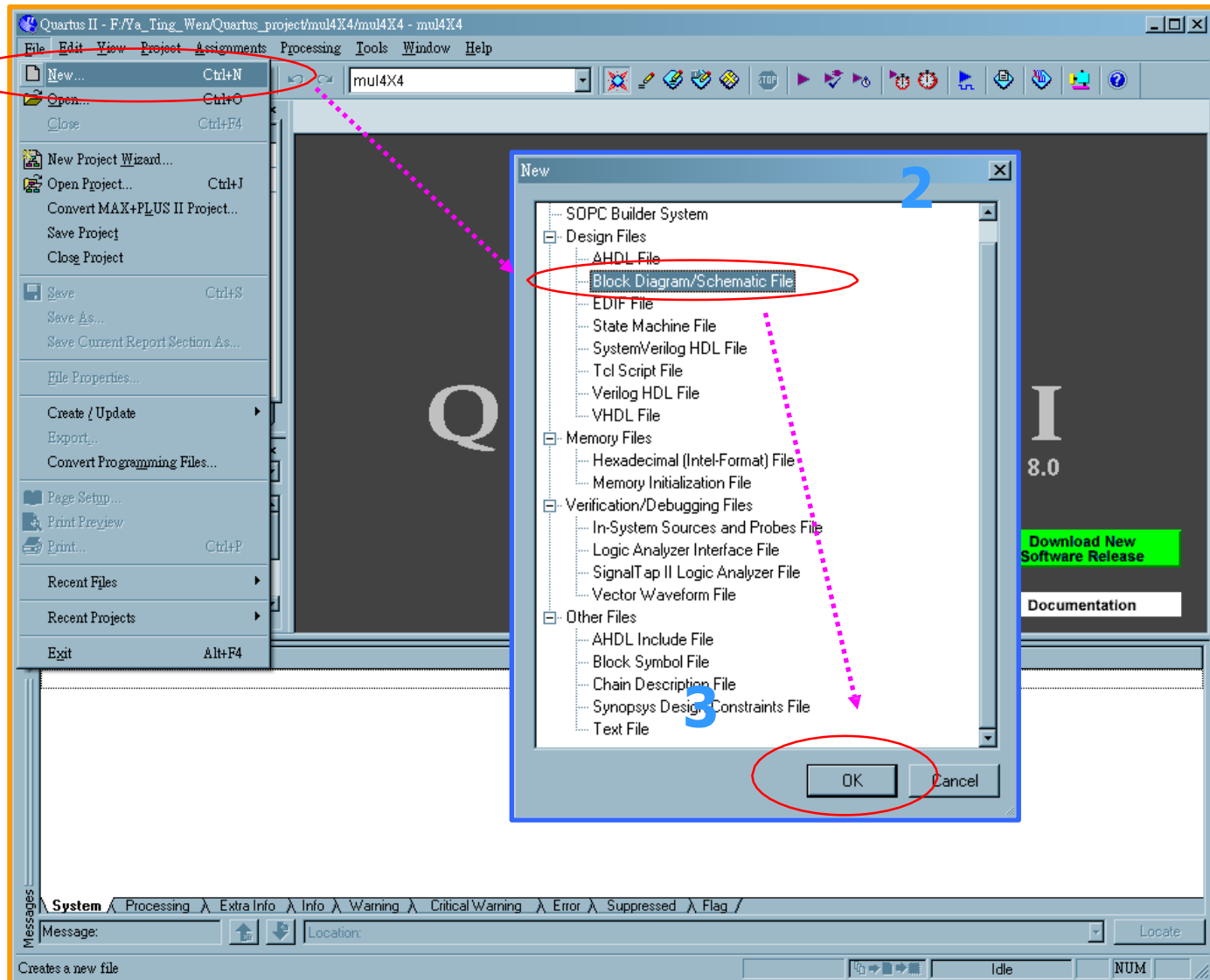
5_01_mul4X4_symbol

1



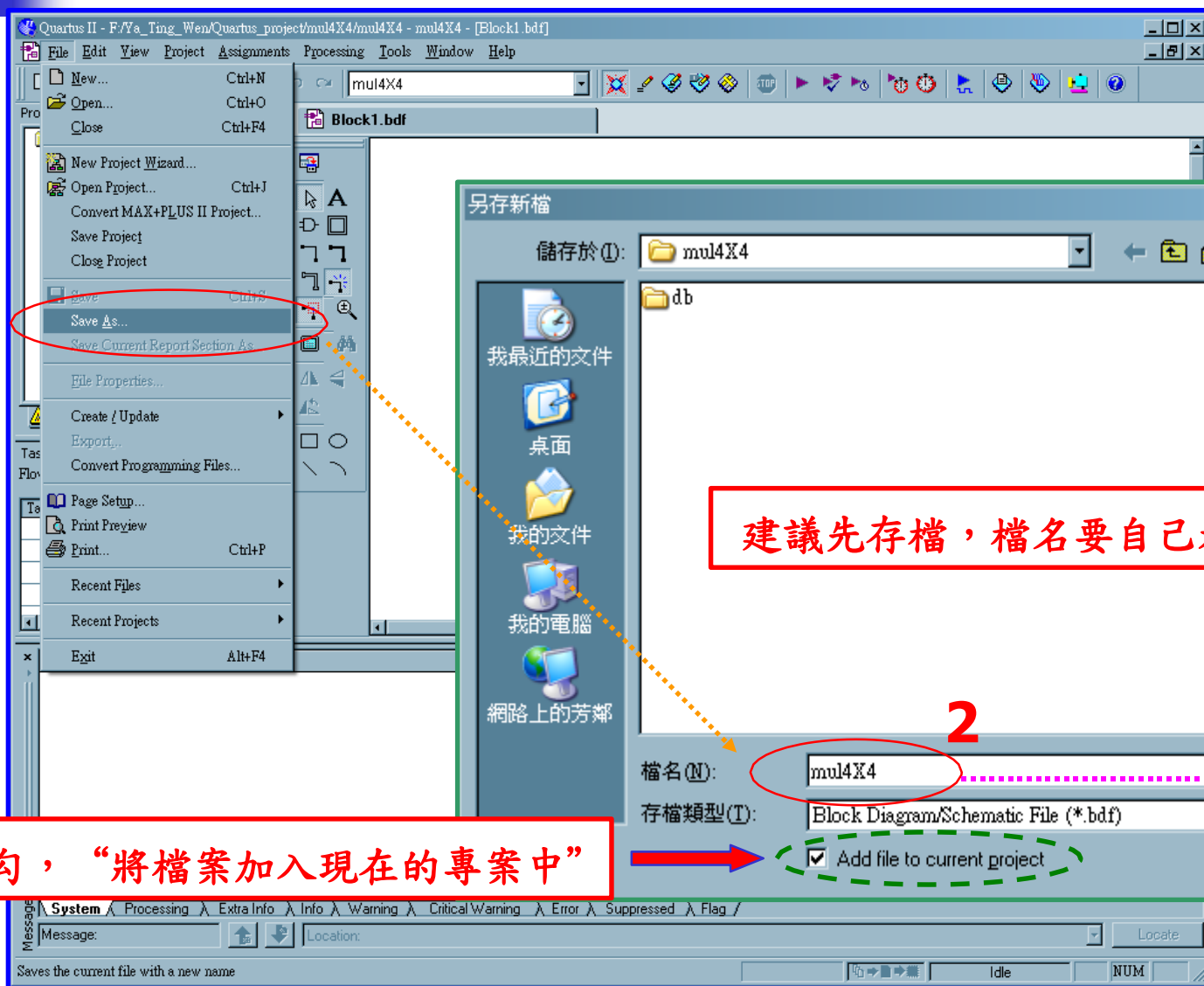
創建 Symbol 之方式_ Step 2

1



創建 Symbol 之方式_ Step 3

1



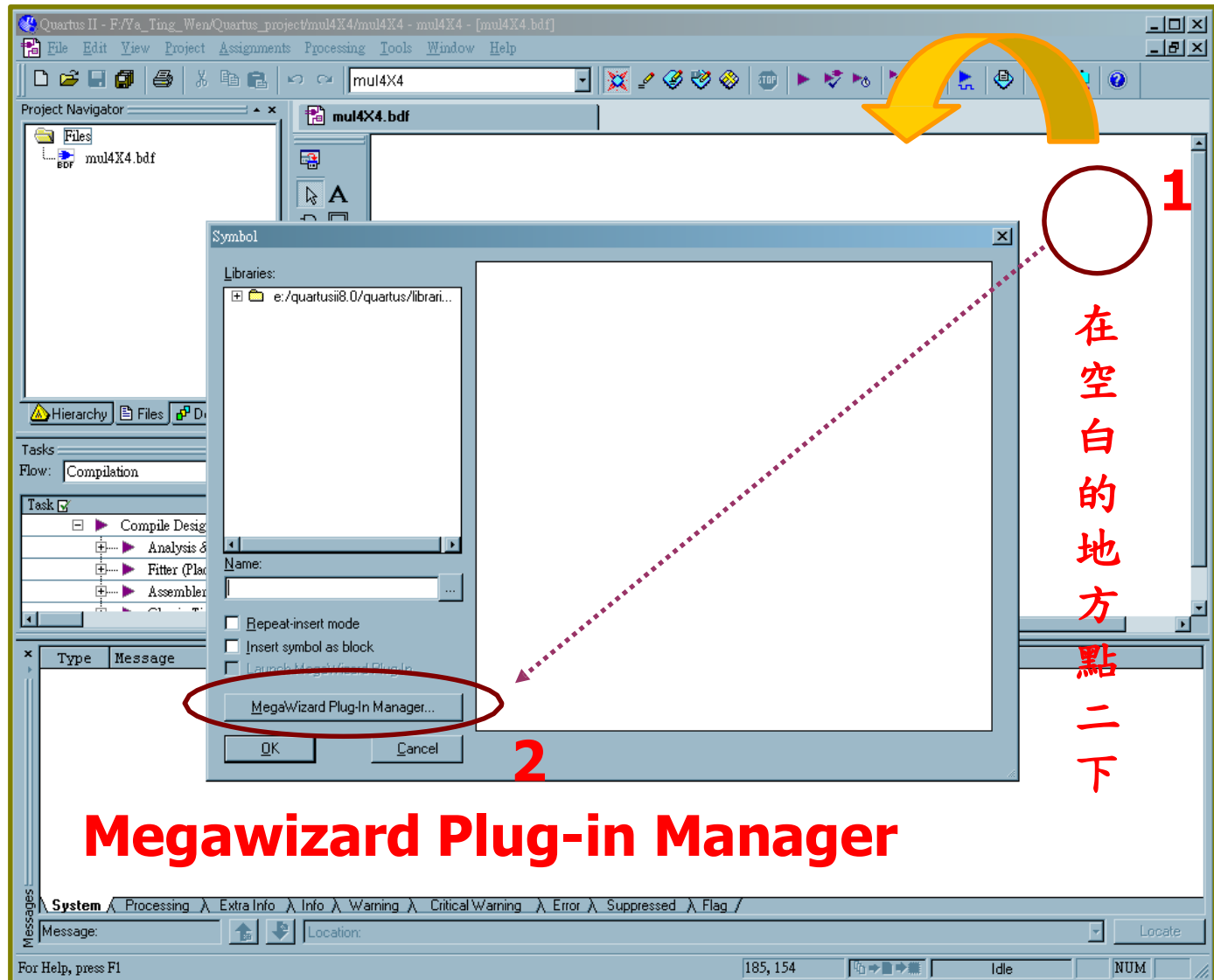
建議先存檔，檔名要自己看得懂

2

3

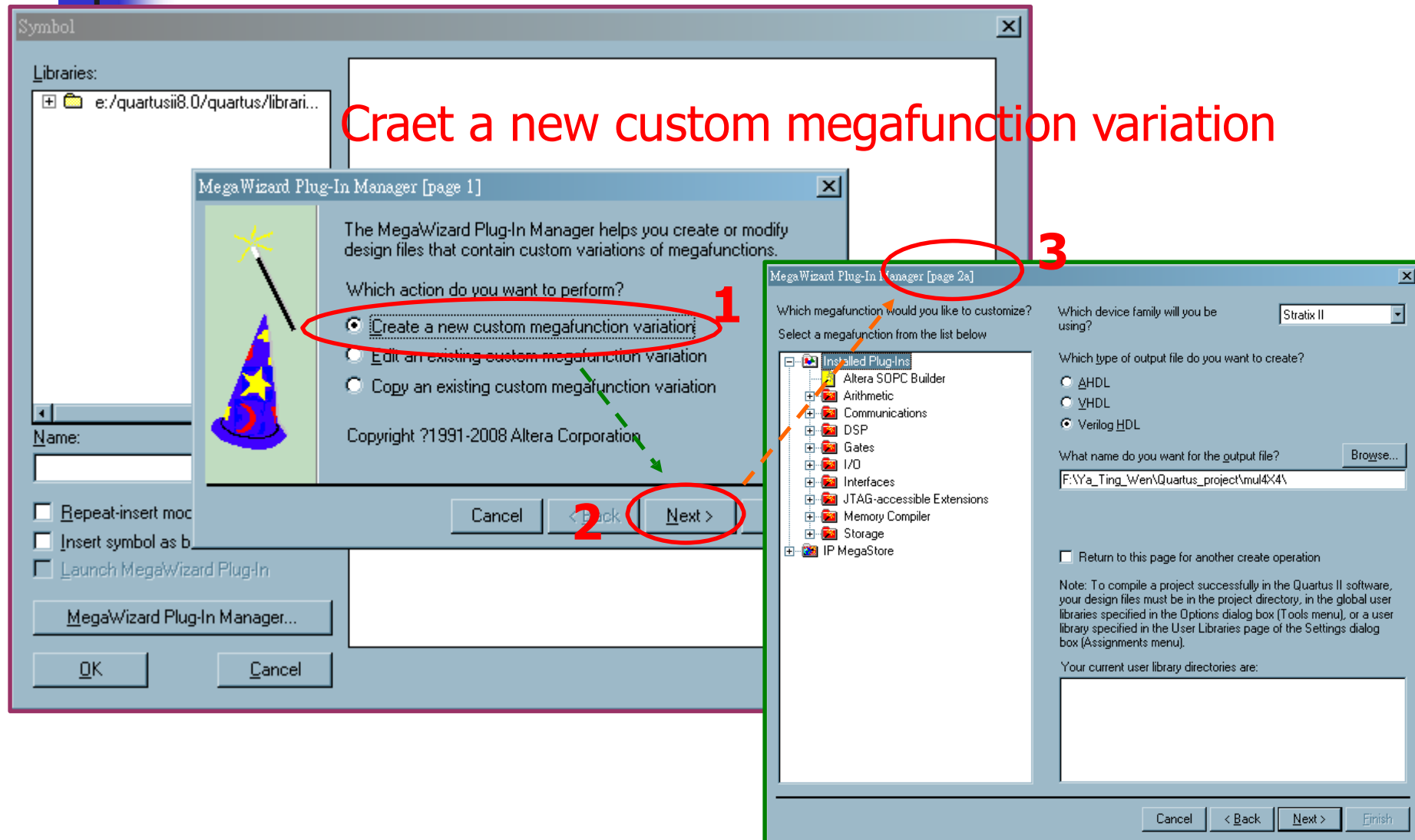
要記得勾，『將檔案加入現在的專案中』

創建 Symbol 之方式_ Step 4

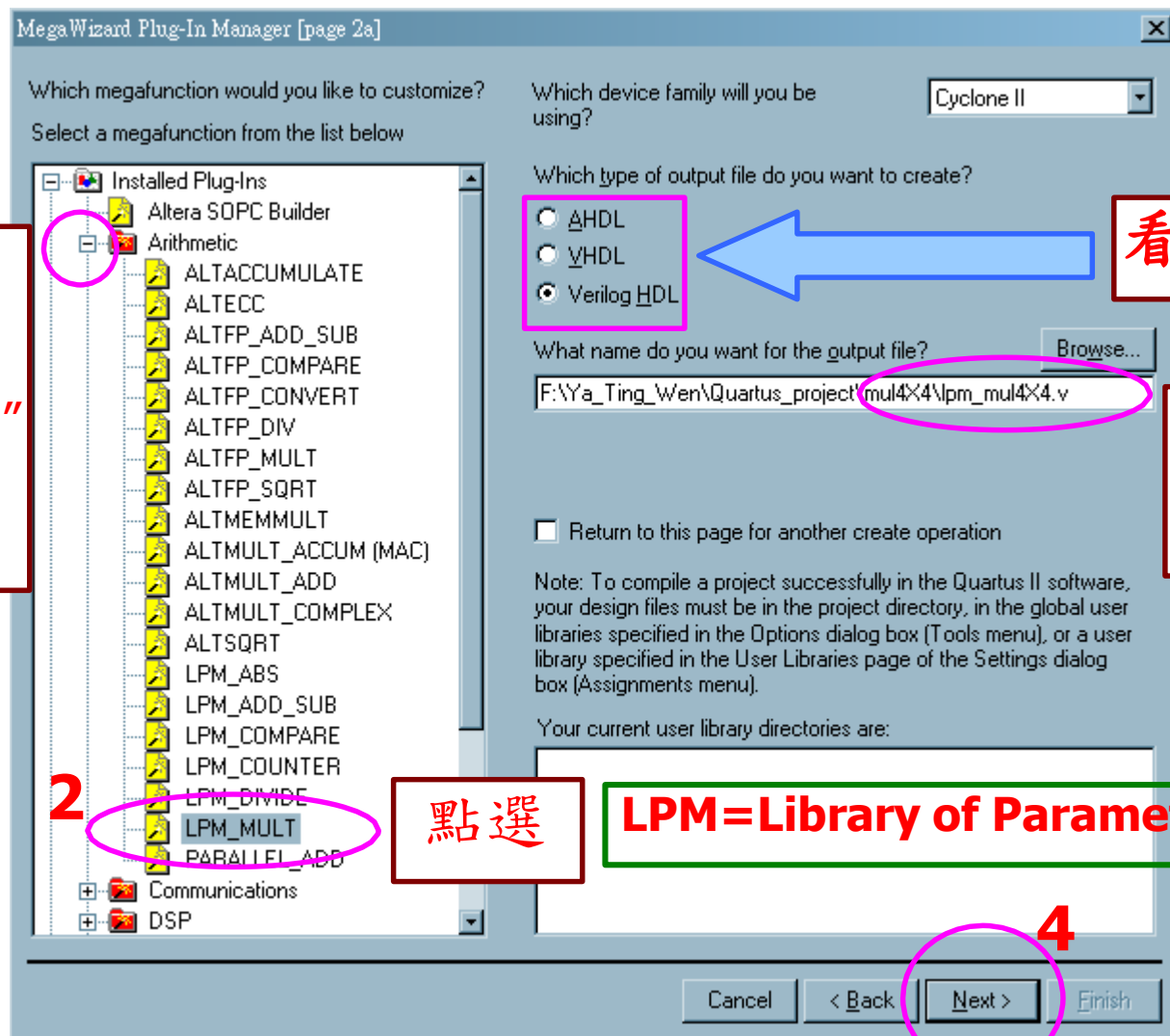


創建 Symbol 之方式_ Step 5

Cræet a new custom megafunction variation



創建 Symbol 之方式_ Step 6

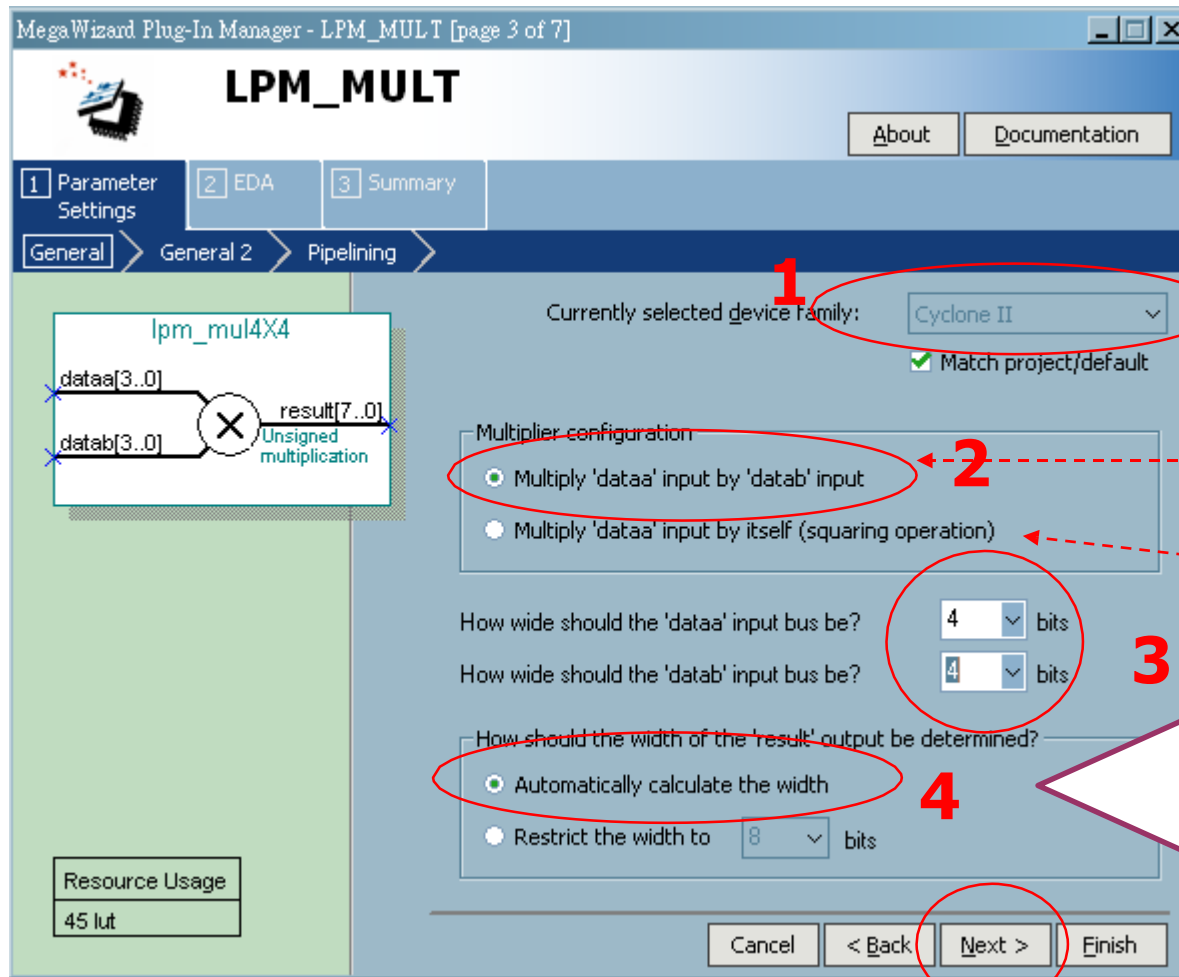


3
看要存成什麼語言

存什麼檔名
(自己取.v)

點選
LPM=Library of Parameterized Module

創建 Symbol 之方式_ Step 7



看設定有無跑掉

$a \times b$

a^2

設定輸入位元數

預設就可以了

5

創建 Symbol 之方式_ Step 8

MegaWizard Plug-In Manager - LPM_MULT [page 4 of 7]

LPM_MULT

About Documentation

1 Parameter Settings 2 EDA 3 Summary

General > General 2 > Pipelining >

lpm_mul4X4

dataa[3..0] datab[3..0] result[7..0]

Unsigned multiplication

Does the 'datab' input bus have a constant value?

☒ No **1**

☐ Yes, the value is

Which type of multiplication do you want?

☒ Unsigned **2**

☐ Signed

Which multiplier implementation should be used?

☒ Use the default implementation **3**

☐ Use dedicated multiplier circuitry (Not available for all families)

☐ Use logic elements

Resource Usage

45 lut

Cancel < Back **Next >** **4** Finish

創建 Symbol 之方式_ Step 9

MegaWizard Plug-In Manager - LPM_MULT [page 5 of 7]

LPM_MULT

About Documentation

1 Parameter Settings 2 EDA 3 Summary

General > General 2 > Pipelining >

lpm_mul4X4

dataa[3..0]

datab[3..0]

result[7..0]

Unsigned multiplication

Resource Usage

45 lut

Do you want to pipeline the function?

☒ No 1

☐ Yes, I want an output latency of clock cycles

☐ Create an asynchronous Clear input

☐ Create a Clock Enable input

Delay time = ?

Which type of optimization do you want?

☒ Default 2

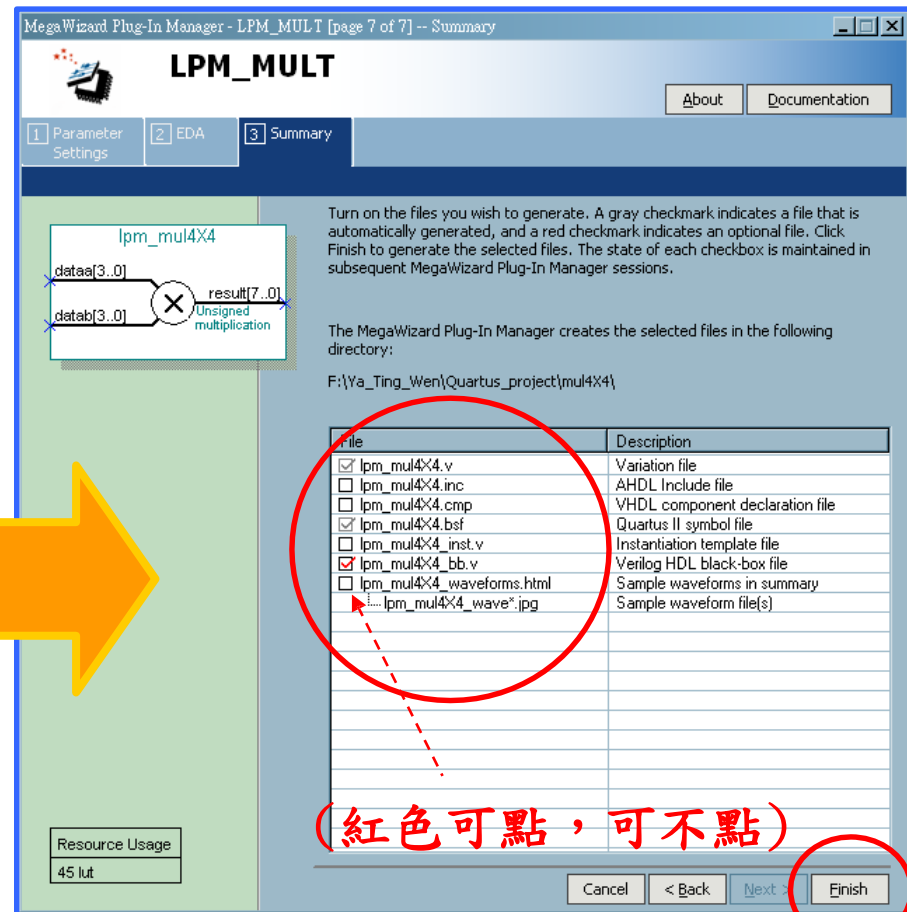
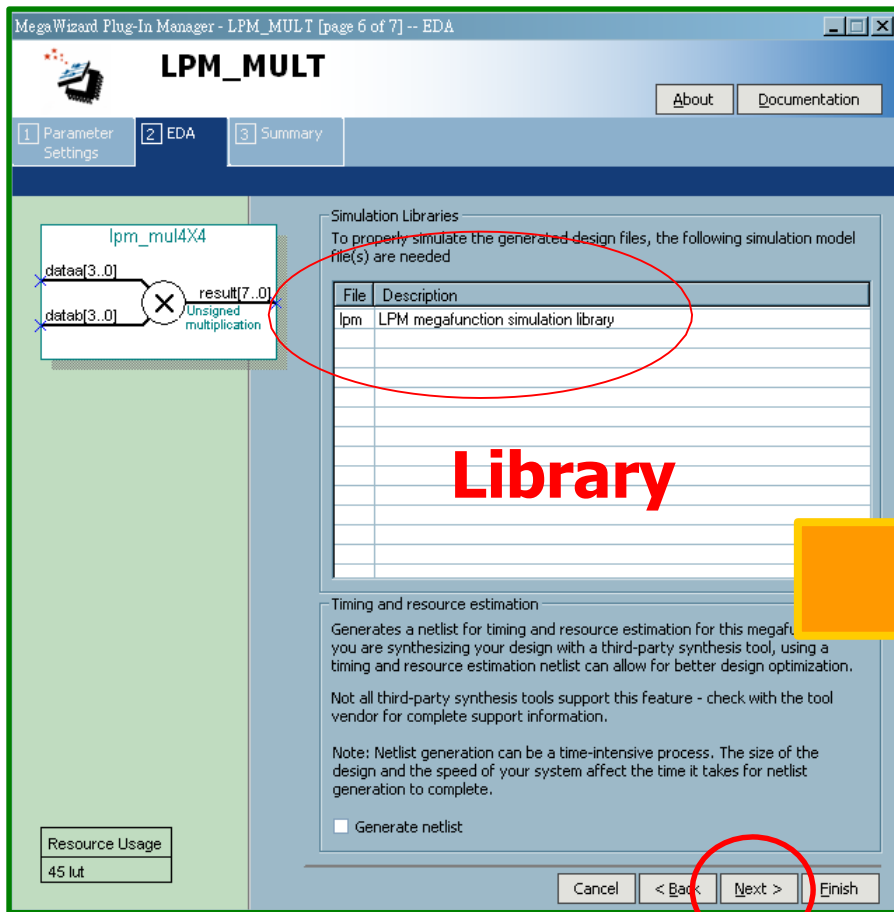
☐ Speed

☐ Area

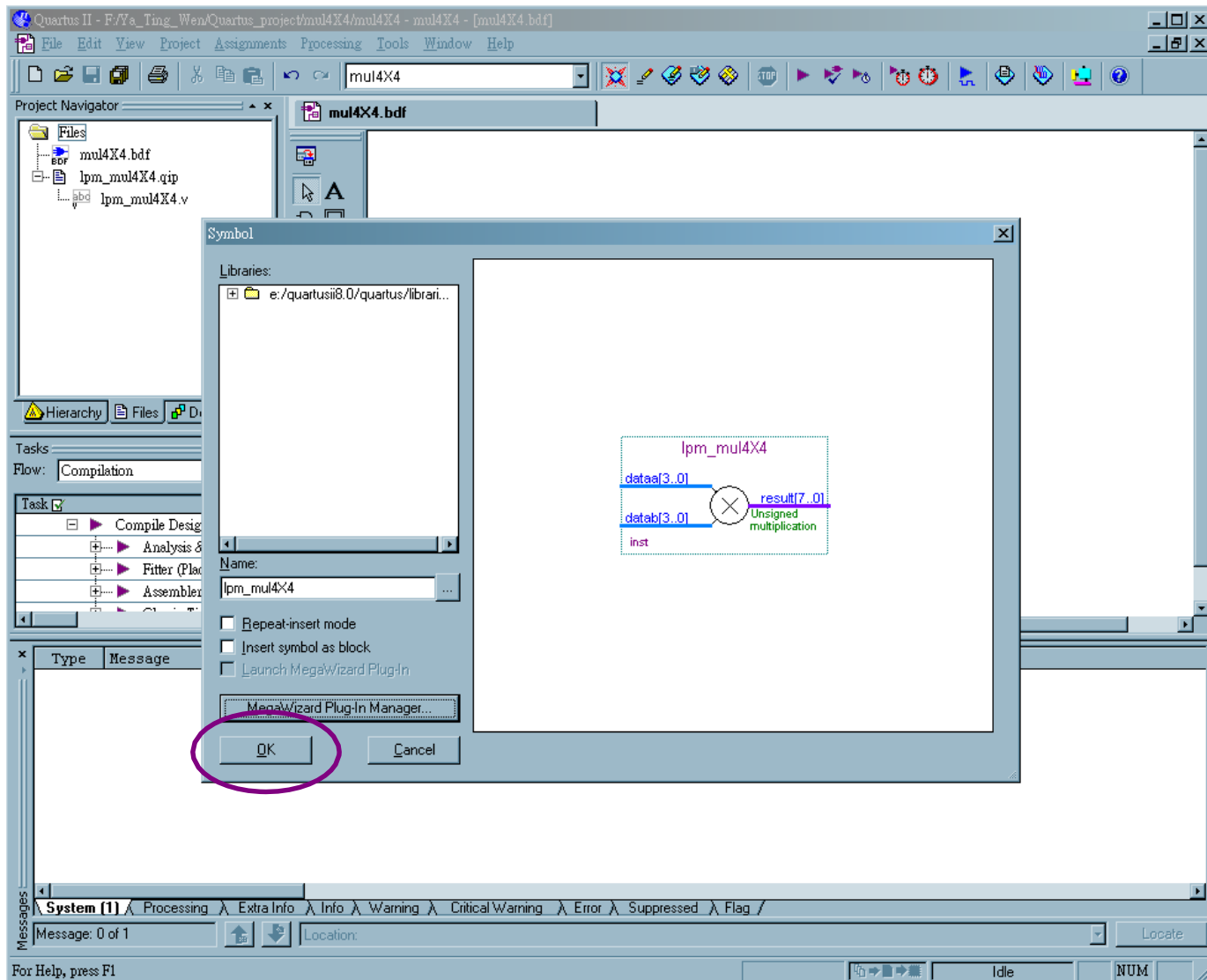
3

Cancel < Back Next > Finish

創建 Symbol 之方式_ Step 10



創建 Symbol 之方式_ Step 11

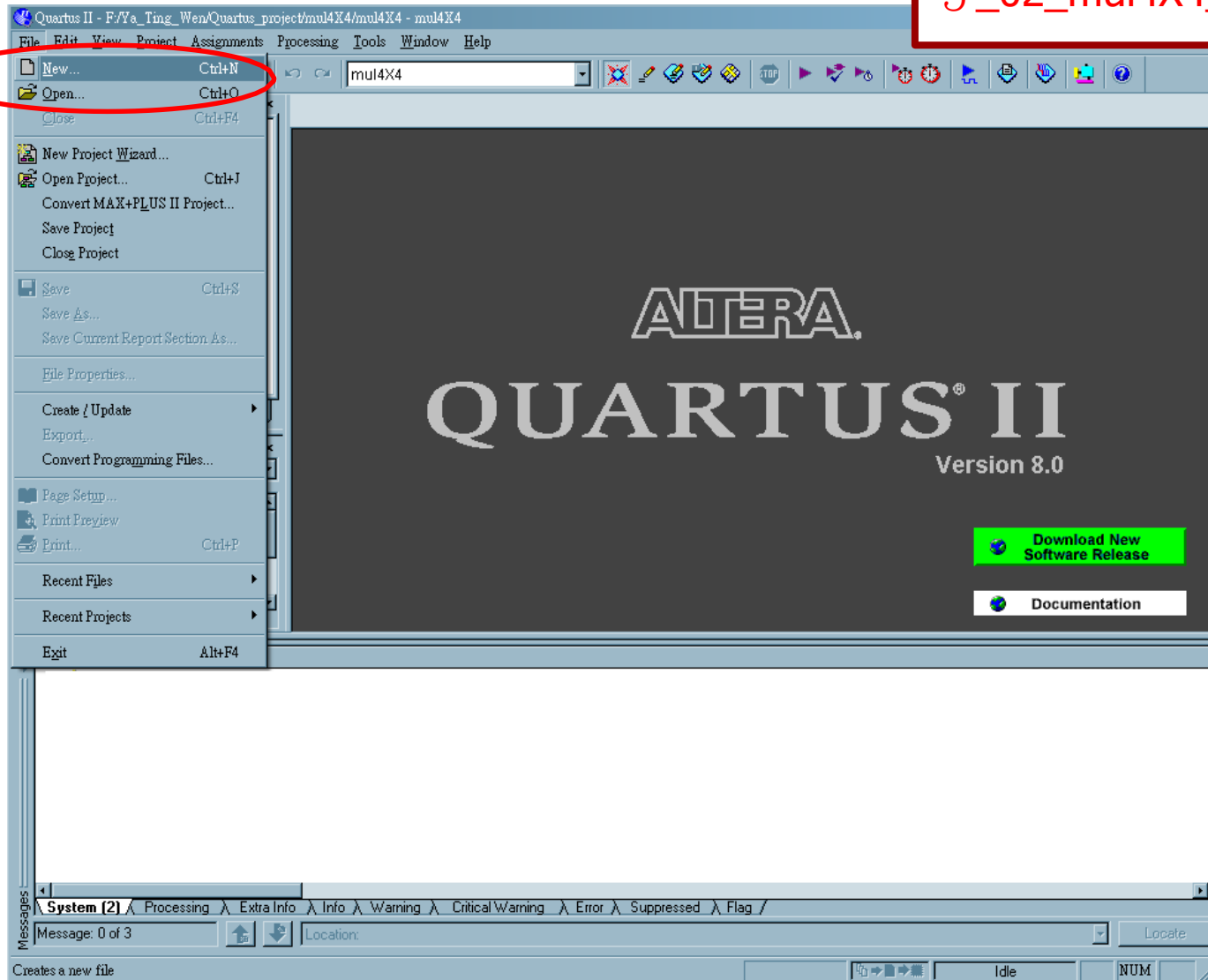


創建 Symbol 之方式_ Step 12

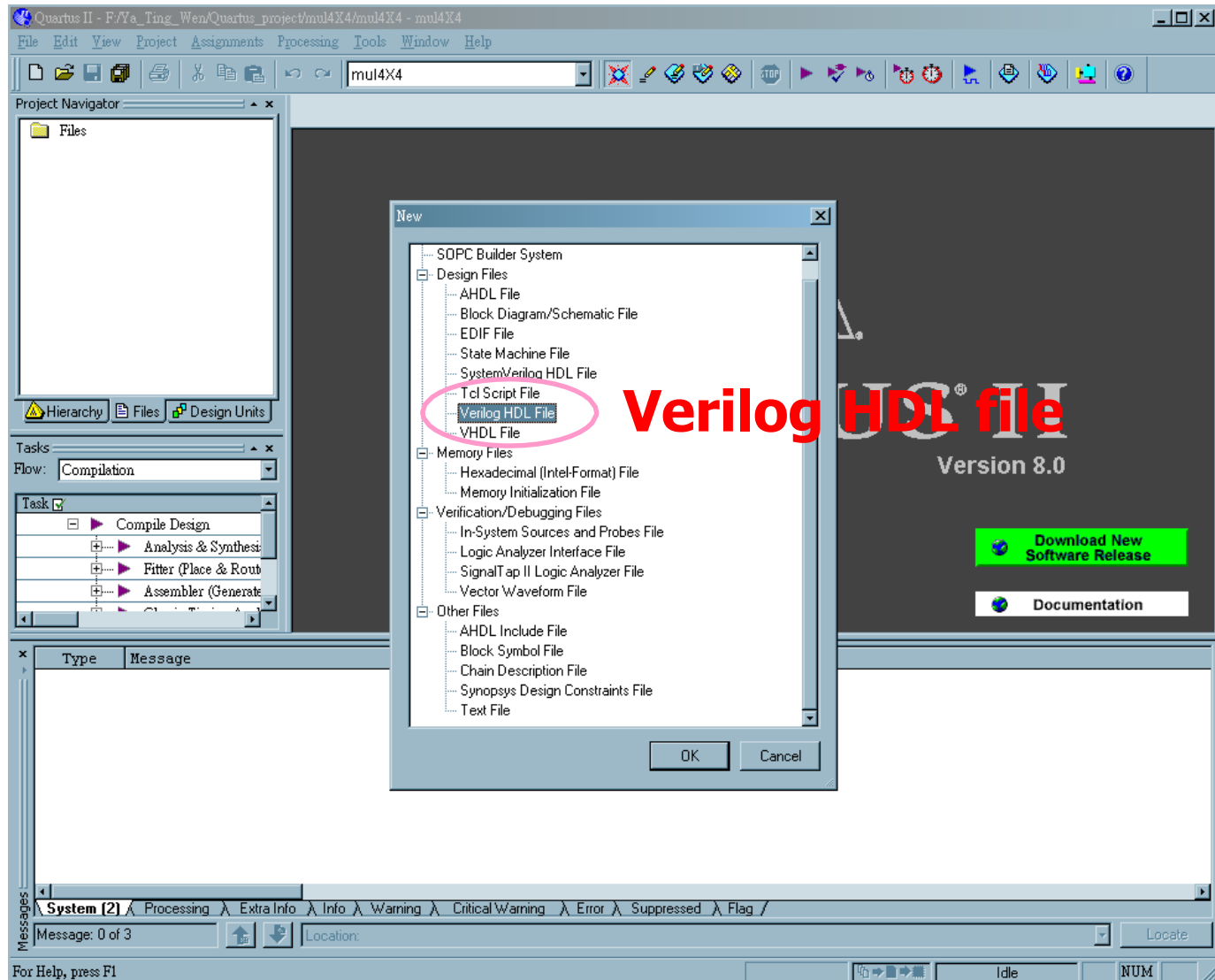
加入腳位即可模擬驗證(採用自動加入方式)
(按右鍵=>Add Input and Output Pins=> Compiler)

* Verilog 之方式_Step 1

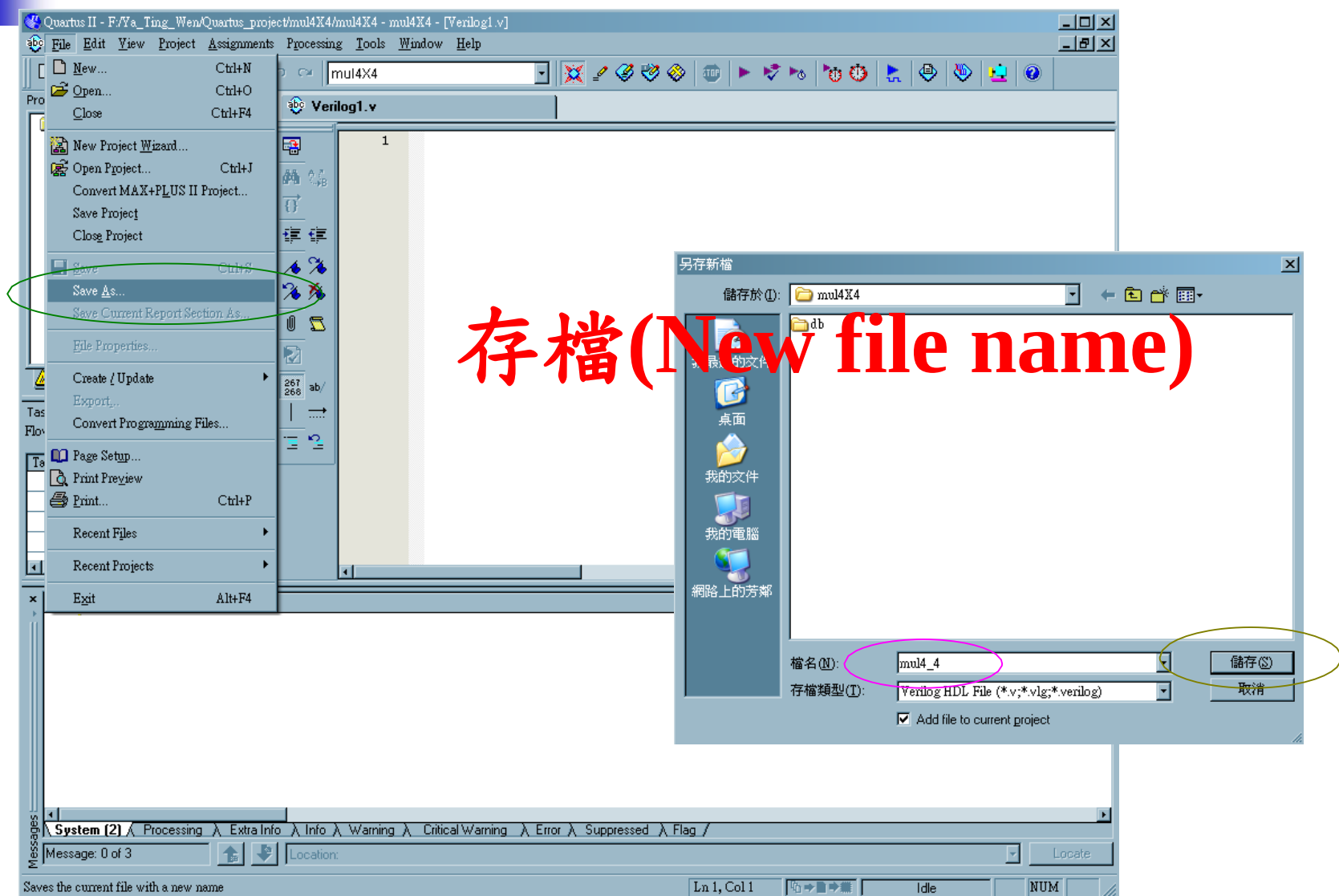
5_02_mul4X4_Template



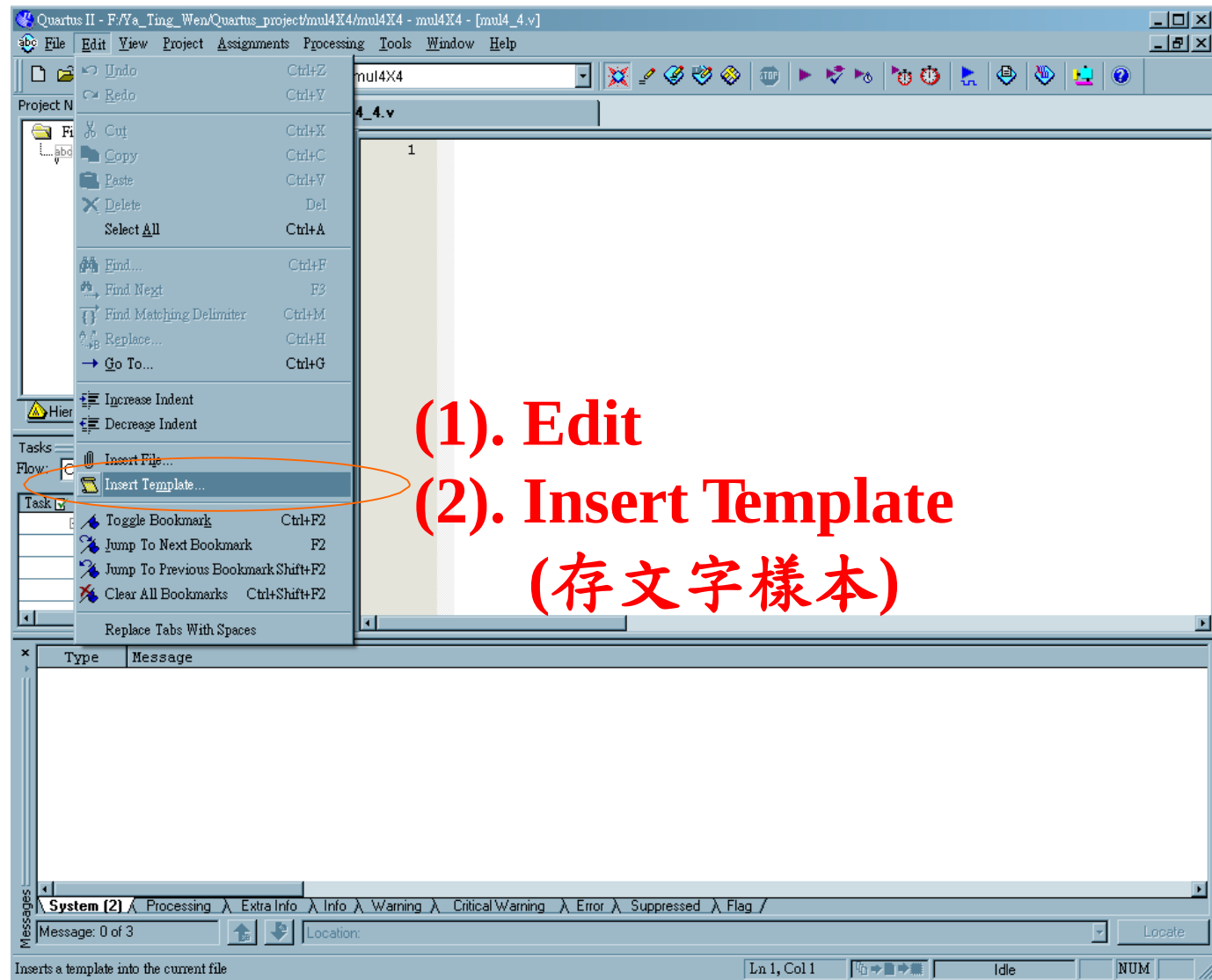
* Verilog 之方式_Step 2



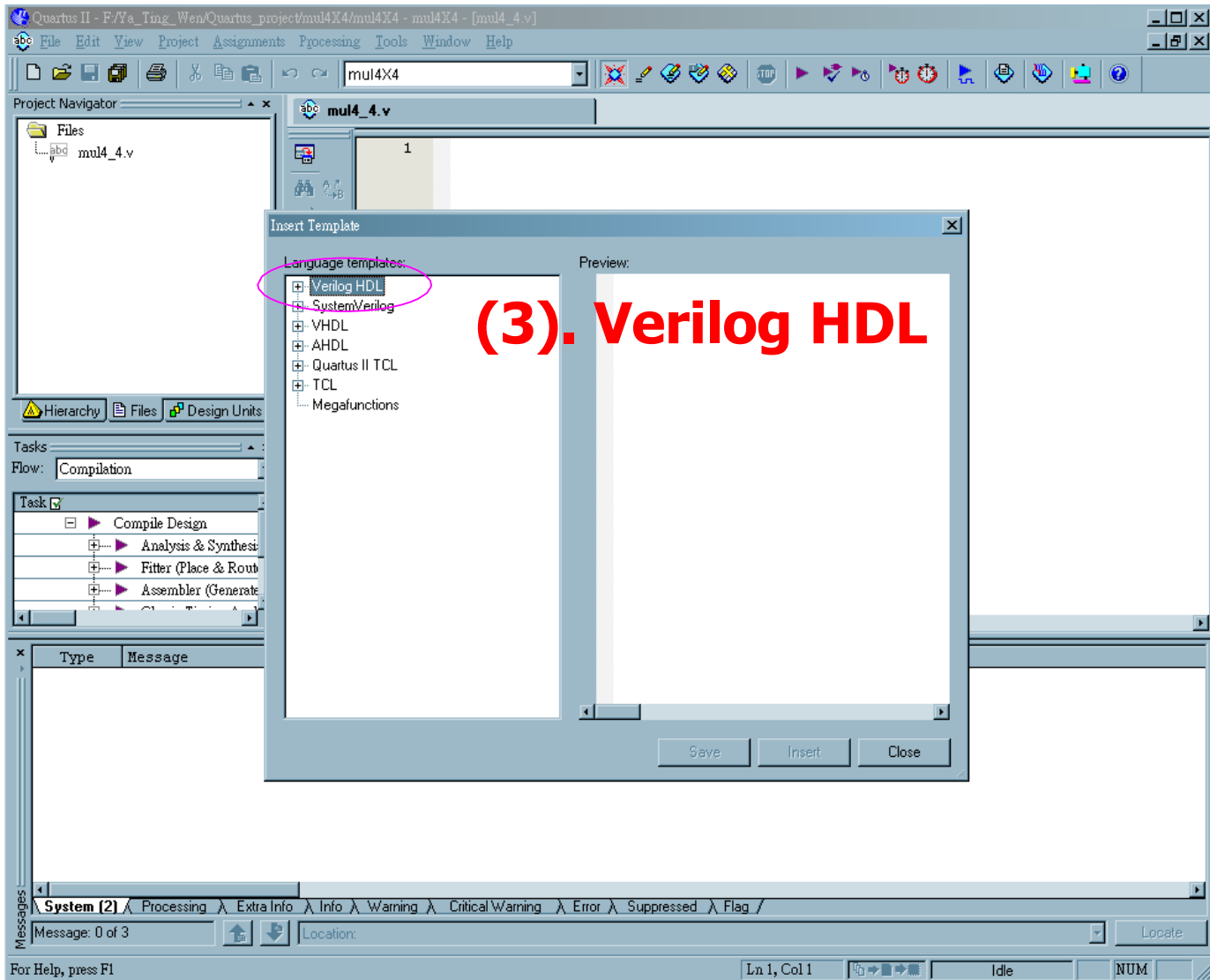
* Verilog 之方式_ Step 3



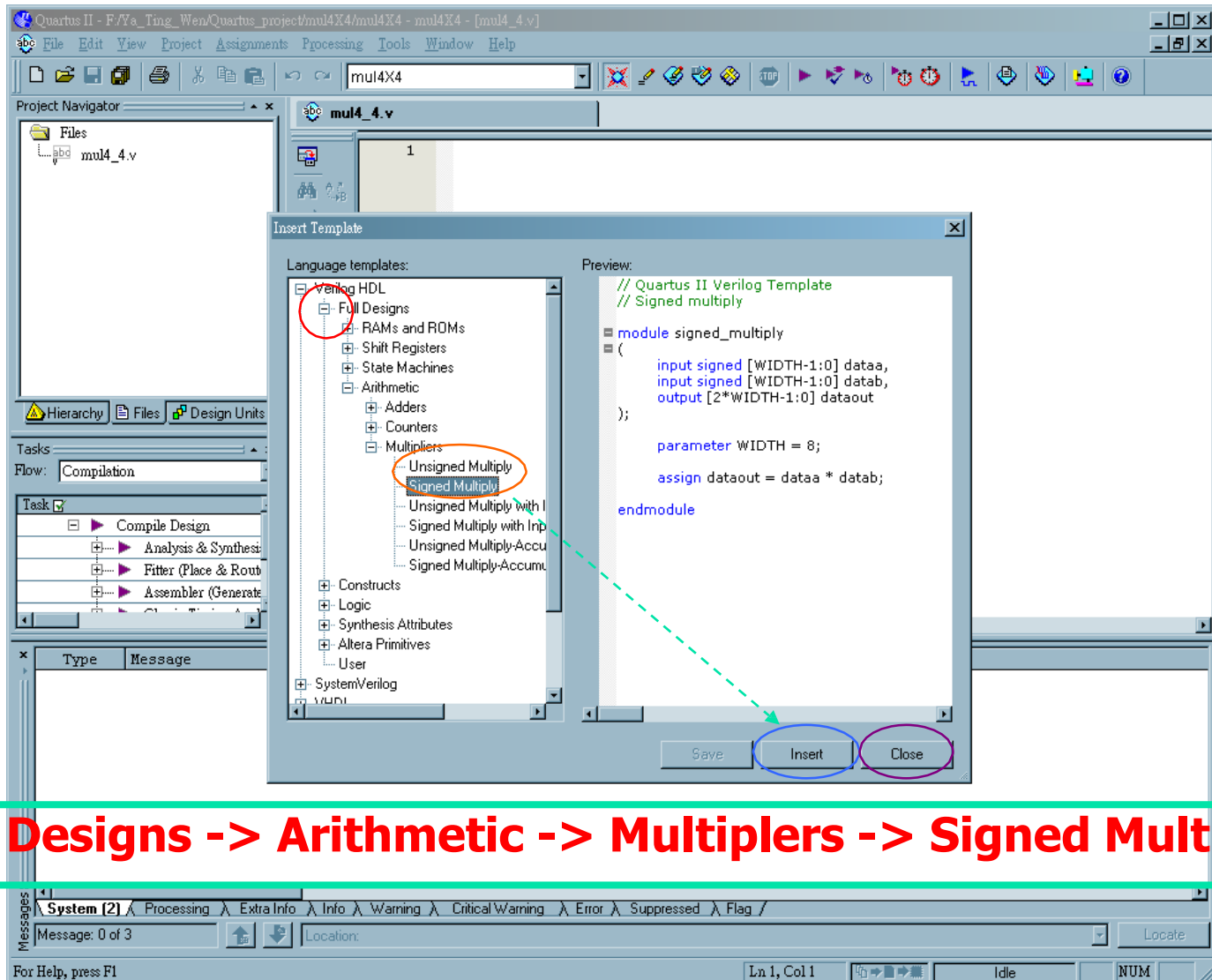
* Verilog 之方式_ Step 4



* Verilog 之方式_ Step 5



* Verilog 之方式_ Step 6



(4). Full Designs -> Arithmetic -> Multipliers -> Signed Multiply

* Verilog 之方式_Step 7

The screenshot shows the Quartus II Verilog editor interface. The Project Navigator on the left shows a file named 'mul4_4.v'. The main editor window displays the Verilog code for a module named 'mul4X4'. A yellow arrow points from the file name 'mul4_4.v' to the module name 'mul4X4' in the code, with the text '名字要一樣' (Names must be the same). A blue arrow points from the code to a modified version of the code, with the text '改成' (Change to). The modified code is shown in a separate window, where the module name is 'mul4_4' and the inputs are 'in_A' and 'in_B'. A red circle highlights the module name 'mul4_4' in the modified code. A green arrow points from the text '依自己需求修改程式內容 (注意修正檔名及“,”=>“;”)' (Modify program content according to your needs (note the correction of file name and “,”=>“;”)) to the modified code.

名字要一樣

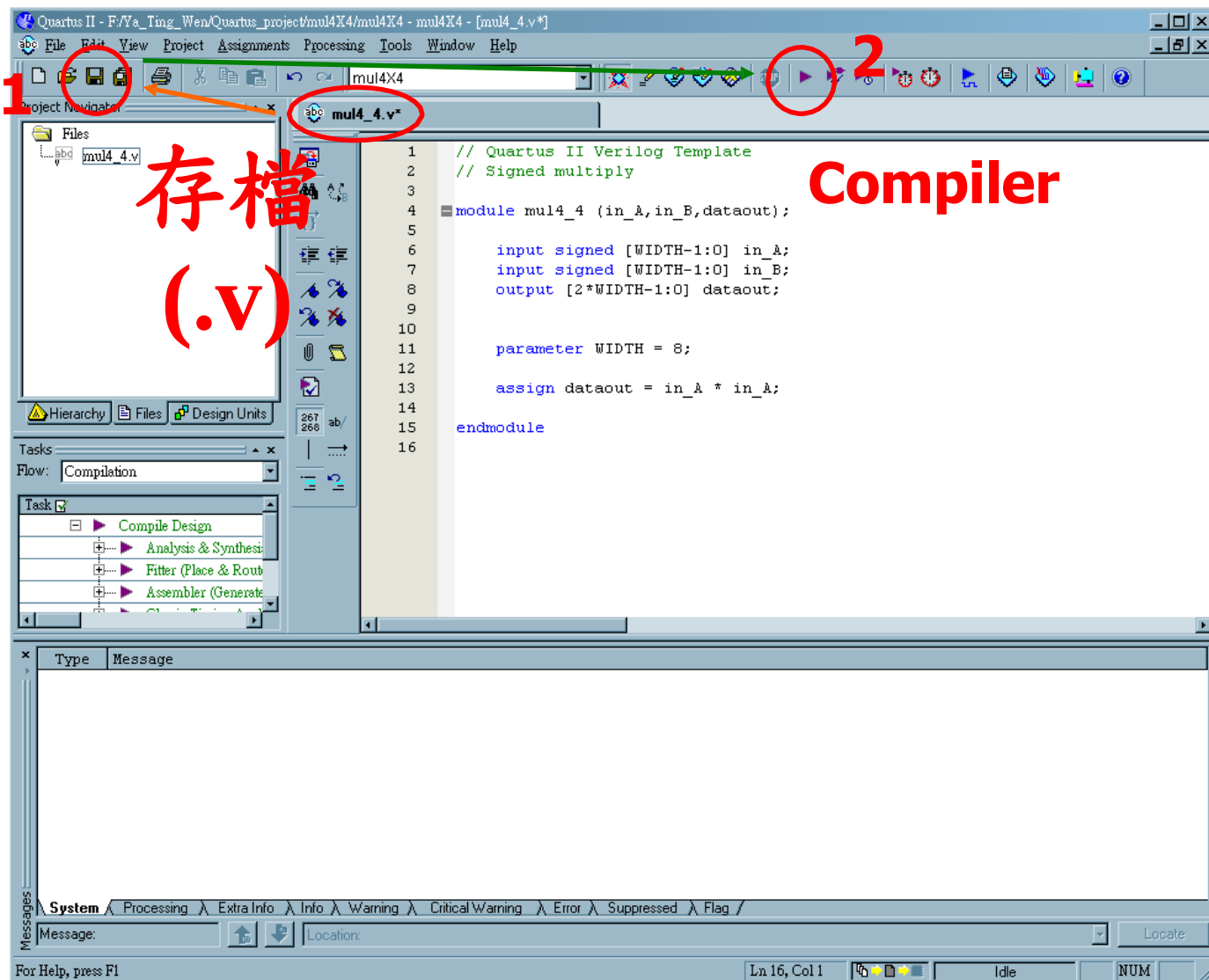
```
1 // Quartus II Verilog Template
2 Signed multiply
3 module mul4X4
4
5
6     input signed [WIDTH-1:0] dataa,
7     input signed [WIDTH-1:0] datab,
8     output [2*WIDTH-1:0] dataout
9
10
11     parameter WIDTH = 8;
12
13     assign dataout = dataa * datab;
14
15 endmodule
```

改成

```
1 // Quartus II Verilog Template
2 // Signed multiply
3
4 module mul4_4 (in_A, in_B, dataout);
5
6     input signed [WIDTH-1:0] in_A;
7     input signed [WIDTH-1:0] in_B;
8     output [2*WIDTH-1:0] dataout;
9
10
11     parameter WIDTH = 8;
12
13     assign dataout = in_A * in_A;
14
15 endmodule
```

依自己需求修改程式內容
(注意修正檔名及“,”=>“;”)

* Verilog 之方式_ Step 8



* Verilog 之方式_ Step 9

The screenshot displays the Quartus II IDE interface. The main window shows the 'Compilation Report - Flow Summary' for the 'mul4X4' project. The flow summary indicates a successful compilation on May 04, 2009, using Quartus II 8.0. The device is a Cyclone II EP2C35F672C8. The compilation process includes steps like Analysis & Synthesis, Fitter, and Assembler, all of which are marked as successful. A dialog box titled 'Quartus II' is overlaid on the report, displaying the message 'Full Compilation was successful (21 warnings)' and a '確定' (OK) button, which is circled in red. The bottom panel shows the 'Messages' window with detailed compilation logs, including commands for the assembler and timing analyzer, and a summary of 0 errors and 21 warnings.

Project Navigator: mul4_4.v

Compilation Report - Flow Summary

Flow Summary

Flow Status	Successful - Mon May 04 20:24:18 2009
Quartus II Version	8.0 Build 215 05/29/2008 SJ Full Version
Revision Name	mul4X4
Top-level Entity Name	mul4_4
Family	Cyclone II
Device	EP2C35F672C8
Timing Models	Final
Met timing requirements	Yes
Total logic elements	0 / 33,216 (0 %)
Total combinational functions	0 / 33,216 (0 %)
Total combinational functions	0 / 33,216 (0 %)
Total combinational functions	0
Total combinational functions	32 / 475 (7 %)
Total combinational functions	0
Total combinational functions	0 / 483,840 (0 %)
Total combinational functions	1 / 70 (1 %)
Total combinational functions	0 / 4 (0 %)

Quartus II

Full Compilation was successful (21 warnings)

確定

Messages

Type: Message

Info: Command: quartus_asm --read_settings_files=off --write_settings_files=off mul4X4 -c mul4X4

Info: Writing out detailed assembly data for power analysis

Info: Assembler is generating device programming files

Info: Quartus II Assembler was successful. 0 errors, 0 warnings

Info: *****

Info: Running Quartus II Classic Timing Analyzer

Info: Command: quartus_tan --read_settings_files=off --write_settings_files=off mul4X4 -c mul4X4 --timing_analysis_only

Info: Longest tpd from source pin "in_A[6]" to destination pin "dataout[9]" is 20.859 ns

Info: Quartus II Classic Timing Analyzer was successful. 0 errors, 0 warnings

Info: Quartus II Full Compilation was successful. 0 errors, 21 warnings

System (4) Processing (58) Extra Info Info (50) Warning (8) Critical Warning Error Suppressed (6) Flag /

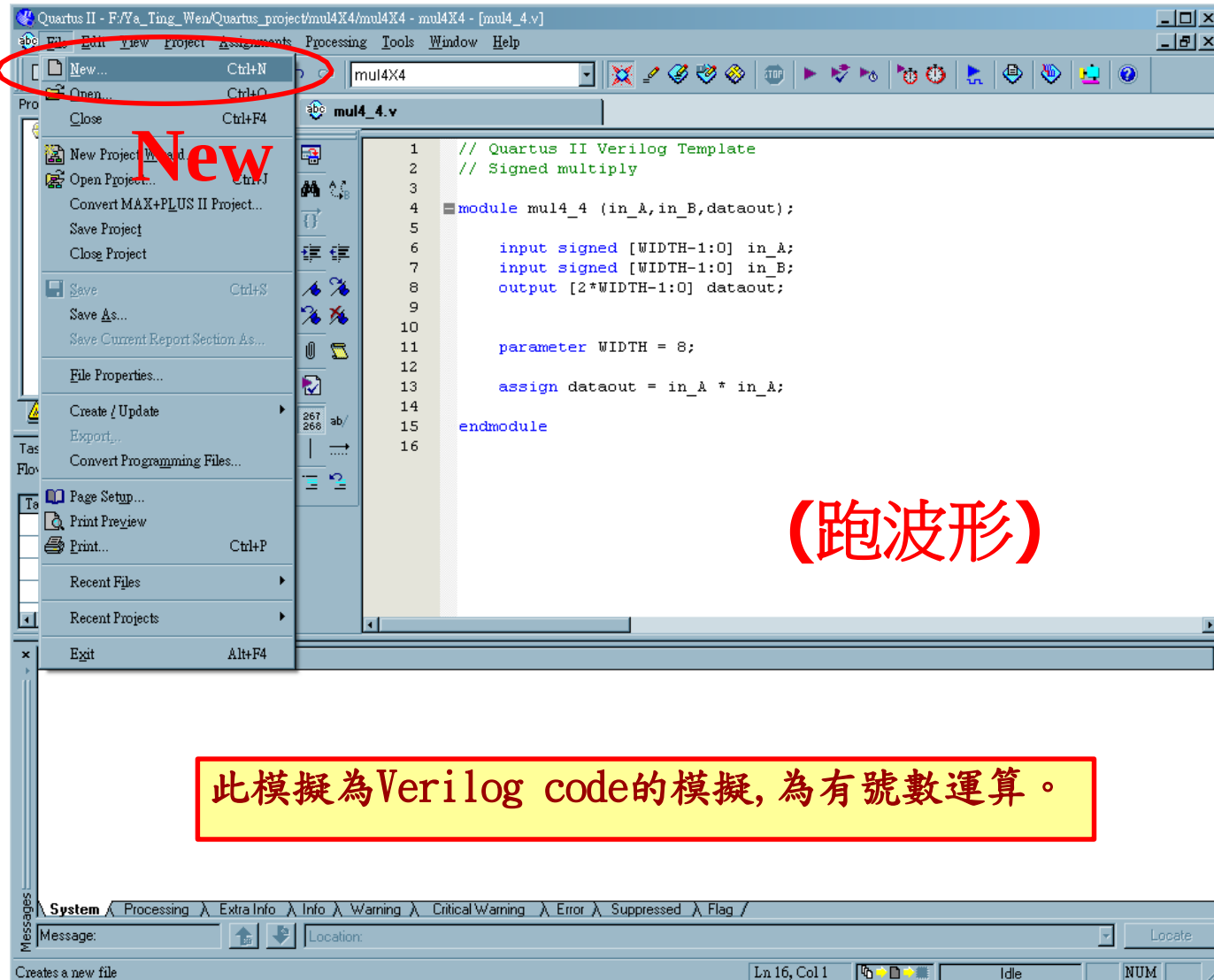
Message: 0 of 189

Location: Locate

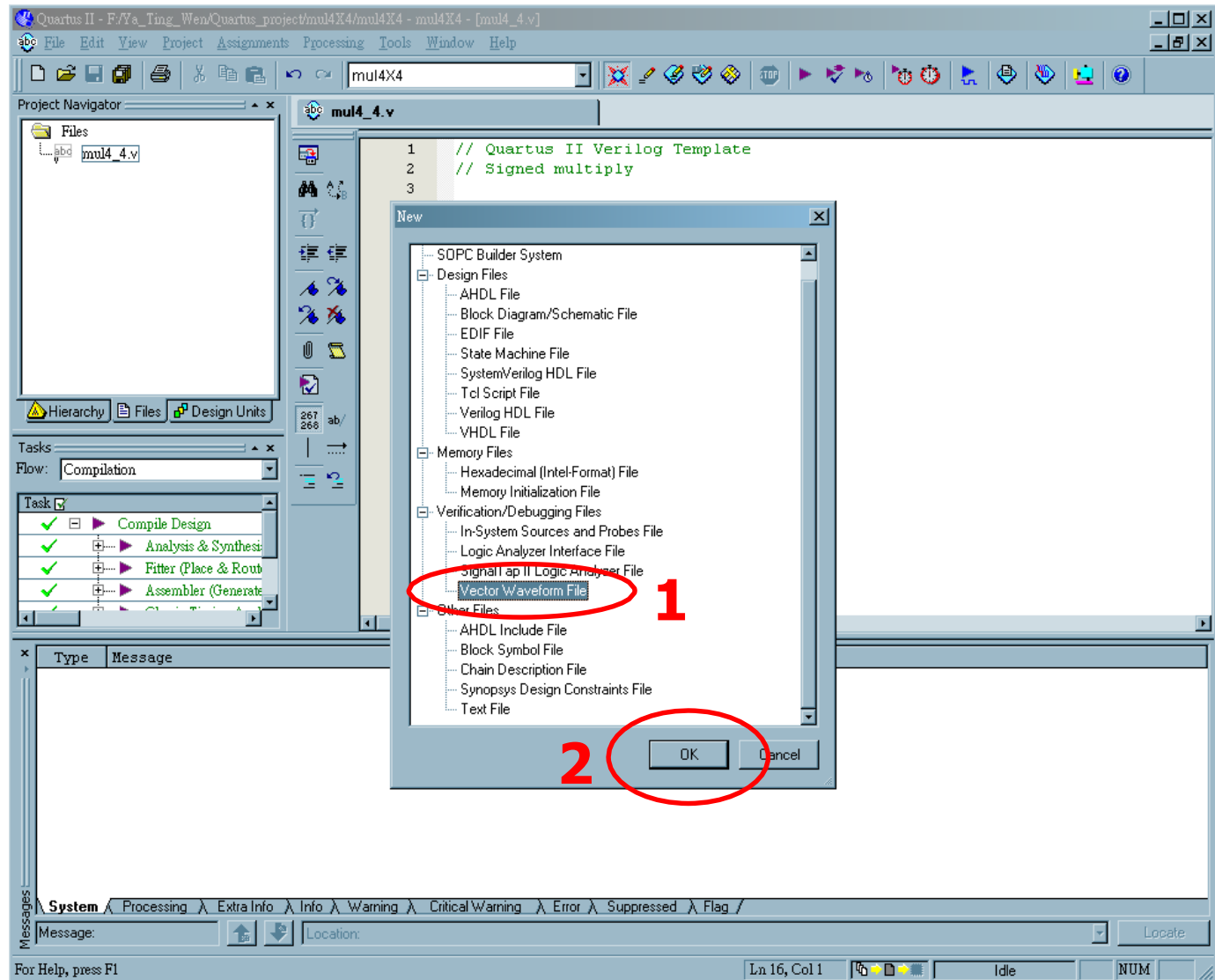
For Help, press F1

Idle NUM

乘法器模擬_Step 1



乘法器模擬_Step 2



乘法器模擬_Step 3

1. In the **Insert Node or Bus** dialog, select **Node Finder...** from the **Value type** dropdown menu.

2. In the **Node Finder** dialog, click the **List** button.

3. In the **Messages** window, select the **System** tab. The output shows the simulation results for the multiplier circuit. The text "選擇要觀看的腳位" (Select the pins to be observed) is written in red.

4. In the **Node Finder** dialog, click the **OK** button.

5. In the **Insert Node or Bus** dialog, click the **Node Finder...** button.

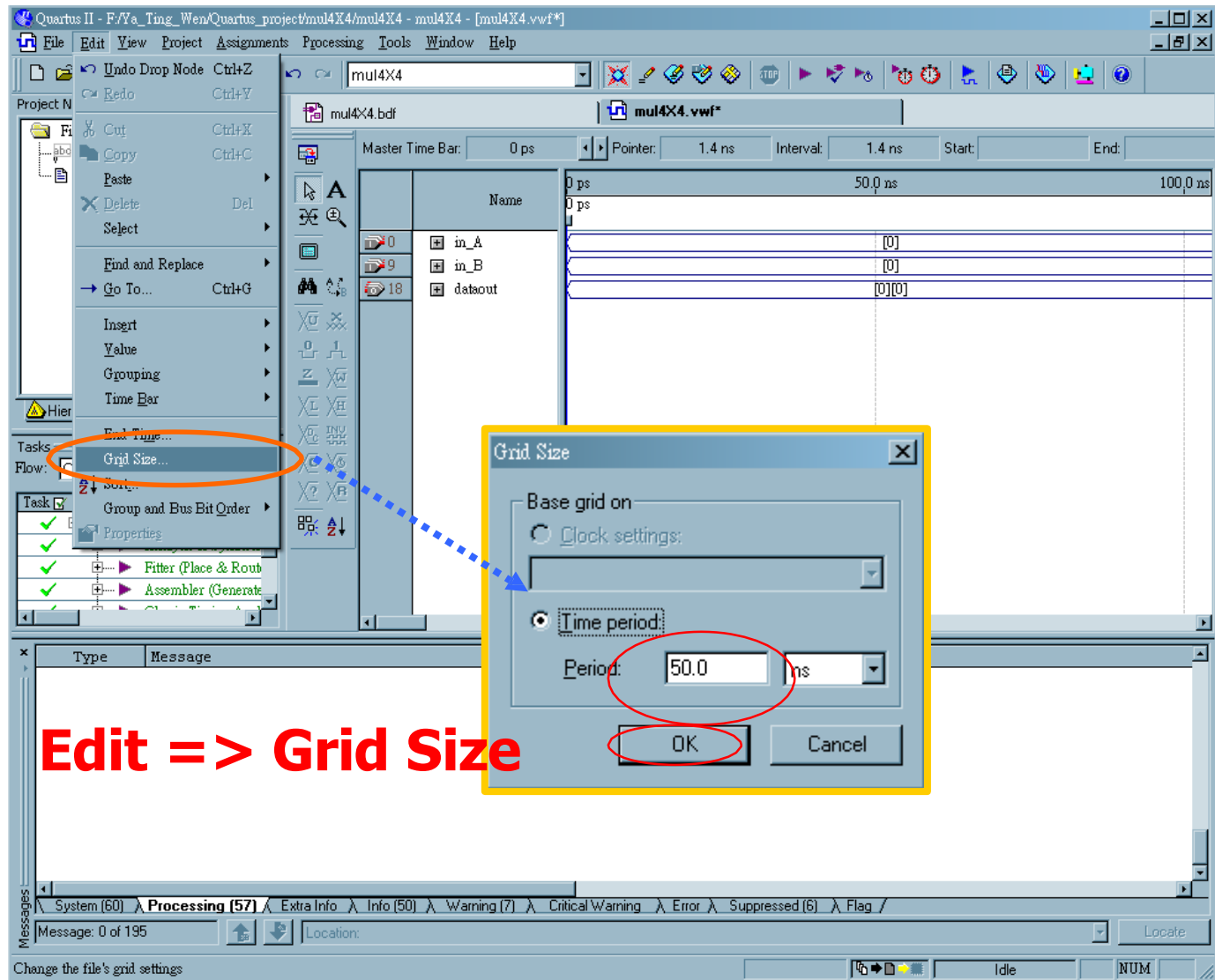
The **Node Finder** dialog shows the following nodes found:

Name	Assignments
dataout	Unassigned
dataout[0]	Unassigned
dataout[1]	Unassigned
dataout[2]	Unassigned
dataout[3]	Unassigned
dataout[4]	Unassigned
dataout[5]	Unassigned
dataout[6]	Unassigned
dataout[7]	Unassigned
dataout[8]	Unassigned
dataout[9]	Unassigned
dataout[10]	Unassigned
dataout[11]	Unassigned
dataout[12]	Unassigned
dataout[13]	Unassigned
dataout[14]	Unassigned
dataout[15]	Unassigned
in_A	Unassigned
in_A[0]	Unassigned
in_A[1]	Unassigned

The **Selected Nodes** window shows the following nodes selected:

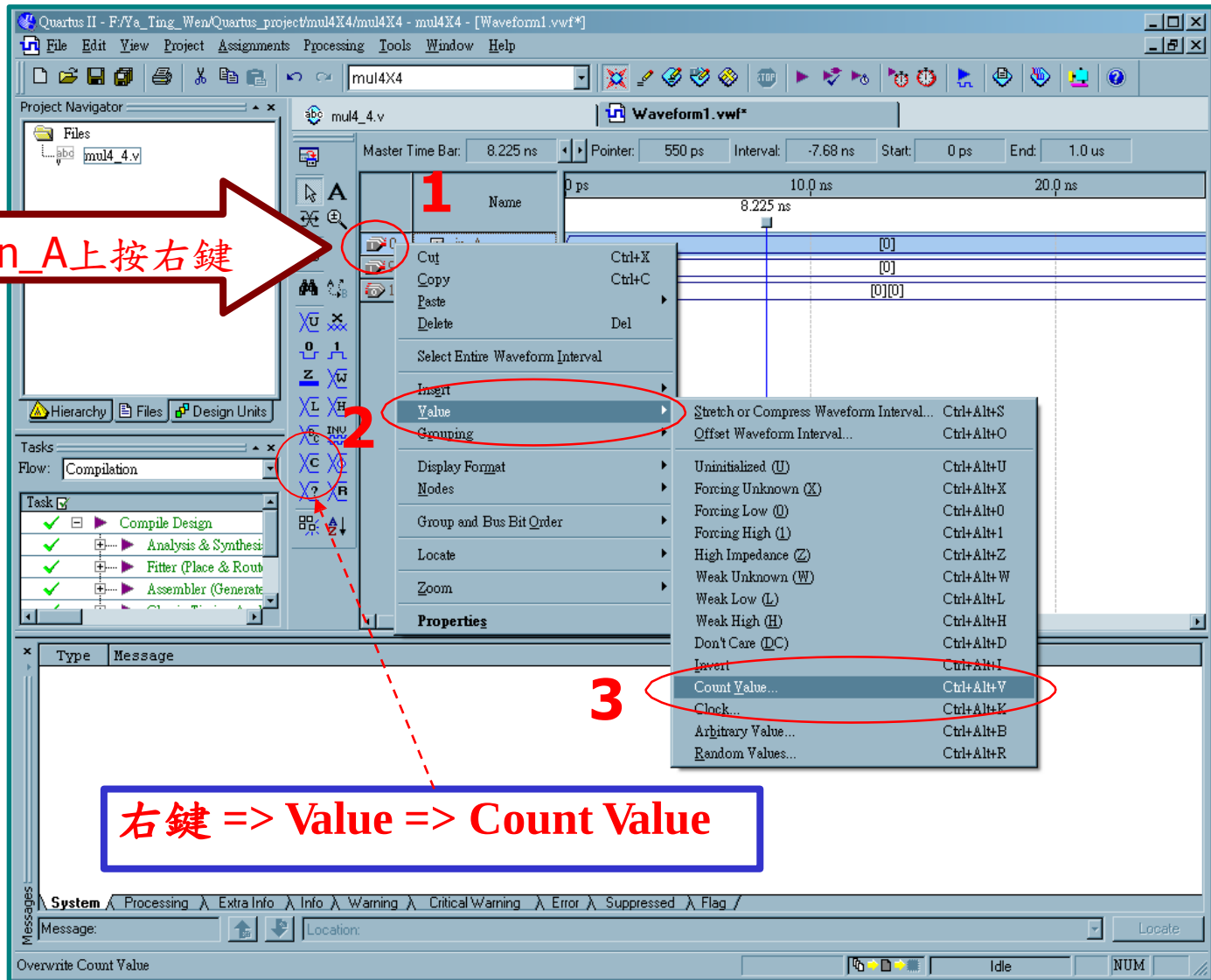
Name	Assignments
Imul4_4dataout	Unassigned
Imul4_4lin_A	Unassigned
Imul4_4lin_B	Unassigned

乘法器模擬_Step 4

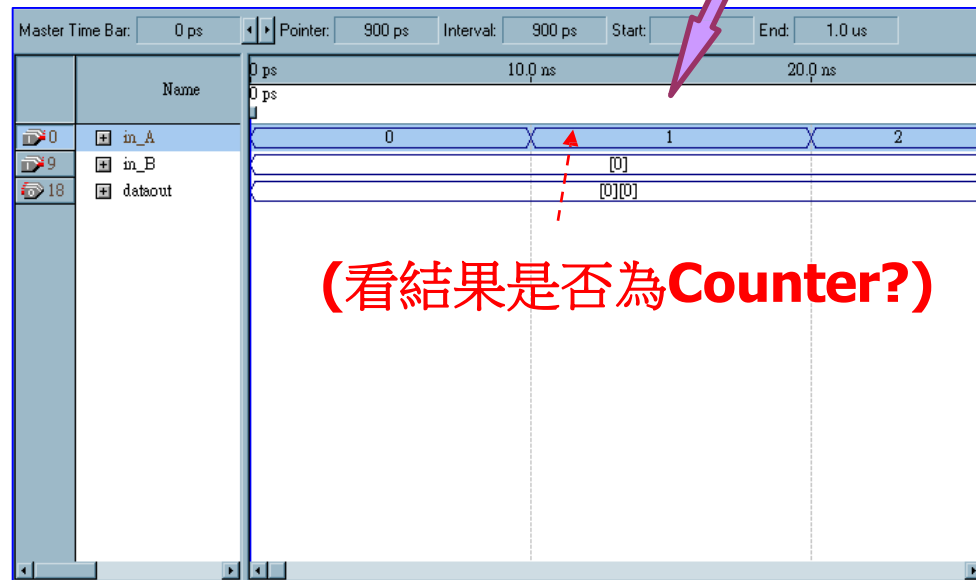
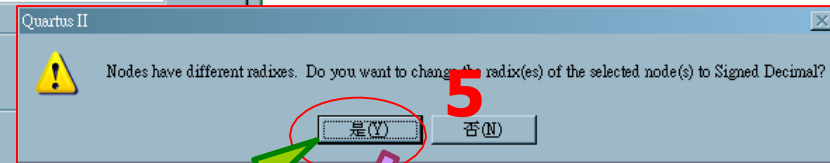
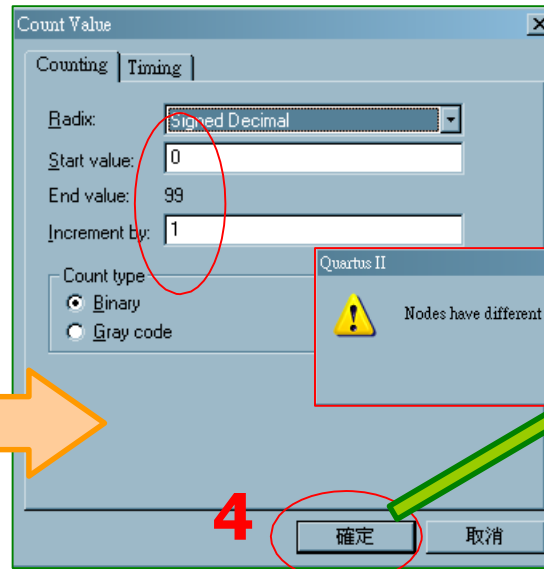
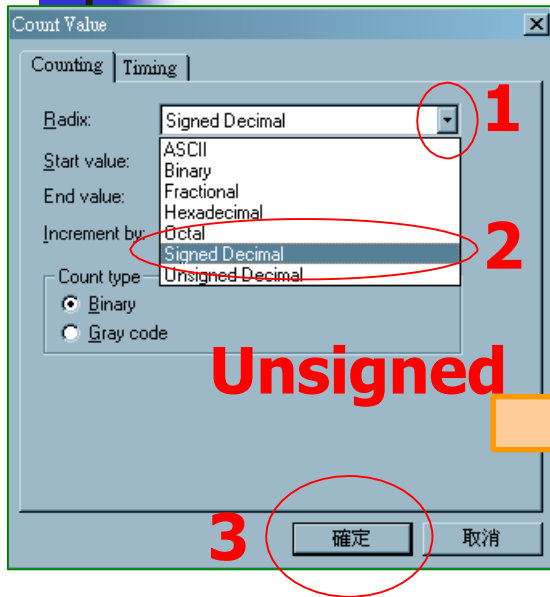


乘法器模擬_Step 5

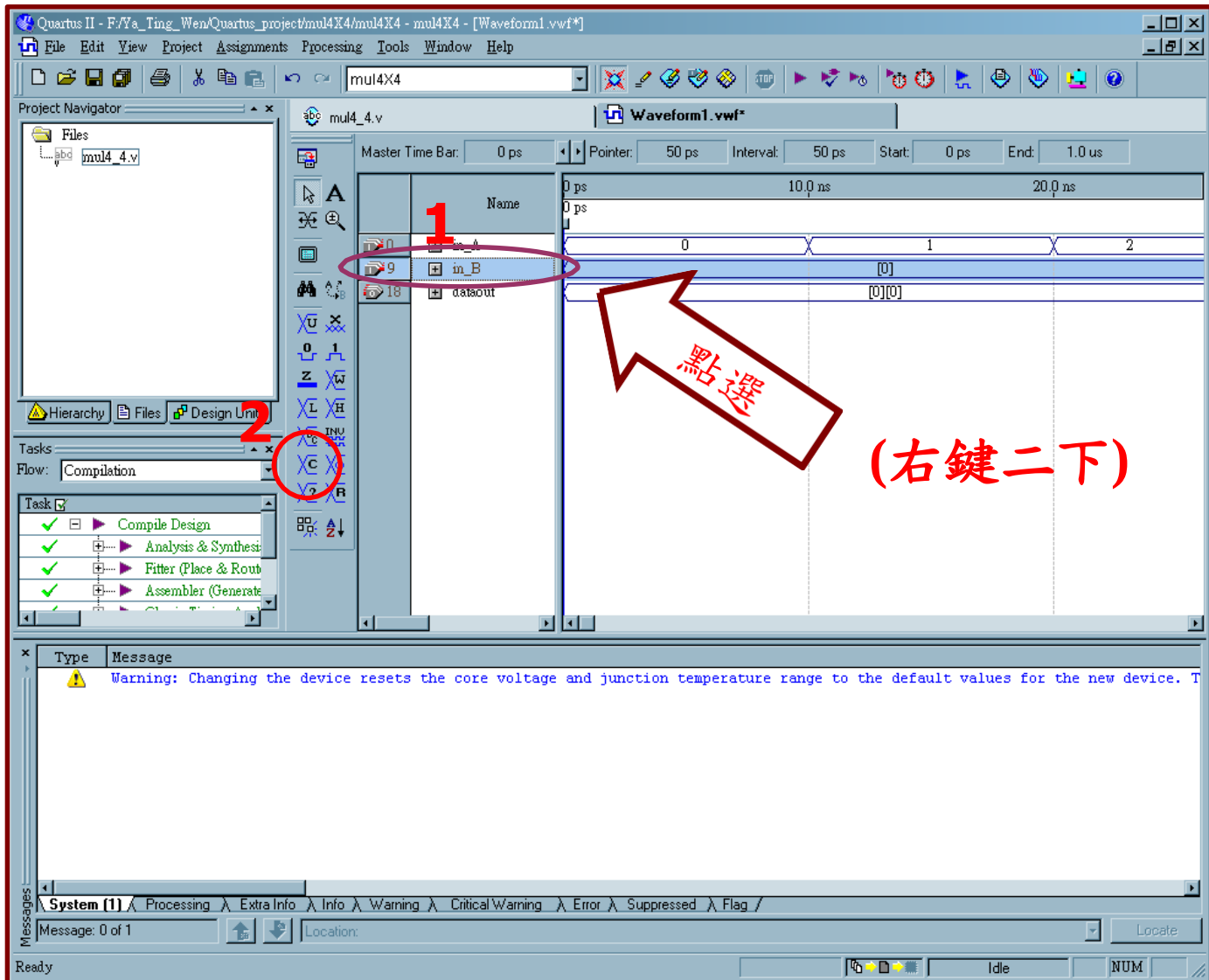
在in_A上按右鍵

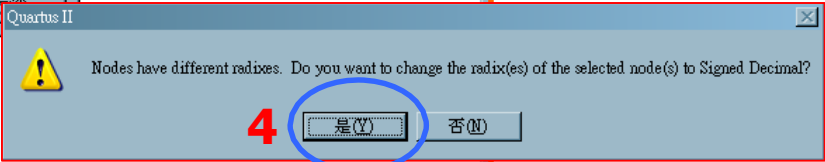


乘法器模擬_Step 6



乘法器模擬_Step 7





乘法器模擬_Step 9

The screenshot shows the Quartus II interface with the following components:

- File Menu:** The 'File' menu is open, and the 'Save As...' option is highlighted with a red circle and the number 1.
- Project Navigator:** The 'Files' pane shows the project structure, including 'mul4_4.v' and 'mul4_4.vwf'.
- Waveform Editor:** The 'Waveform1.vwf*' file is open, showing a timing diagram with signals 'in_A', 'in_B', and 'dataout'.
- Save As Dialog:** A '另存新檔' (Save As) dialog box is open, showing the file name 'mul4X4' and the file type 'Vector Waveform File (*.vwf)'. The 'Save' button is circled in green and labeled with the number 3.

Red annotations include:

- A red circle around the 'File' menu with the number 1.
- A red circle around the 'Waveform1.vwf*' file in the Project Navigator with the number 2.
- A red circle around the 'Save' button in the 'Save As' dialog with the number 3.

Additional text in the image includes:

- '存檔 (原檔名)' (Save (original filename)) in red text.
- 'mul4X4' in the file name field of the 'Save As' dialog.
- 'Vector Waveform File (*.vwf)' in the file type dropdown of the 'Save As' dialog.

乘法器模擬_Step 10

Quartus II - F:\Ya_Ting_Wen\Quartus_project\mul4X4\mul4X4 - [mul4X4.vwf]

File Edit View Project Assignments Processing Tools Window Help

mul4X4

Project Navigator

- Files
 - mul4_4.v
 - mul4X4.vwf

mul4_4.v mul4X4.vwf

Tasks

Flow: Compilation

Task

- ✓ Compile Design
- ✓ Analysis & Synthesis
- ✓ Fitter (Place & Route)
- ✓ Assembler (Generate)

Master Time Bar: 0 ps Pointer: 0 ps Interval: 0 ps Start: End:

0 ps 10.0 ns 20.0 ns

0 ps

in_A

in_B

dataout

0 1 2

5 6 7

[0][0]

Type Message

Warning: Changing the device resets the core voltage and junction temperature range to the default values for the new device. T

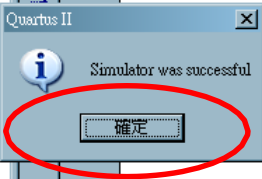
Info: Vector file mul4X4.vwf is saved in text format. You can compress it into Compressed Vector Waveform File format in order

System [2] Processing Extra Info Info Warning Critical Warning Error Suppressed Flag /

Message: 0 of 2 Location: Locate

For Help, press F1

Idle NUM



乘法器模擬_Step 12

在dataout按右鍵或點二下

1

2

Simulation Waveforms
Simulation mode: Timing

Master Time Bar: 0 ps | Pointer: 4.41 ns | Interval: 4.41 ns | Start: 0 ps | End: 1.0 us

Name	0 ps	20.0 ns	40.0 ns	60.0 ns	80.0 ns	100.0 ns	120.0 ns
in_A	0	1	2	3	4	5	6
in_B	5	6	7	8	9	10	11
dataout	0	1	2	3	4	5	6

Tasks
Flow: Compilation

- Task
- Compile Design
- Analysis & Synthesis
- Fitter (Place & Route)
- Assembler (Generate)

Messages

Type	Message
Info	Info: Running Quartus II Simulator
Info	Info: Command: quartus_sim --read_settings_files=on -
Info	Info: Using vector source file "F:/Ya_Ting_Wen/Quartu
Info	Info: Option to preserve fewer signal transitions to
Info	Info: Simulation partitioned into 1 sub-simulations
Info	Info: Simulation coverage is 73.21 %
Info	Info: Number of transitions in simulation is 1504
Info	Info: Quartus II Simulator was successful. 0 errors, 0 warnings

Message: 0 of 16

Location:

Location

Sets the properties of the current selection

乘法器模擬_Step 13

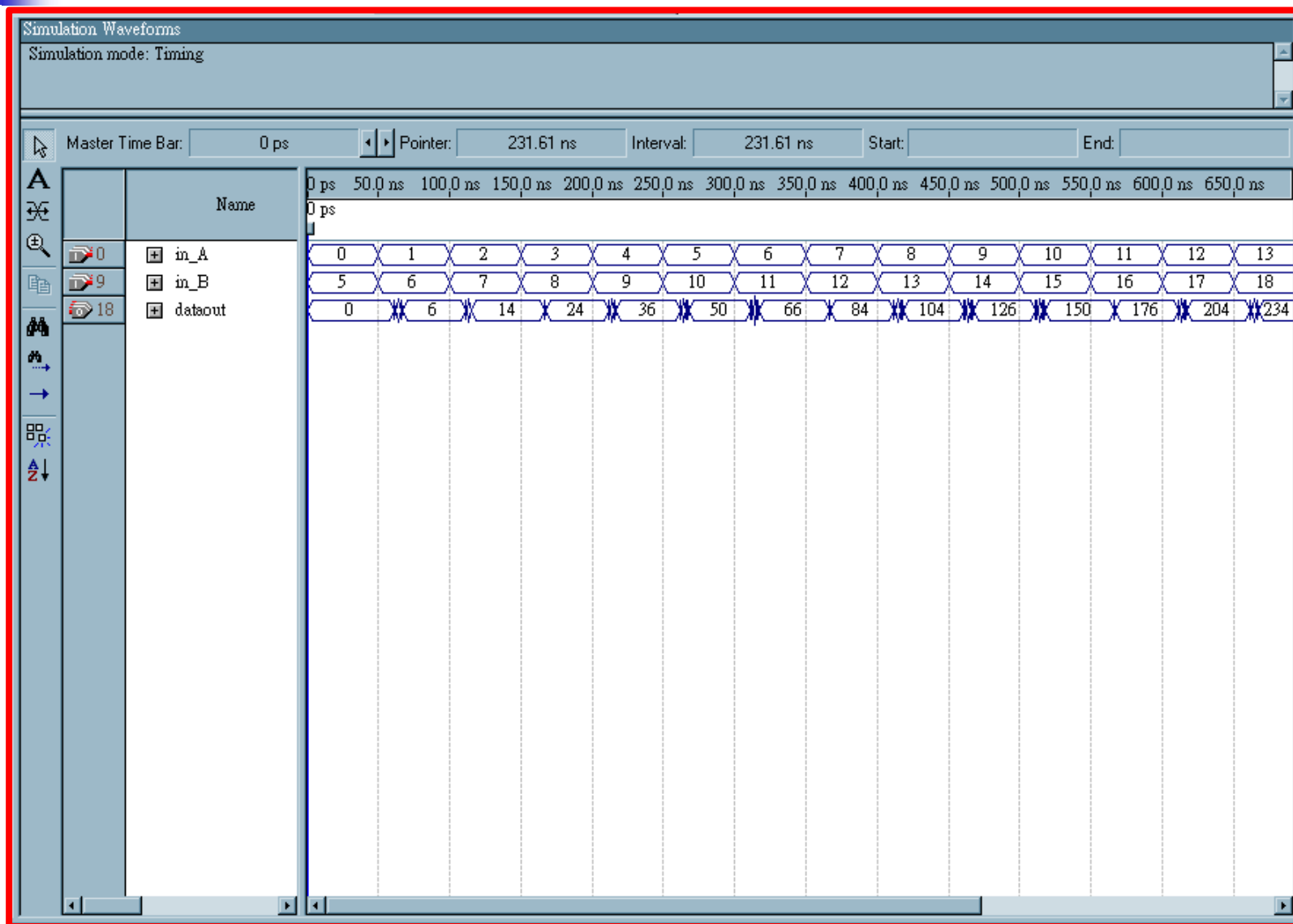
The screenshot displays the Quartus II simulation environment. The main window shows the 'Simulation Waveforms' for 'mul4X4.vwf' in 'Timing' mode. The 'Node Properties' dialog is open for the node 'dataout', showing the following settings:

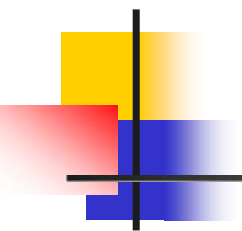
- Name: dataout
- Type: OUTPUT
- Value type: 9-Level
- Radix: ASCII
- Bus width: (empty)
- Display group: (empty)
- Selected Radix: **Unsigned Decimal** (highlighted with a red circle and the text 'Unsigned Decimal' in red)
- Other Radix options: ASCII, Binary, Fractional, Hexadecimal, Octal, Signed Decimal, Signed Decimal

The background shows the 'Project Navigator' with files 'mul4_4.v' and 'mul4X4.vwf'. The 'Messages' window at the bottom displays simulation results:

```
Info: *****
Info: Running Quartus II Simulator
Info: Command: quartus_sim --read_settings_files=on --write_settings_files=off mul4X4 -c mul4X4
Info: Using vector source file "F:/Ya_Ting_Wen/Quartus_project/mul4X4/mul4X4.vwf"
Info: Option to preserve fewer signal transitions to reduce memory requirements is enabled
Info: Simulation partitioned into 1 sub-simulations
Info: Simulation coverage is 73.21 %
Info: Number of transitions in simulation is 1504
Info: Quartus II Simulator was successful. 0 errors, 0 warnings
```

乘法器模擬_Step 14





Introduction to IP (1)

- 介紹

- 乘法器

 - ✓ 創建 Symbol 之方式

 - ✓ 以 Verilog 之方式

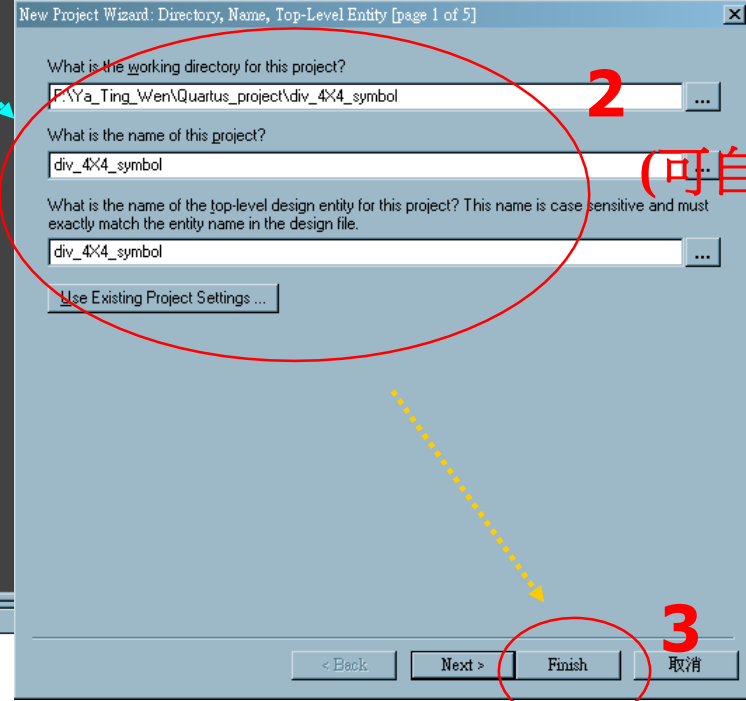
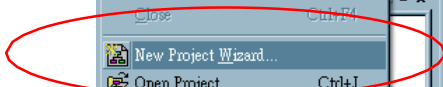
 - ✓ 模擬

- 除法器使用

 - ✓ 創建 **Symbol** 之方式

 - ✓ 模擬

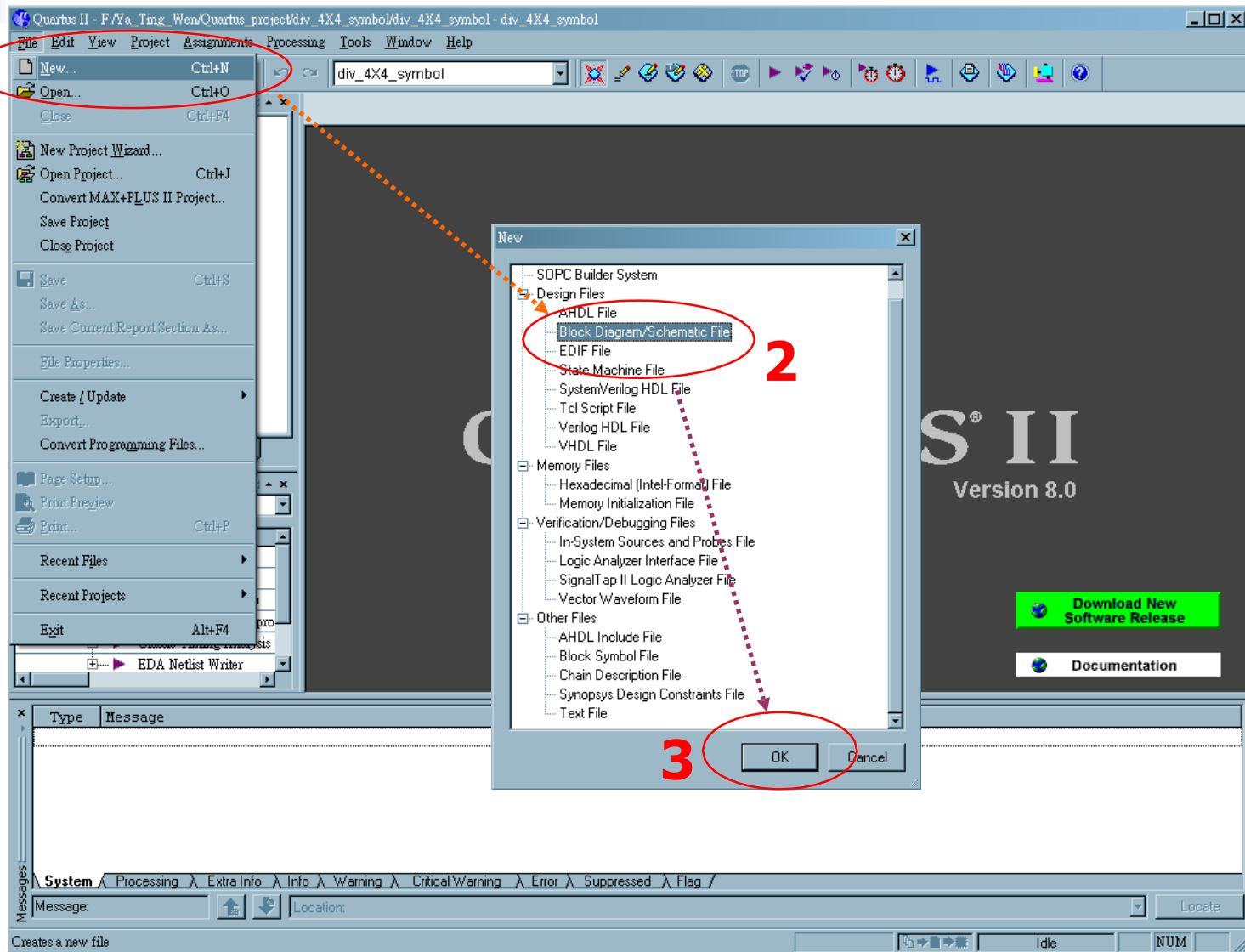
5_03_div_4X4_symbol



3

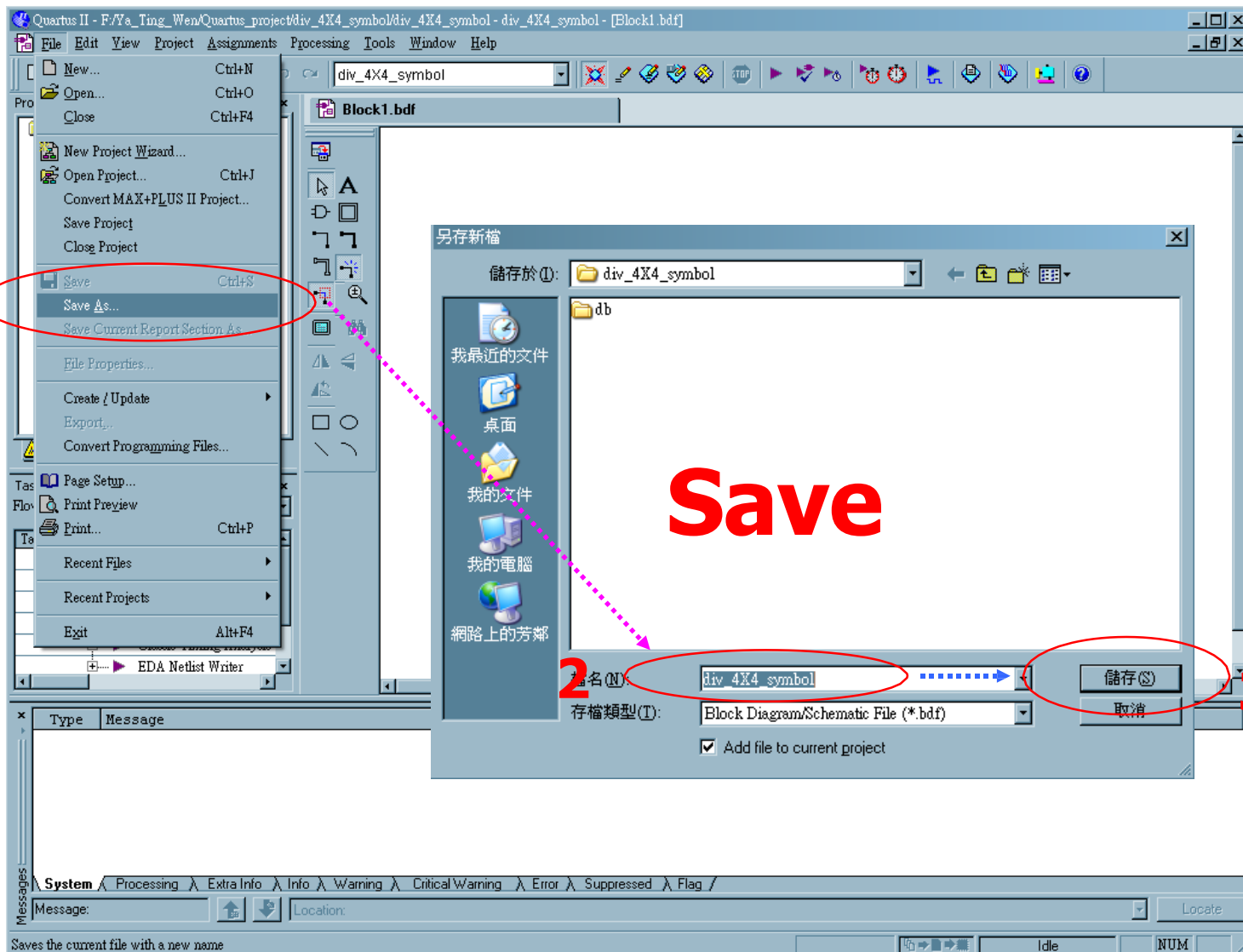
創建 Symbol 之方式_Step 2

1

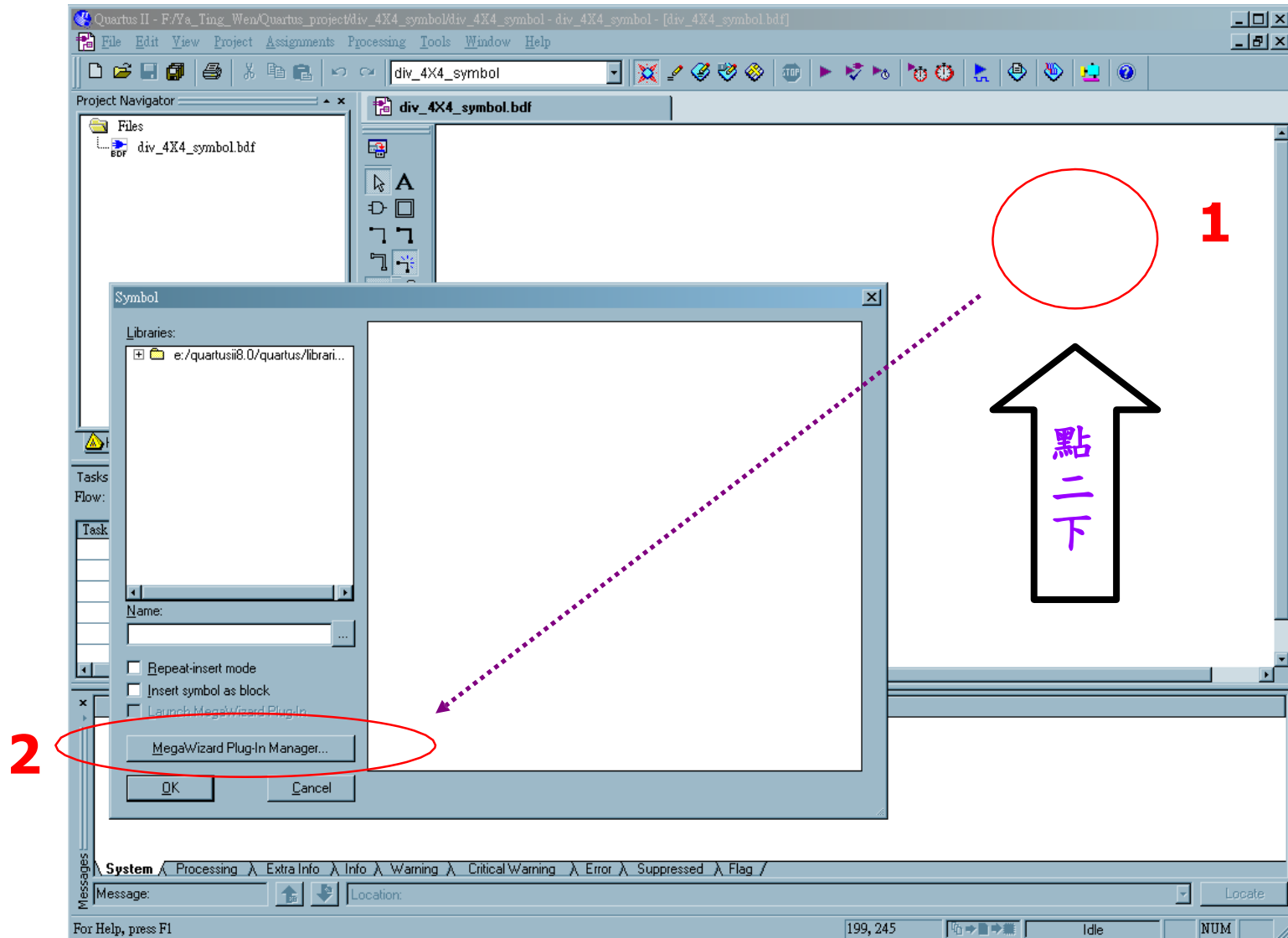


創建 Symbol 之方式_Step 3

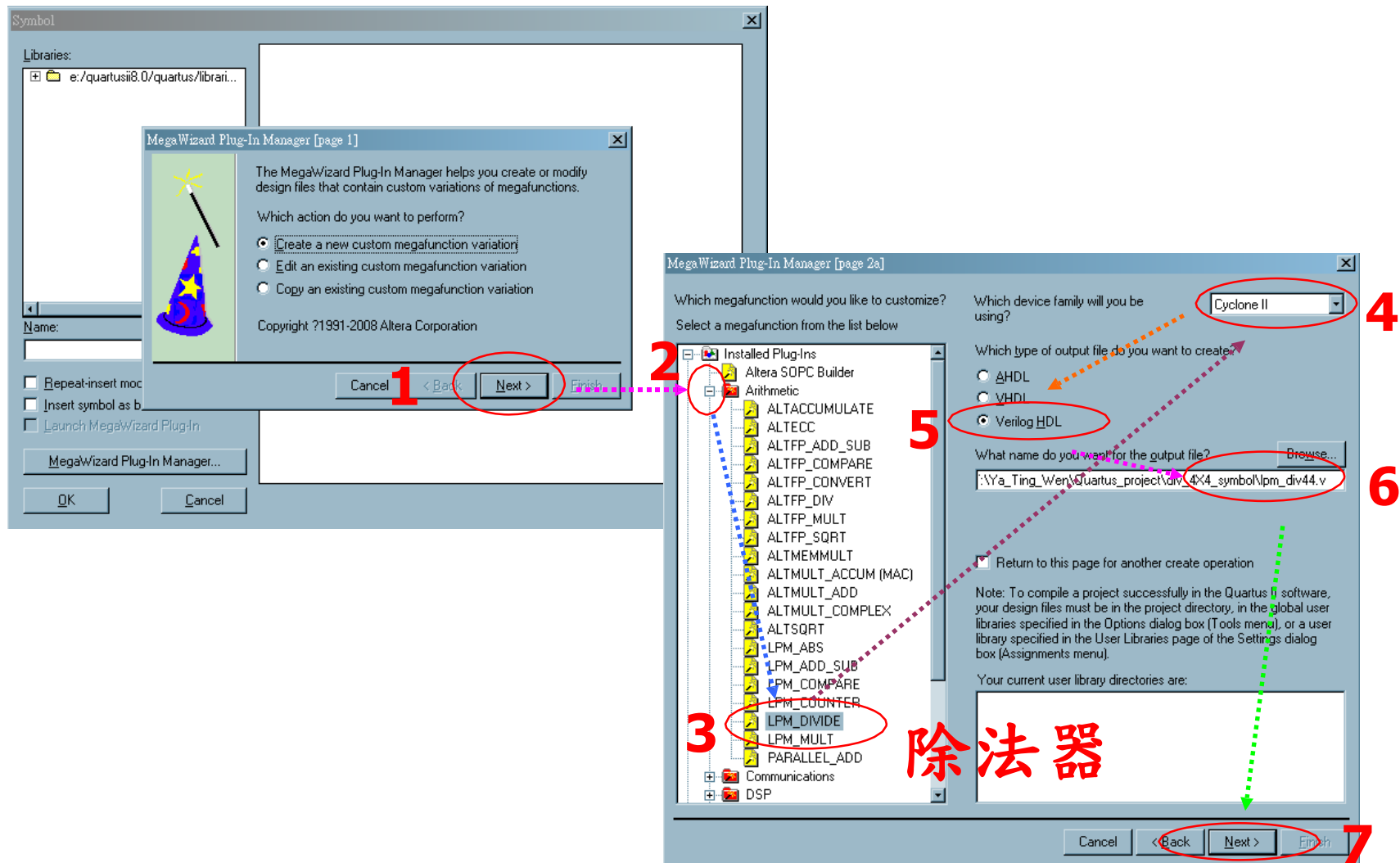
1



創建 Symbol 之方式_Step 4



創建 Symbol 之方式_Step 5



創建 Symbol 之方式_Step 6

MegaWizard Plug-In Manager - LPM_DIVIDE [page 3 of 6]

LPM_DIVIDE

About Documentation

1 Parameter Settings 2 EDA 3 Summary

General General2

Currently selected device family: Cyclone II

☒ Match project/default

How wide should the 'numerator' input bus be? 4 bits

How wide should the 'denominator' input bus be? 4 bits

1

2

3

4

lpm_div44

number[3..0] quotient[3..0]

denom[3..0] remain[3..0]

Numer is SIGNED

Denom is SIGNED

Resource Usage

...

Numerator Representation

☒ Unsigned

☐ Signed (Two's complement)

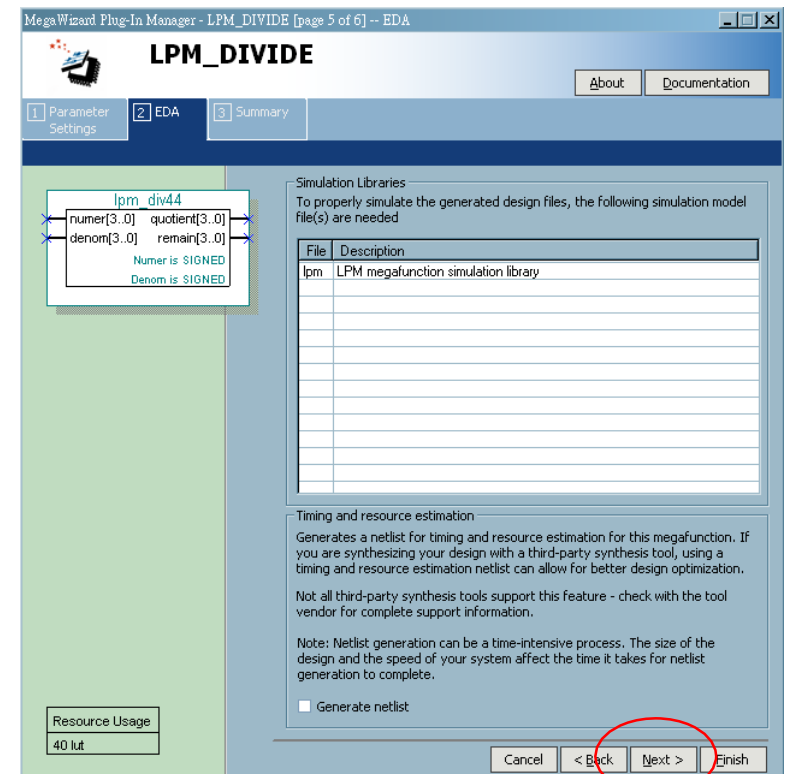
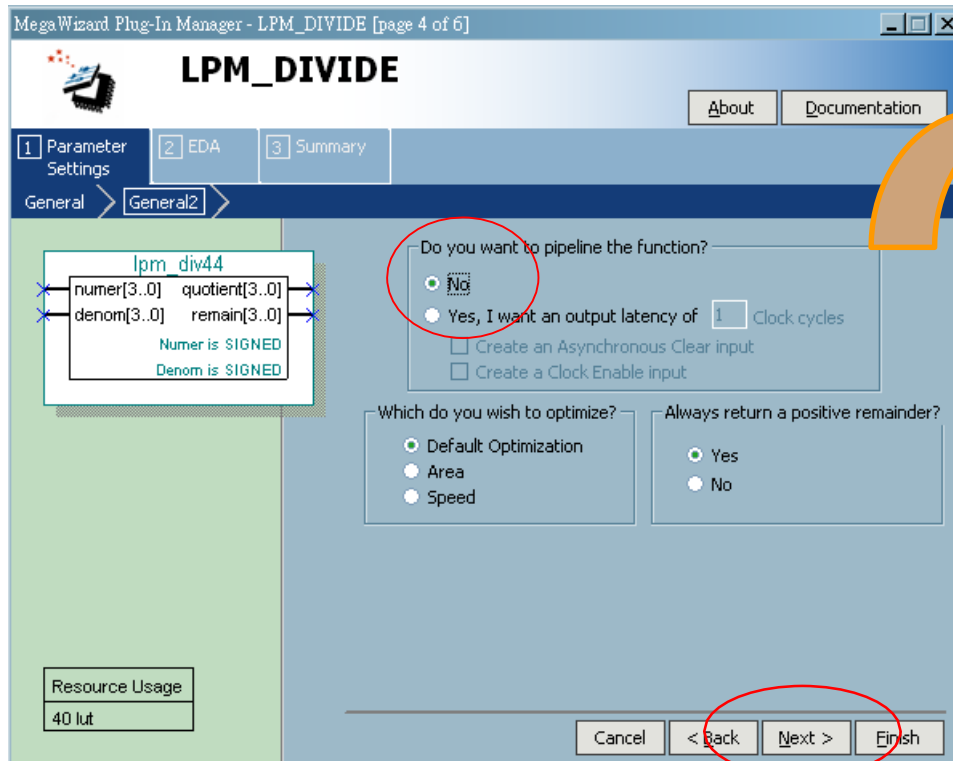
Denominator Representation

☒ Unsigned

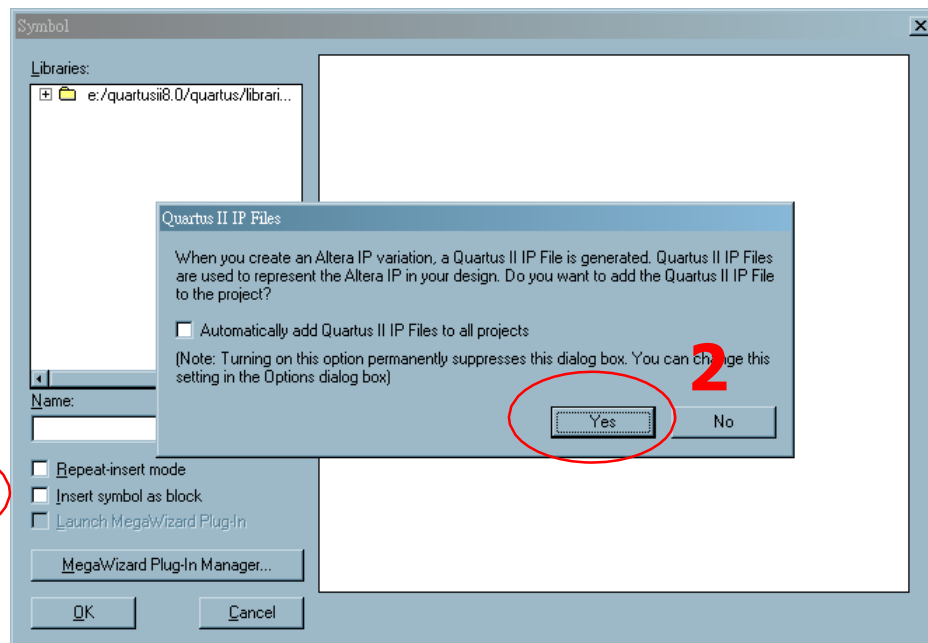
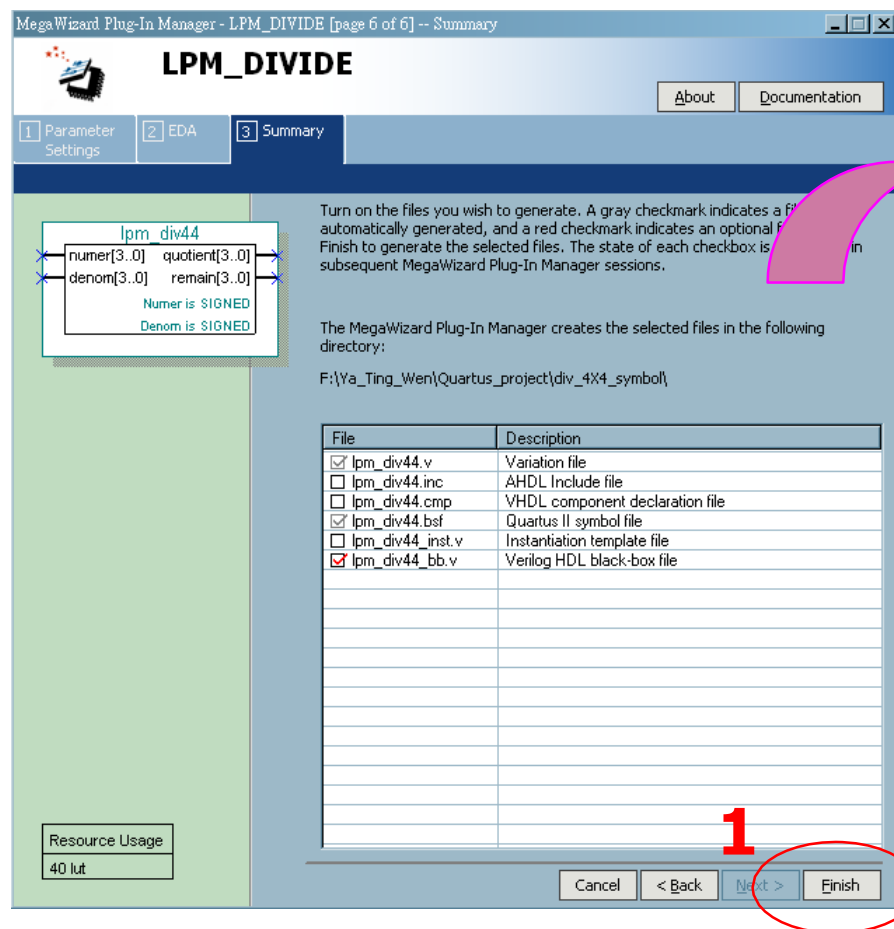
☐ Signed (Two's complement)

Cancel < Back Next > Finish

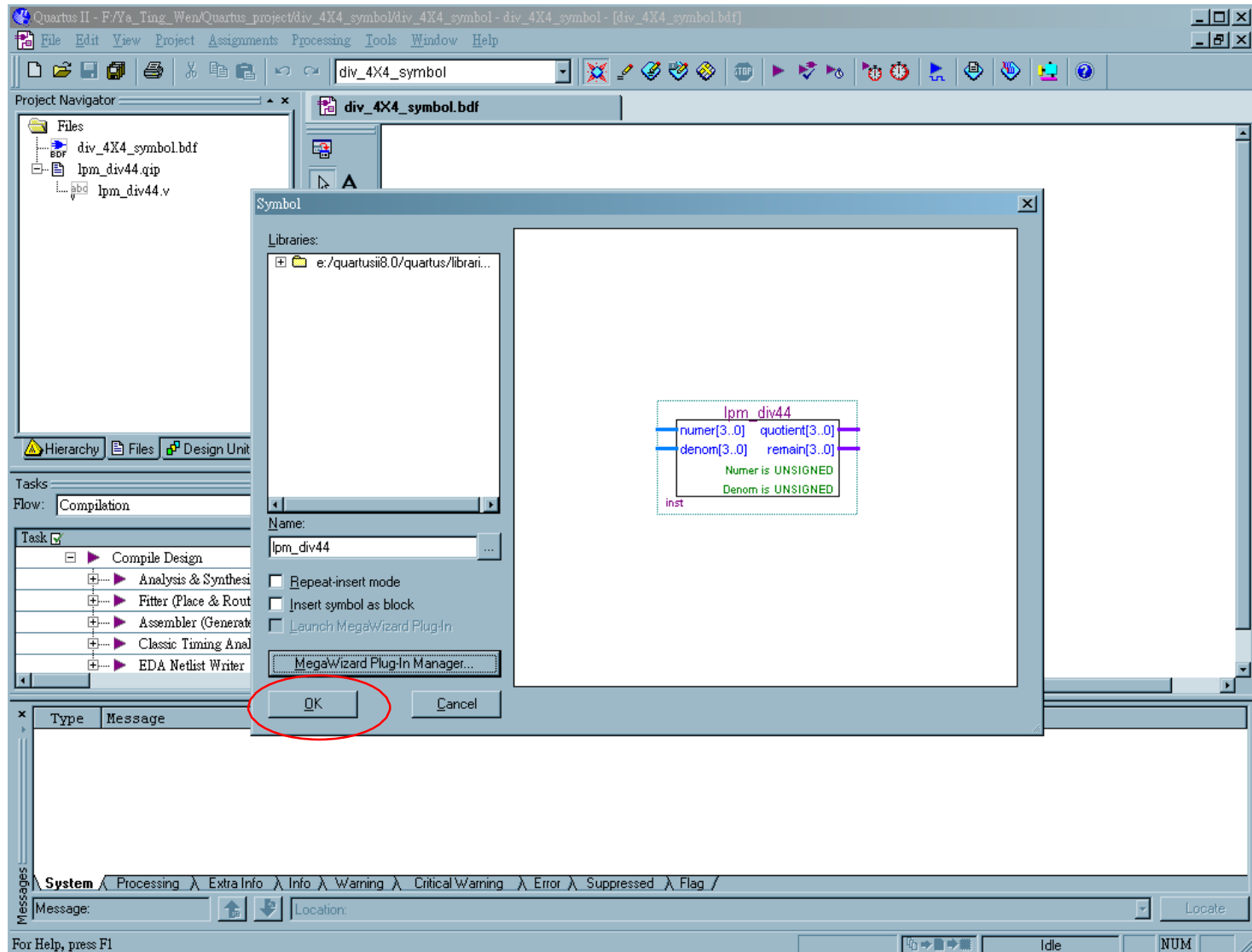
創建 Symbol 之方式_Step 7



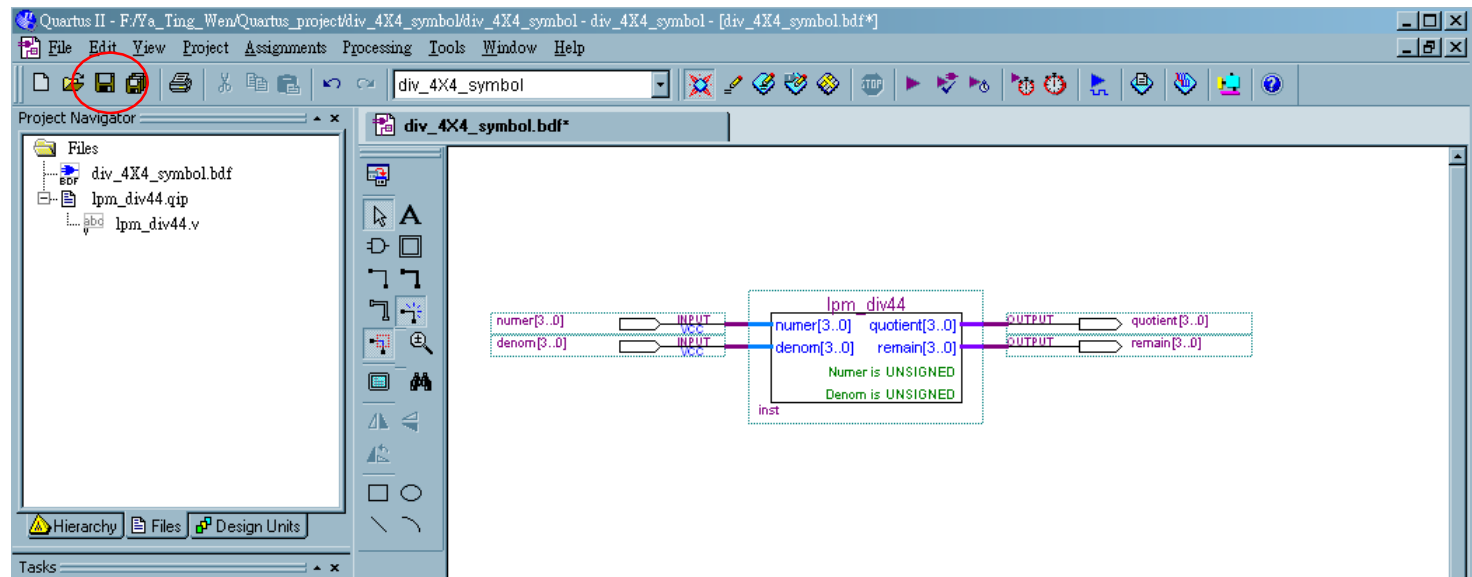
創建 Symbol 之方式_Step 8



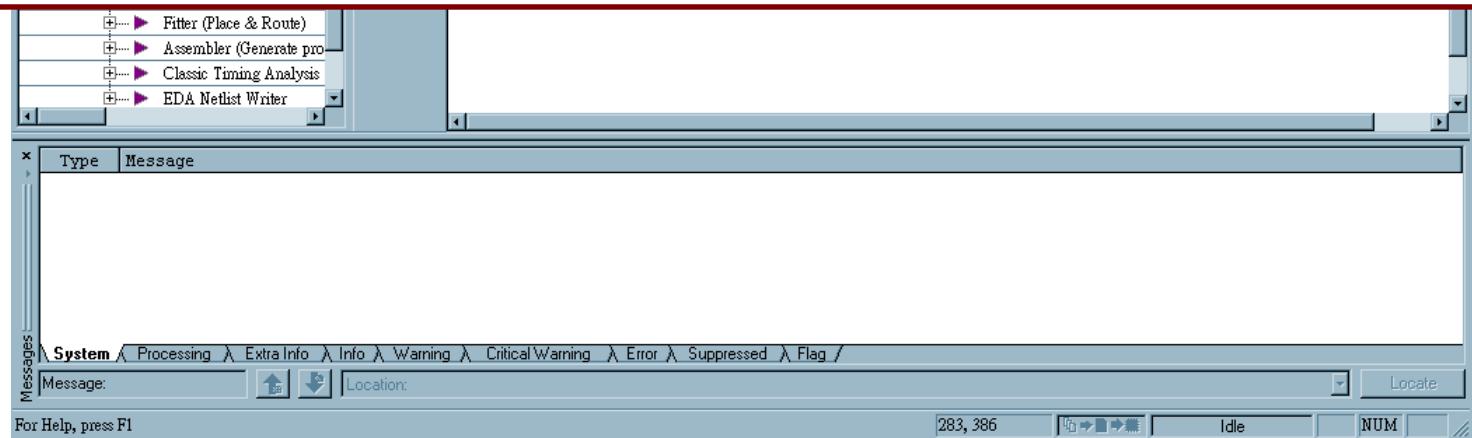
創建 Symbol 之方式_Step 9



創建 Symbol 之方式_Step 10



按右鍵=>自動加入input、output腳位=>compilation=>模擬。



除法器模擬_Step 1

The screenshot displays the Quartus II IDE interface for a project named 'div_4X4_symbol'. The main workspace shows a schematic diagram of a 4-bit divider using the 'lpm_div44' block. The inputs are 'numer[3..0]' and 'denom[3..0]', both labeled as 'INPUT'. The outputs are 'quotient[3..0]' and 'remain[3..0]', both labeled as 'OUTPUT'. The block is labeled 'inst' and includes the text 'Numer is UNSIGNED' and 'Denom is UNSIGNED'. The Project Navigator on the left shows the file structure: 'div_4X4_symbol.bdf', 'lpm_div44.qip', and 'lpm_div44.v'. The Tasks pane on the bottom left shows the 'Compilation' flow with tasks like 'Compile Design', 'Analysis & Synthesis', 'Fitter (Place & Route)', 'Assembler (Generate pro)', 'Classic Timing Analysis', and 'EDA Netlist Writer'. The Messages pane at the bottom is empty, and the status bar at the bottom shows '283, 386' and 'Idle'.

除法器模擬_Step 2

The screenshot displays the Quartus II IDE interface. The main window shows the 'Compilation Report - Flow Summary' for the project 'div_4X4_symbol'. A modal dialog box titled 'Quartus II' is overlaid on the report, displaying the message 'Full Compilation was successful (3 warnings)' with a '確定' (OK) button circled in red. The background report shows a successful flow status and various compilation metrics.

Project Navigator:

- Files
 - div_4X4_symbol.bdf
 - div4X4.qip
 - div4X4.v

Compilation Report - Flow Summary:

Flow Status	Successful - Tue May 05 16:41:35 2009
Quartus II Version	8.0 Build 215 05/29/2008 SJ Full Version
Revision Name	div_4X4_symbol
Top-level Entity Name	div_4X4_symbol
Family	Cyclone II
Device	EP2C35F672C8
Timing Models	Final
Met timing requirements	Yes
Total logic elements	24 / 33,216 (< 1 %)
Combinational logic	24 / 33,216 (< 1 %)
Registers	0 / 33,216 (0 %)
Block RAMs	0
Embedded Multiplier 9-bit elements	16 / 475 (3 %)
Total PLLs	0
Block RAMs	0 / 483,840 (0 %)
Embedded Multiplier 9-bit elements	0 / 70 (0 %)
Total PLLs	0 / 4 (0 %)

Task List:

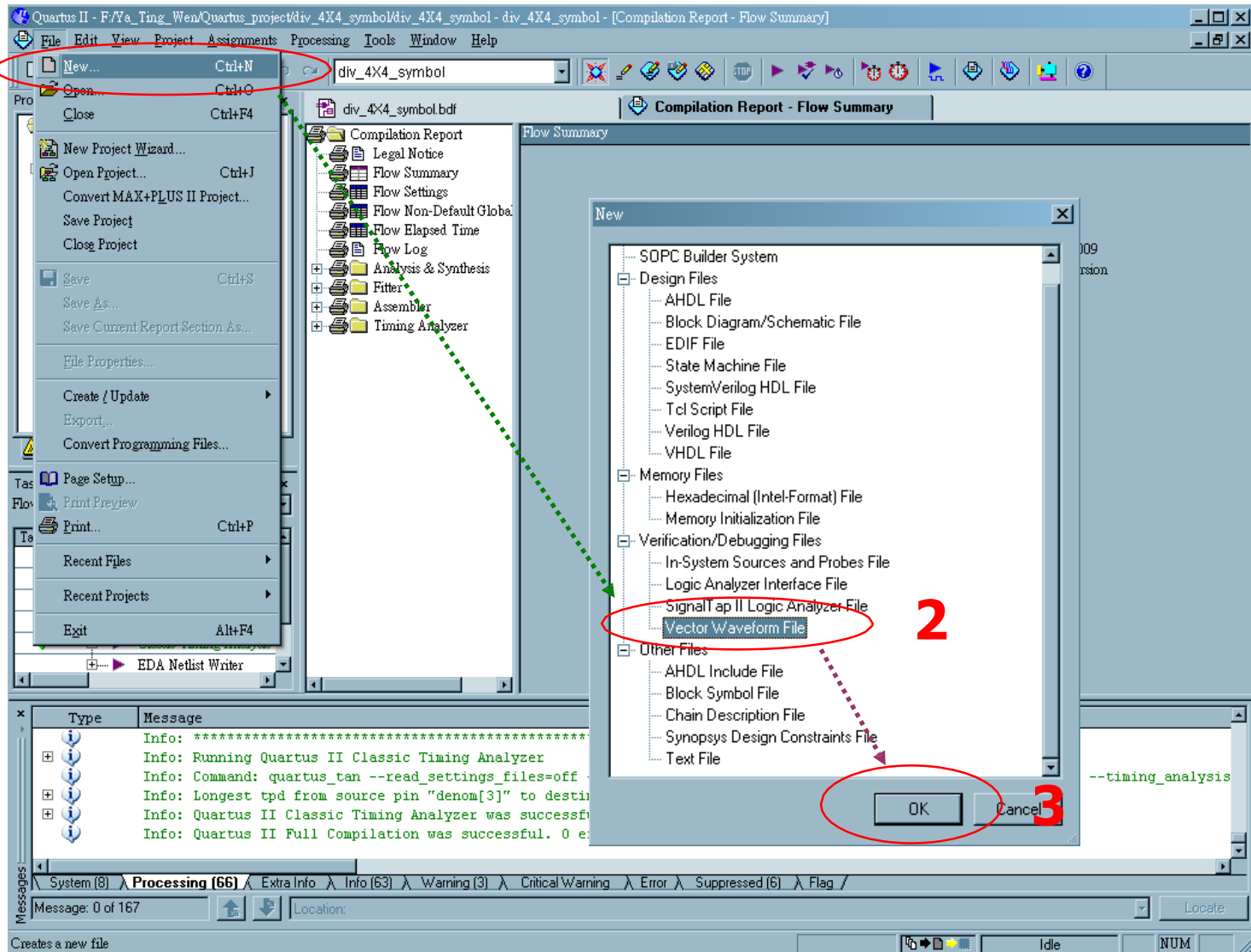
- Compile Design
- Analysis & Synthesis
- Fitter (Place & Route)
- Assembler (Generate pro)
- Classic Timing Analysis
- EDA Netlist Writer

Messages:

```
Info: *****
Info: Running Quartus II Classic Timing Analyzer
Info: Command: quartus_tan --read_settings_files=off --write_settings_files=off div_4X4_symbol -c div_4X4_symbol --timing_analysis
Info: Longest tpd from source pin "denom[3]" to destination pin "remain[3]" is 18.942 ns
Info: Quartus II Classic Timing Analyzer was successful. 0 errors, 0 warnings
Info: Quartus II Full Compilation was successful. 0 errors, 3 warnings
```

除法器模擬_Step 3

1



除法器模擬_Step 4

空白處點二下

1

2

3

Node Finder

Named: [] Filter: Pins: all Customize... List Stop OK Cancel

Look in: [div_4X4_symbol] Include subentities

Nodes Found:

Name	Assignments	T
------	-------------	---

Selected Nodes:

Name	Assignments	T
------	-------------	---

Messages

System [8] Processing [66] E

Message: 0 of 167

For Help, press F1

除法器模擬_Step 5

1 選取所要觀測之腳位

2

3

4

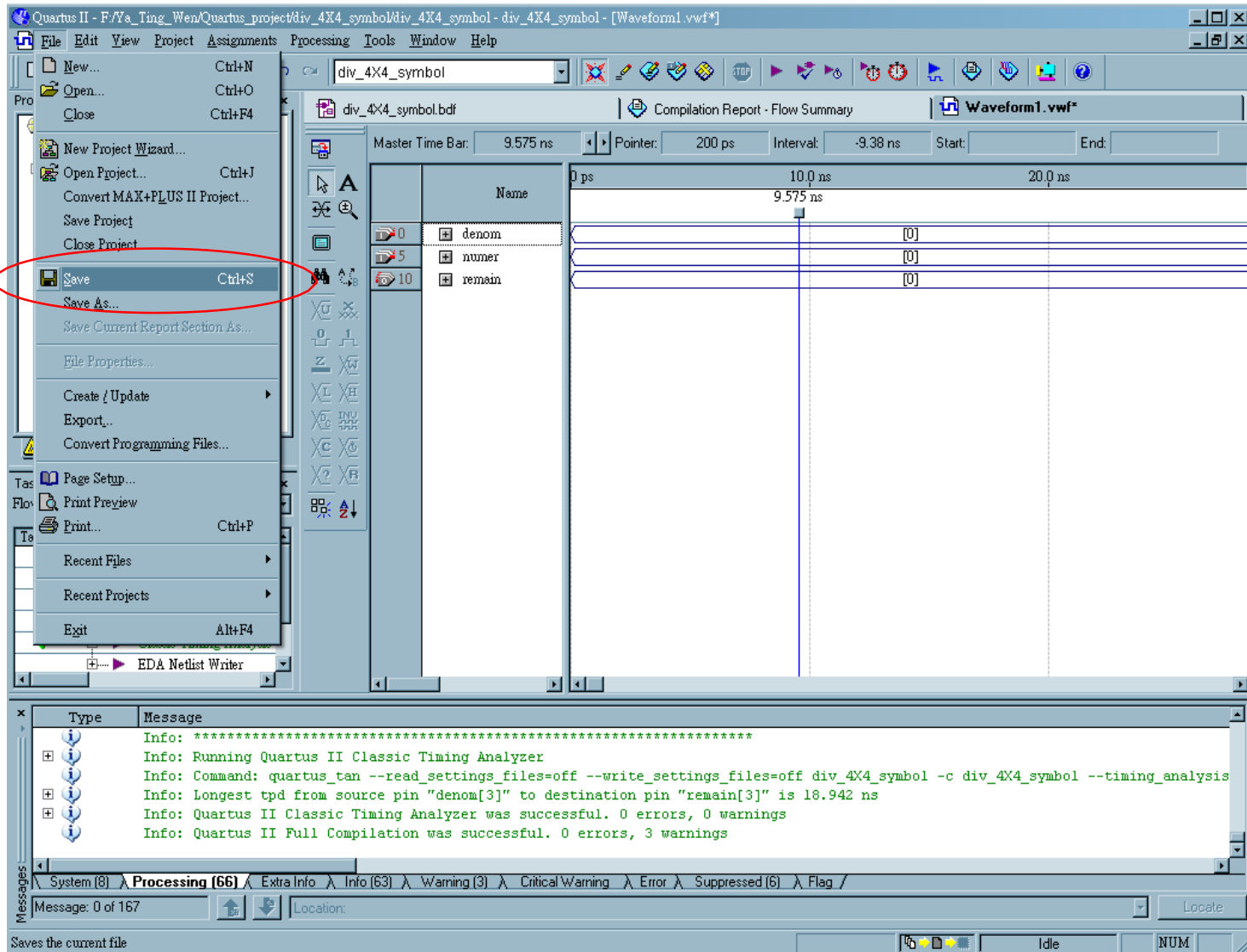
The Node Finder dialog shows the following nodes found:

Name	Assignments
denom	Unassigned
denom[0]	Unassigned
denom[1]	Unassigned
denom[2]	Unassigned
denom[3]	Unassigned
numer	Unassigned
numer[0]	Unassigned
numer[1]	Unassigned
numer[2]	Unassigned
numer[3]	Unassigned
quotient	Unassigned
quotient[0]	Unassigned
quotient[1]	Unassigned
quotient[2]	Unassigned
quotient[3]	Unassigned
remain	Unassigned
remain[0]	Unassigned
remain[1]	Unassigned
remain[2]	Unassigned
remain[3]	Unassigned

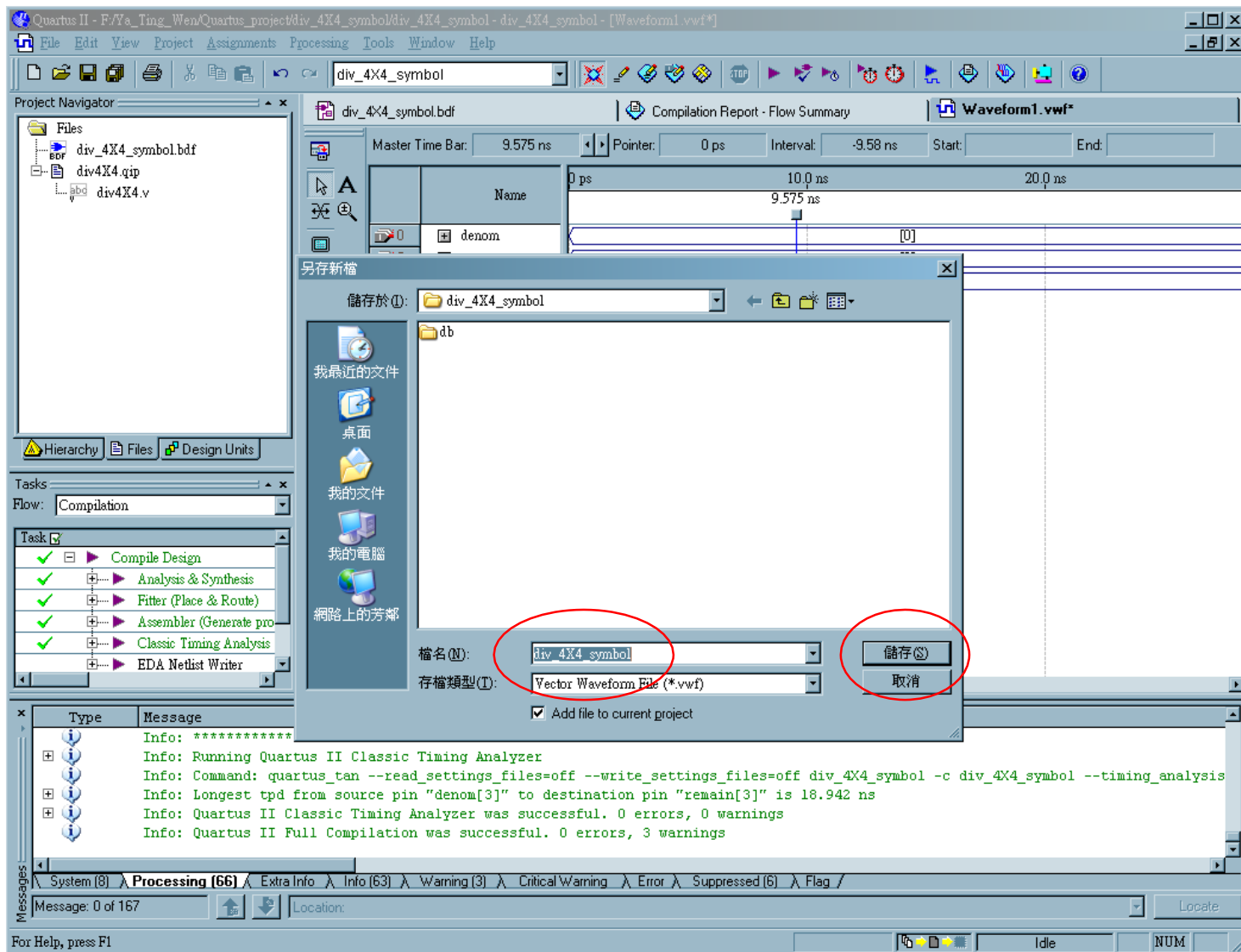
The Insert Node or Bus dialog shows the following settings:

Field	Value
Name	Multiple Items
Type	Multiple Items
Value type	9-Level
Radix	ASCII
Bus width	4
Start index	0

除法器模擬_Step 6



除法器模擬_Step 7



除法器模擬_Step 8

1

2

3

Grid Size

Base grid on

☐ Clock settings:

☒ Time period

Period: 50.0 ns

OK Cancel

Messages

Type	Message
Info	Info: Using vector source file "F:/Ya_Ting_Wen/Quartus_project/div_4X4_symbol/div_4X4_symbol.vwf"
Info	Info: Option to preserve fewer signal transitions to reduce memory requirements is enabled
Info	Info: Simulation partitioned into 1 sub-simulations
Info	Info: Simulation coverage is 100.00 %
Info	Info: Number of transitions in simulation is 5477
Info	Info: Quartus II Simulator was successful. 0 errors, 0 warnings

Change the file's grid settings

Idle NUM

除法器模擬_Step 9

Quartus II - F:\Ya_Ting_Wen\Quartus_project\div_4X4_symbol\div_4X4_symbol - div_4X4_symbol - [div_4X4_symbol.vwf]

File Edit View Project Assignments Processing Tools Window Help

div_4X4_symbol

Project Navigator

- Files
 - div_4X4_symbol.bdf
 - div4X4.qip
 - div4X4.v
 - div_4X4_symbol.vwf

Tasks

Flow: Compilation

Task

- Compile Design
- Analysis & Synthesis
- Fitter (Place & Route)
- Assembler (Generate pro
- Classic Timing Analysis
- EDA Netlist Writer

div_4X4_symbol.bdf

Compilation Report - Flow Summary

div_4X4_symbol.vwf

Master Time Bar: 0 ps Pointer: 0 ps Interval: 0 ps Start: 0 ps End: 1.0 us

Name	0 ps	5.68 ns	11.36 ns	17.04 ns	22.72 ns	28.4 ns
numer	0	0	0	0	0	0
denom	0	0	0	0	0	0
quotient	0	0	0	0	0	0
remain	0	0	0	0	0	0

Context Menu:

- Cut (Ctrl+X)
- Copy (Ctrl+C)
- Paste
- Delete (Del)
- Select Entire Waveform Interval
- Insert
- Value
- Grouping
- Display Format
- Nodes
- Group and Bus Bit Order
- Locate
- Zoom
- Properties

Sub-menu for Value:

- Stretch or Compress Waveform Interval... (Ctrl+Alt+S)
- Offset Waveform Interval... (Ctrl+Alt+O)
- Uninitialized (U) (Ctrl+Alt+U)
- Forcing Unknown (X) (Ctrl+Alt+X)
- Forcing Low (0) (Ctrl+Alt+0)
- Forcing High (1) (Ctrl+Alt+1)
- High Impedance (Z) (Ctrl+Alt+Z)
- Weak Unknown (W) (Ctrl+Alt+W)
- Weak Low (L) (Ctrl+Alt+L)
- Weak High (H) (Ctrl+Alt+H)
- Don't Care (DC) (Ctrl+Alt+D)
- Invert (Ctrl+Alt+I)
- Count Value... (Ctrl+Alt+V)
- Clock... (Ctrl+Alt+K)
- Arbitrary Value... (Ctrl+Alt+B)
- Random Values... (Ctrl+Alt+R)

Messages

Type	Message
Info	Info: *****
Info	Info: Running Quartus II Classic Timing Analyzer
Info	Info: Command: quartus_tan --read_settings_files=off --write_settings_files=off div_4X4_symbol -c div_4X4_symbol --timing_analysis
Info	Info: Longest tpd from source pin "denom[3]" to destination pin "remain[3]" is 18.942 ns
Info	Info: Quartus II Classic Timing Analyzer was successful. 0 errors, 0 warnings
Info	Info: Quartus II Full Compilation was successful. 0 errors, 3 warnings

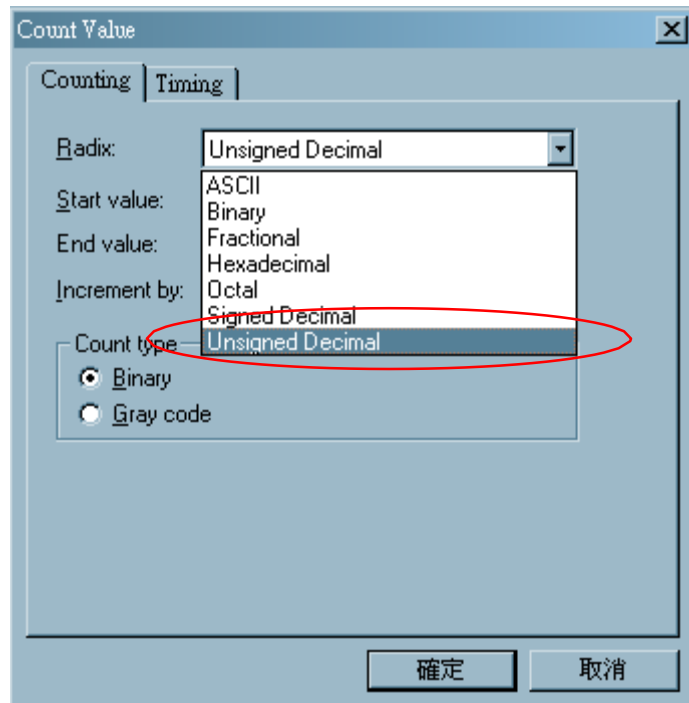
System (12) Processing (64) Extra Info Info (61) Warning (3) Critical Warning Error Suppressed (6) Flag

Message: 0 of 165

Overwrite Count Value

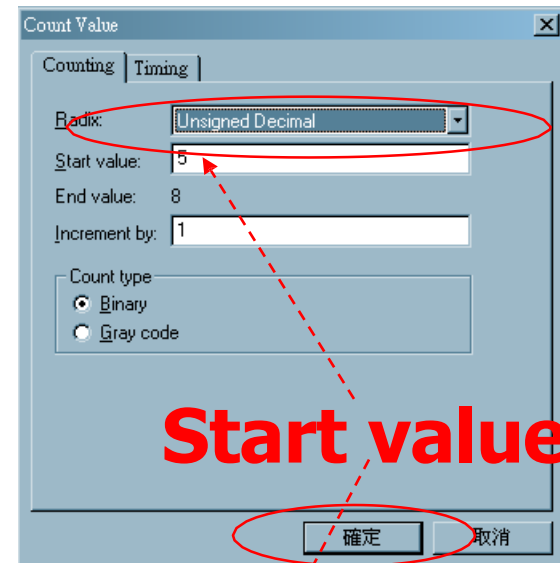
Idle NUM

除法器模擬_Step 10

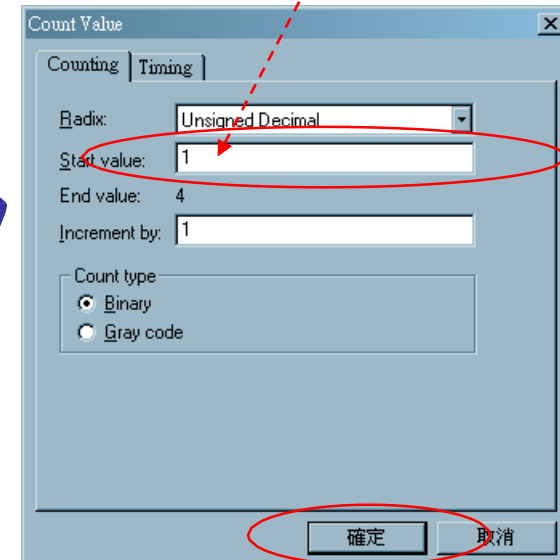


numer

denom



Start value



除法器模擬_Step 11

The screenshot displays the Quartus II software interface for a divider circuit simulation. The Project Navigator on the left shows the project files: div_4X4_symbol.bdf, div4X4.qip, div4X4.v, and div_4X4_symbol.vwf. The Task list shows the simulation steps: Compile Design, Analysis & Synthesis, Fitter (Place & Route), Assembler (Generate pro), Classic Timing Analysis, and EDA Netlist Writer. The Master Time Bar shows a pointer at 20.7 ns and an interval of 20.7 ns. The Name table lists the signals: numer, denom, quotient, and remain. The Message window at the bottom shows the simulation results, including the start and end times of the simulation and the location of the vector file.

Name	Value
numer	5
denom	1
quotient	X
remain	X

Messages

Type	Message
Info	Info: Ended Simulator at Tue May 05 20:37:52 2009 台北標準時間
Info	Info: Started Simulator at Tue May 05 20:46:56 2009 台北標準時間
Info	Info: Ended Simulator at Tue May 05 20:46:59 2009 台北標準時間
Info	Info: Vector file div_4X4_symbol.vwf is saved in text format. You can compress it into Compressed Vector Waveform File format in order to save space.
Info	Info: Started Simulator at Tue May 05 20:52:19 2009 台北標準時間
Info	Info: Ended Simulator at Tue May 05 20:52:22 2009 台北標準時間
Info	Info: Vector file div_4X4 symbol.vwf is saved in text format. You can compress it into Compressed Vector Waveform File format in order to save space.

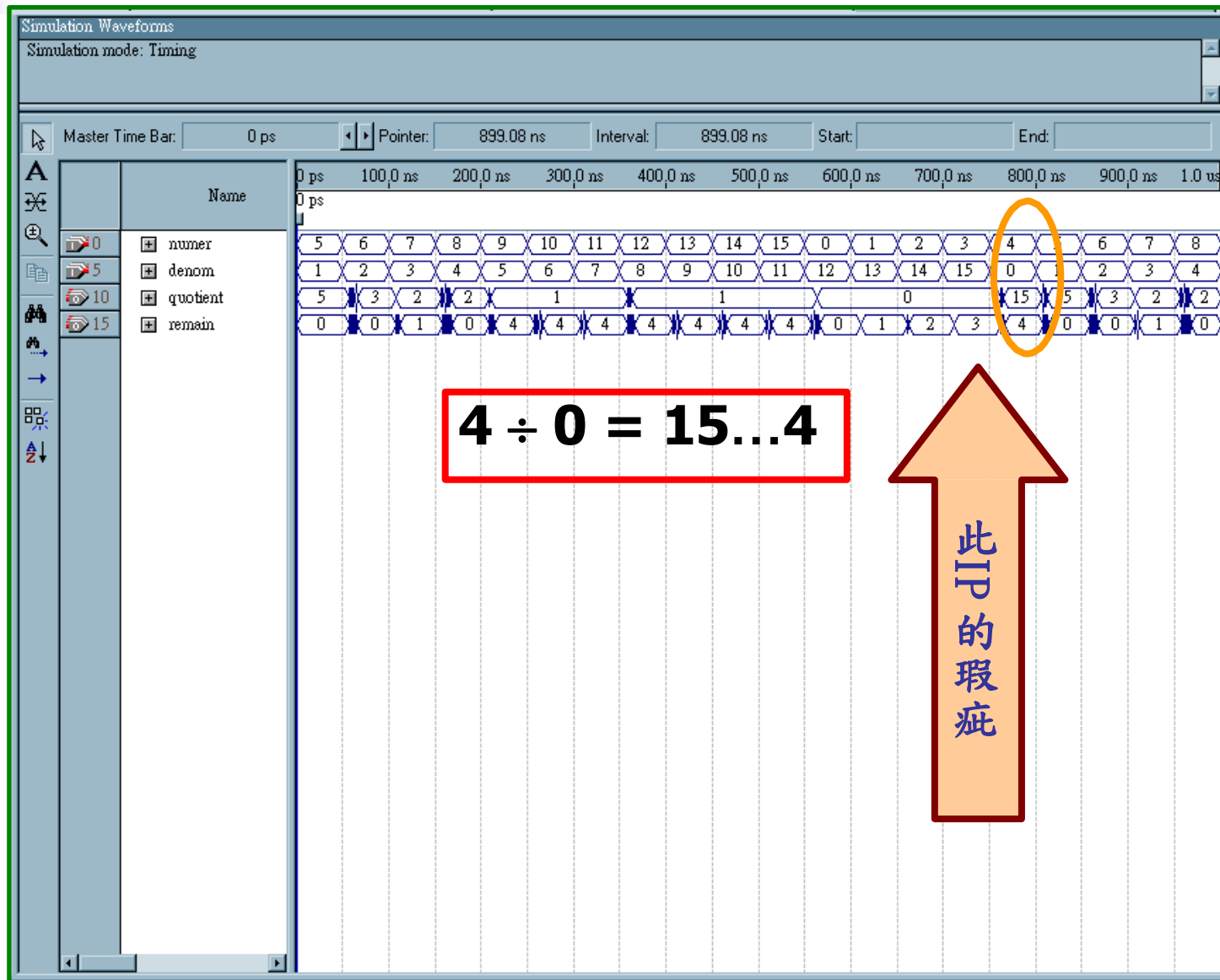
System [21] Processing [9] Extra Info [9] Info [9] Warning [9] Critical Warning [9] Error [9] Suppressed [9] Flag [9]

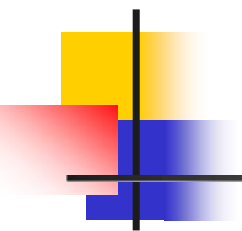
Message: 0 of 22

Location:

For Help, press F1

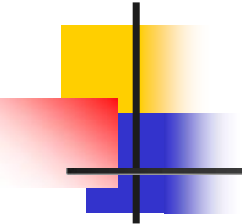
除法器模擬_Step 12





實作：無號數之四則運算

(四位元)IP應用



實驗目的與項目

5_04_Lab_ALU_LPM
5_05_Lab_IPAPPs_Gates
5_06_Lab_IPAPPs_Verilog

● 實驗目的

- 讓學習者更加熟悉IP使用。
- 此次實驗全以Block Diagram方式實作。

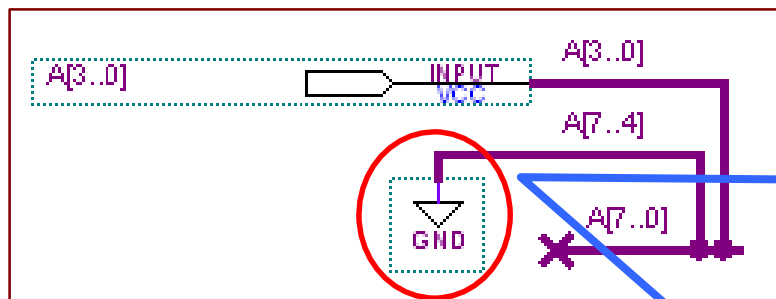
● 實驗項目

- 加、減法器 (IP應用)
- 乘法器 (IP應用)
- 除法器 (IP應用)
- 多工器 (LPM, Gates, and Verilog)

多工器_輸出表

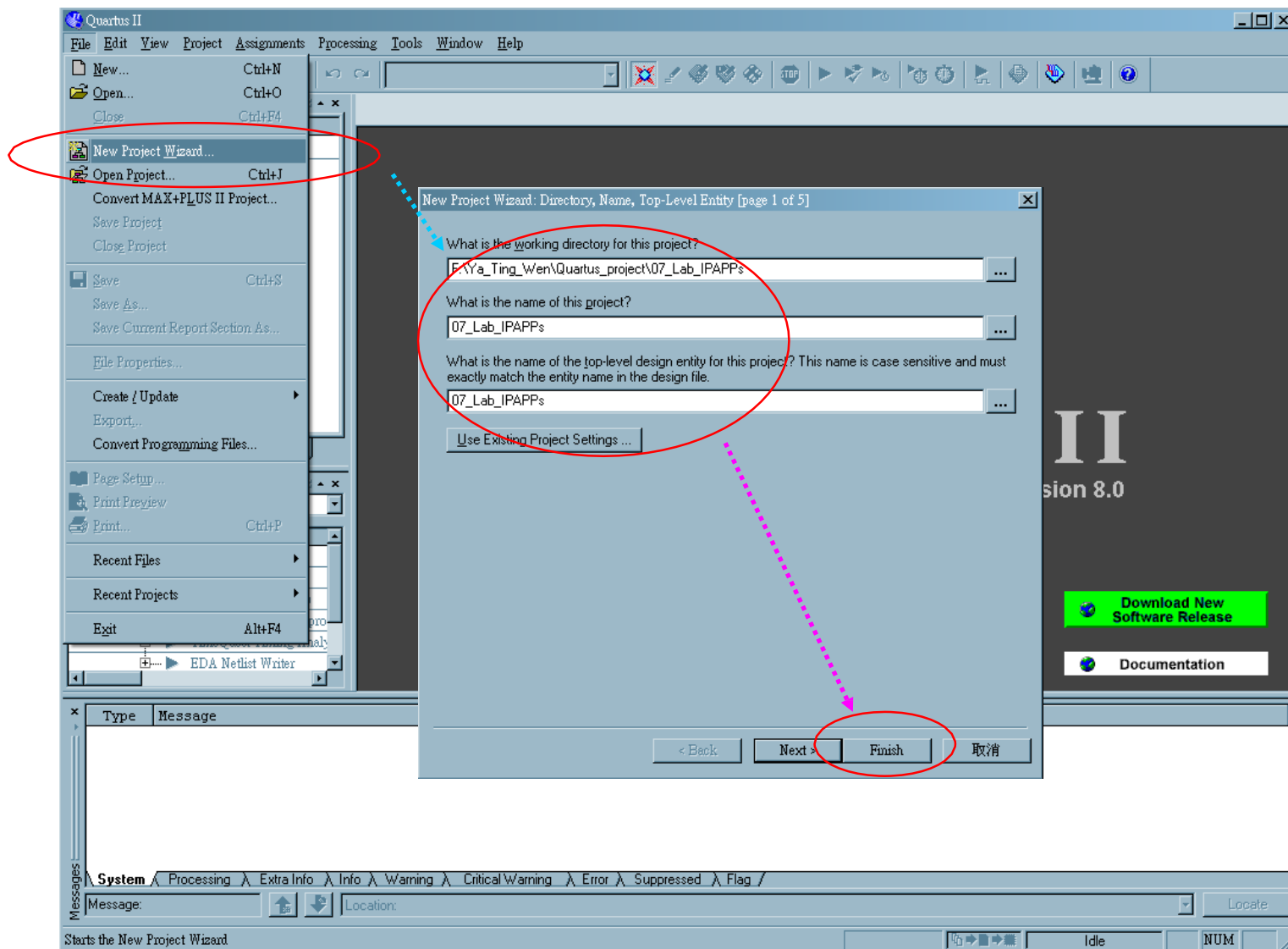
控制線		輸出線	四則運算
S[1]	S[0]	Y[7..0]	
0	0	A[7..0]	-
0	1	B[7..0]	+
1	0	C[7..0]	×
1	1	D[7..0]	÷

多工器_腳位說明

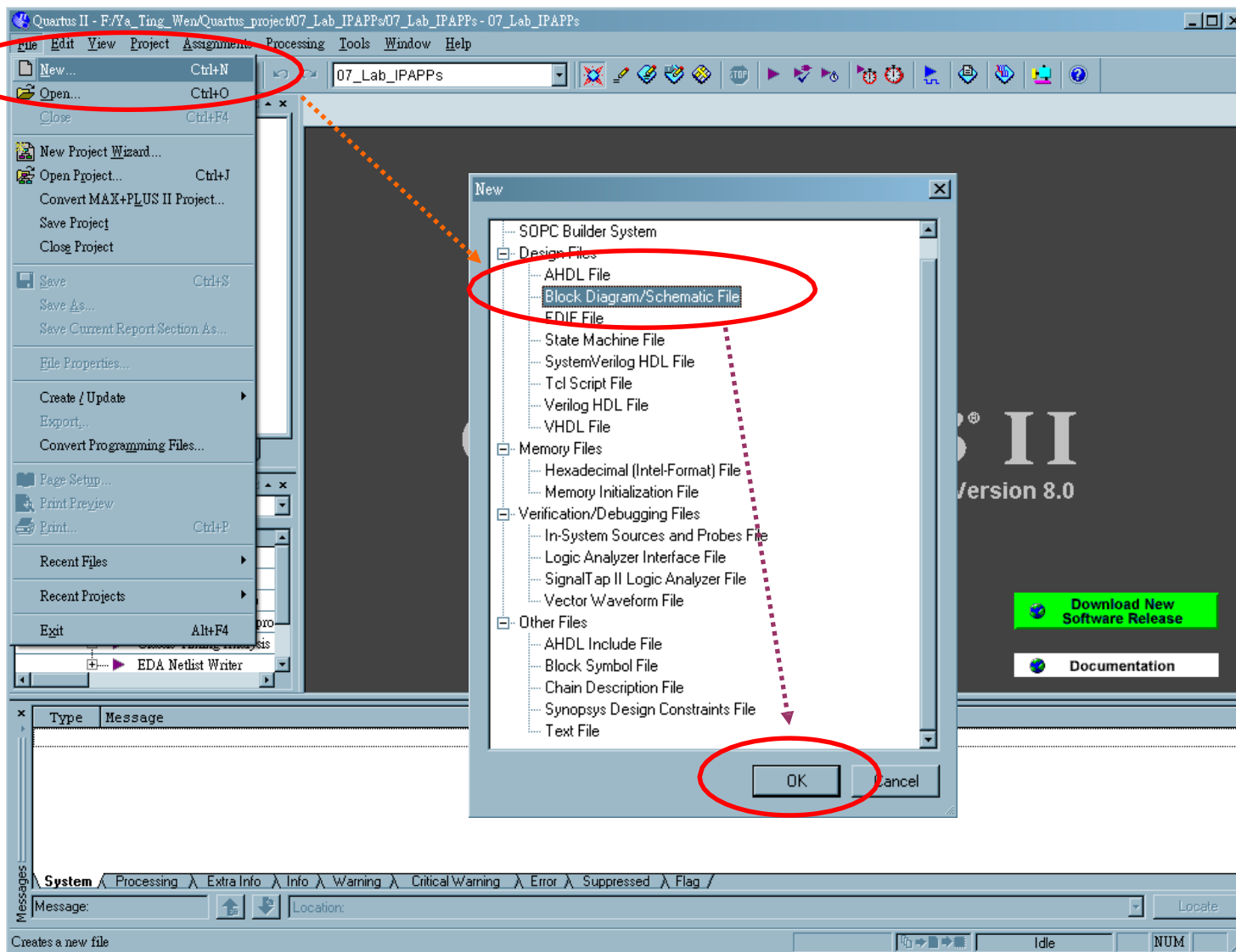


- 因為不能空接腳位，所以在多工器的輸入端，只要少於8位元之腳位，沒使用到的就全都拉向接地端。
- 在後面實作時，會用到這個觀念，一定要接地。

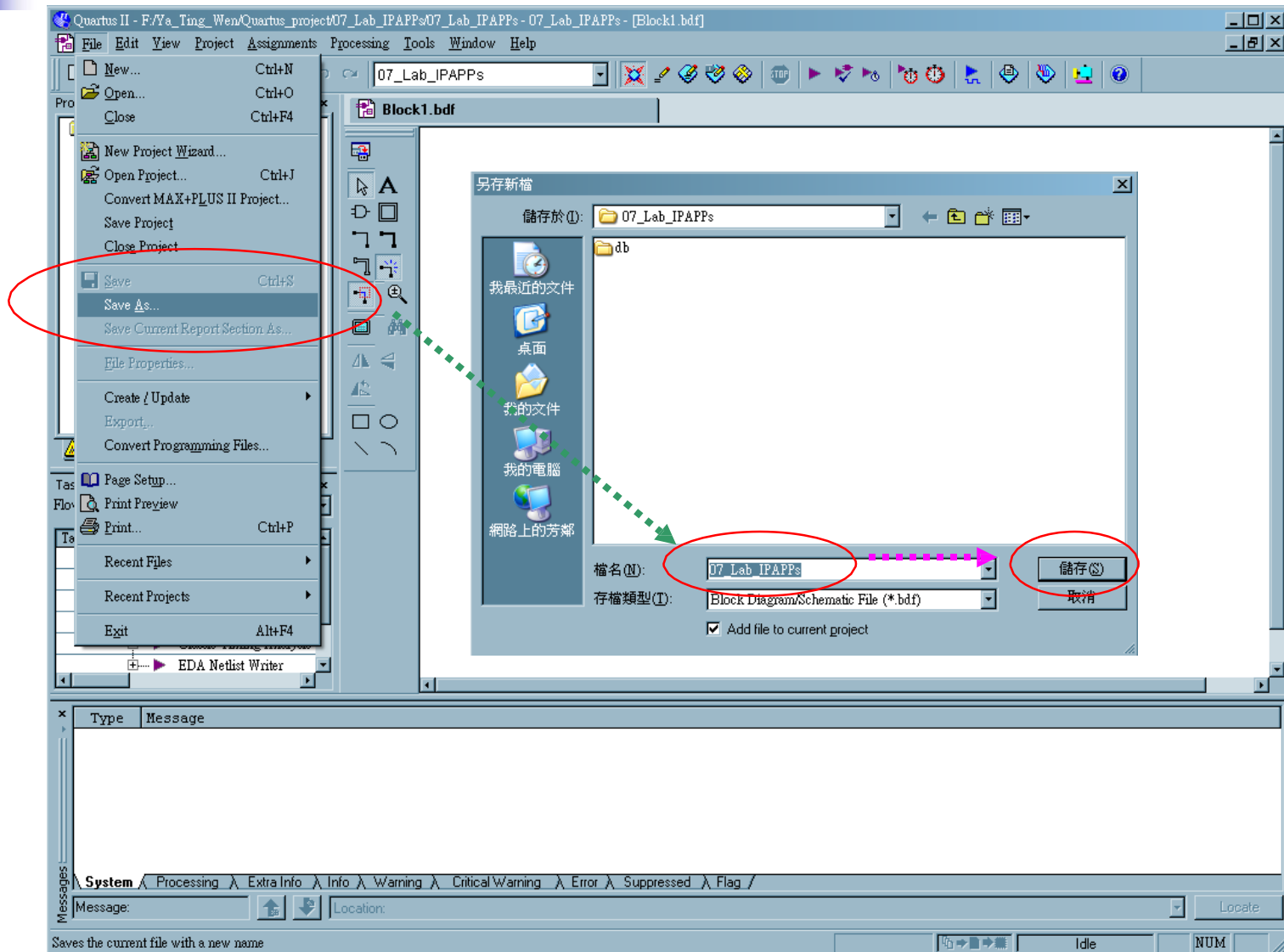
無號數之四則運算 Step 1



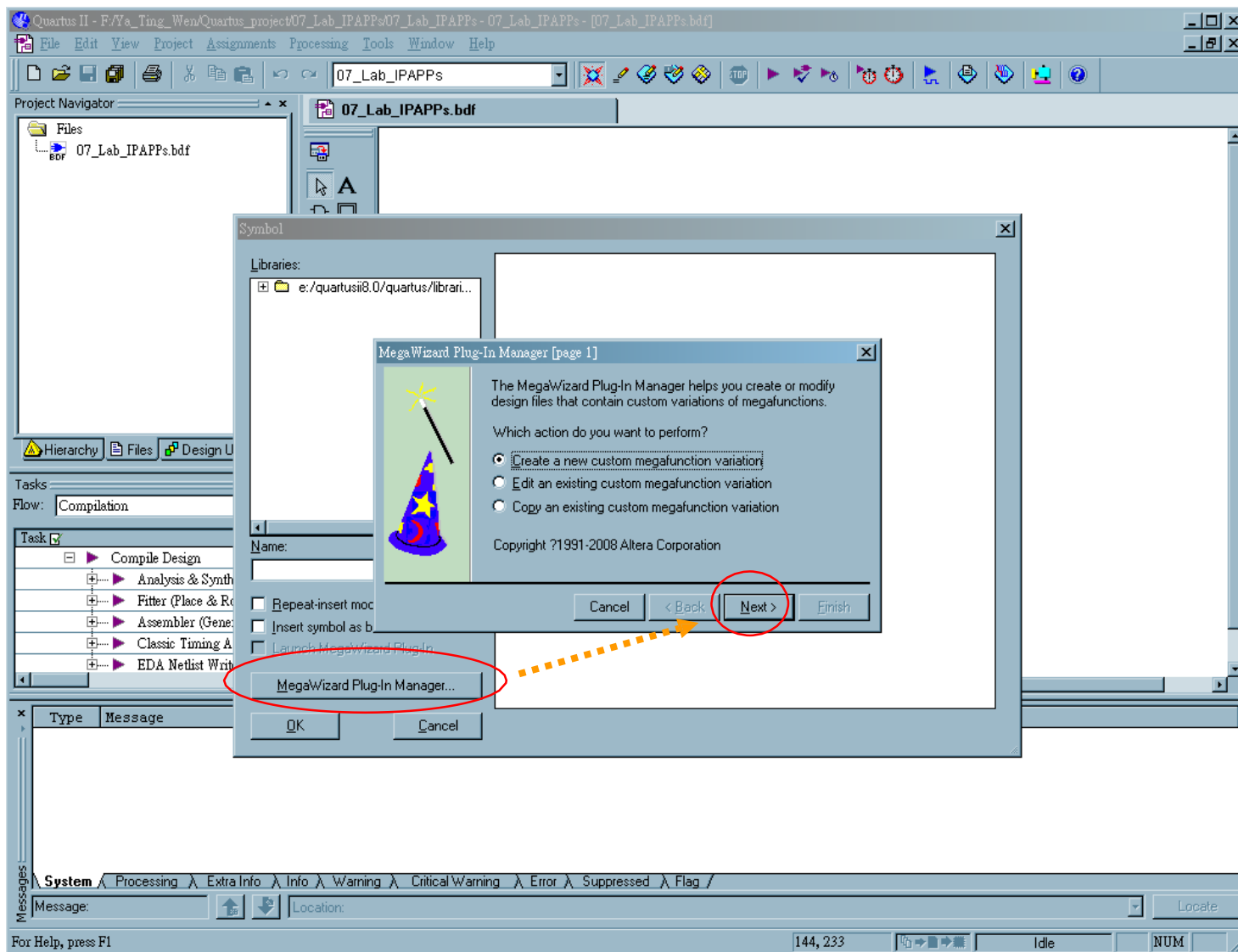
無號數之四則運算 Step 2



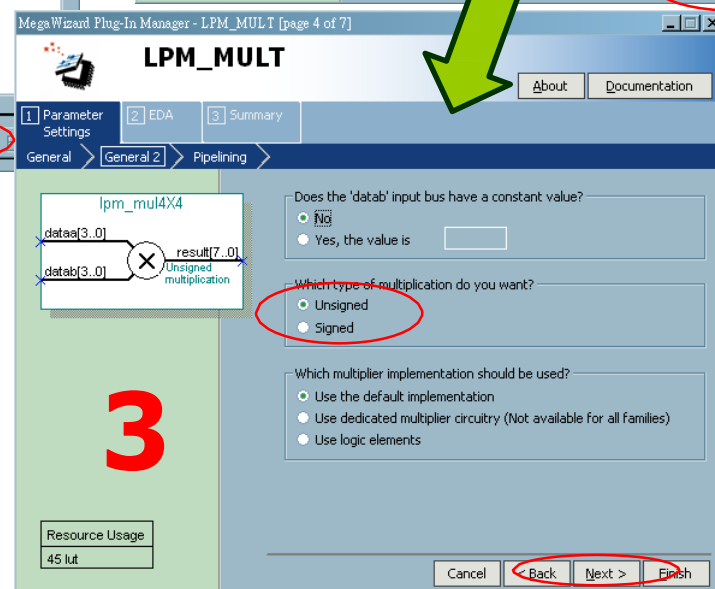
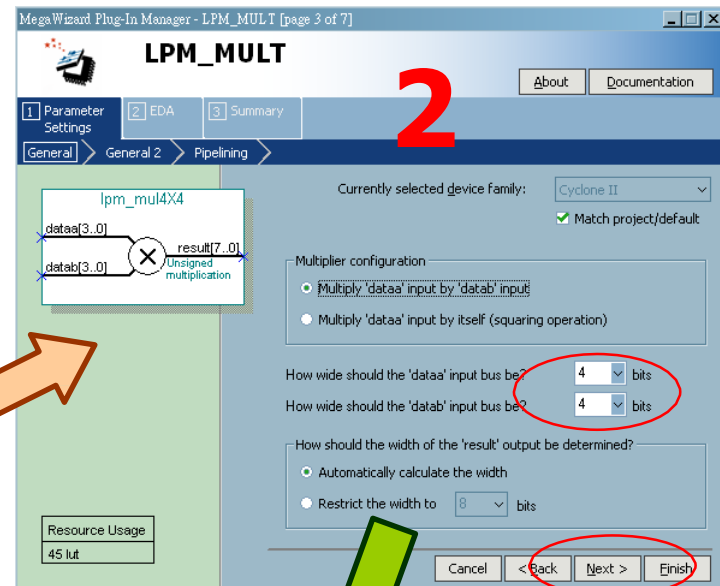
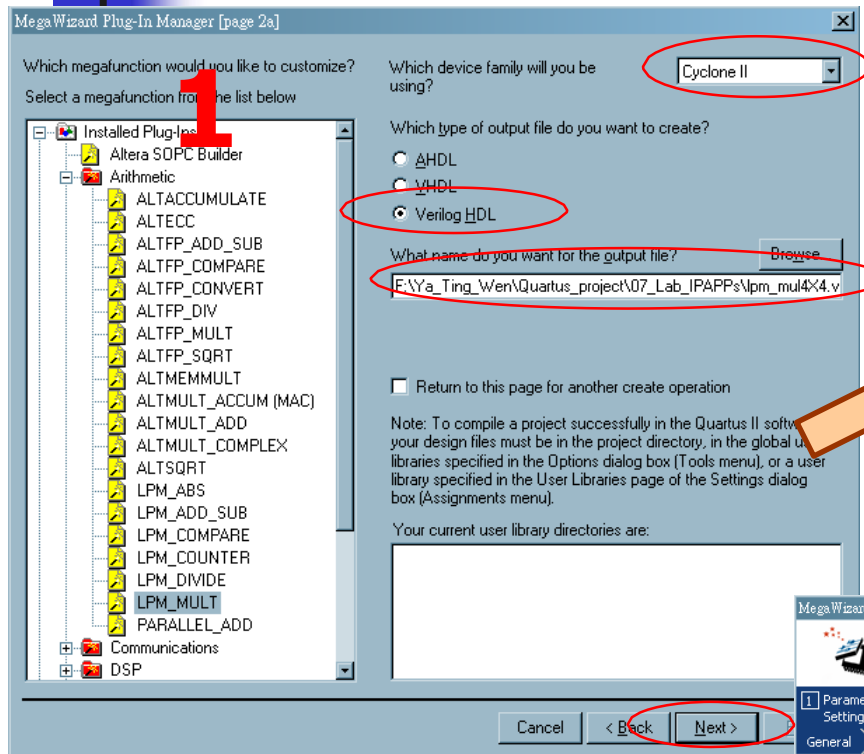
無號數之四則運算 Step 3



無號數之四則運算 Step 4



無號數之四則運算 Step 5



無號數之四則運算 Step 6

The image displays three sequential screenshots of the MegaWizard Plug-In Manager (LPM_MULT) interface, illustrating the steps to configure a 4x4 multiplier.

Step 1: The "Parameter Settings" tab is active. The "General" section shows "lpm_mul4x4" with inputs "dataa[3..0]" and "datab[3..0]", and output "result[7..0]". The "Do you want to pipeline the function?" section has "No" selected. The "Which type of optimization do you want?" section has "Default" selected. The "Resource Usage" section shows "45 lut". The "Next >" button is circled in red.

Step 2: The "EDA" tab is active. It shows a block diagram of the multiplier and a table of simulation libraries. The "Simulation Libraries" section lists "lpm" as the "LPM megafunction simulation library". The "Next >" button is circled in red.

Step 3: The "Summary" tab is active. It shows a list of files to be generated, including "lpm_mul4x4.v", "lpm_mul4x4.inc", "lpm_mul4x4.cmp", "lpm_mul4x4.bsf", "lpm_mul4x4_inst.v", "lpm_mul4x4_bb.v", "lpm_mul4x4_waveforms.html", and "lpm_mul4x4_waveform.jpg". The "Next >" button is circled in red.

Red numbers 1, 2, and 3 are overlaid on the screenshots to indicate the sequence. A purple arrow points from Step 1 to Step 2, and a blue arrow points from Step 2 to Step 3.

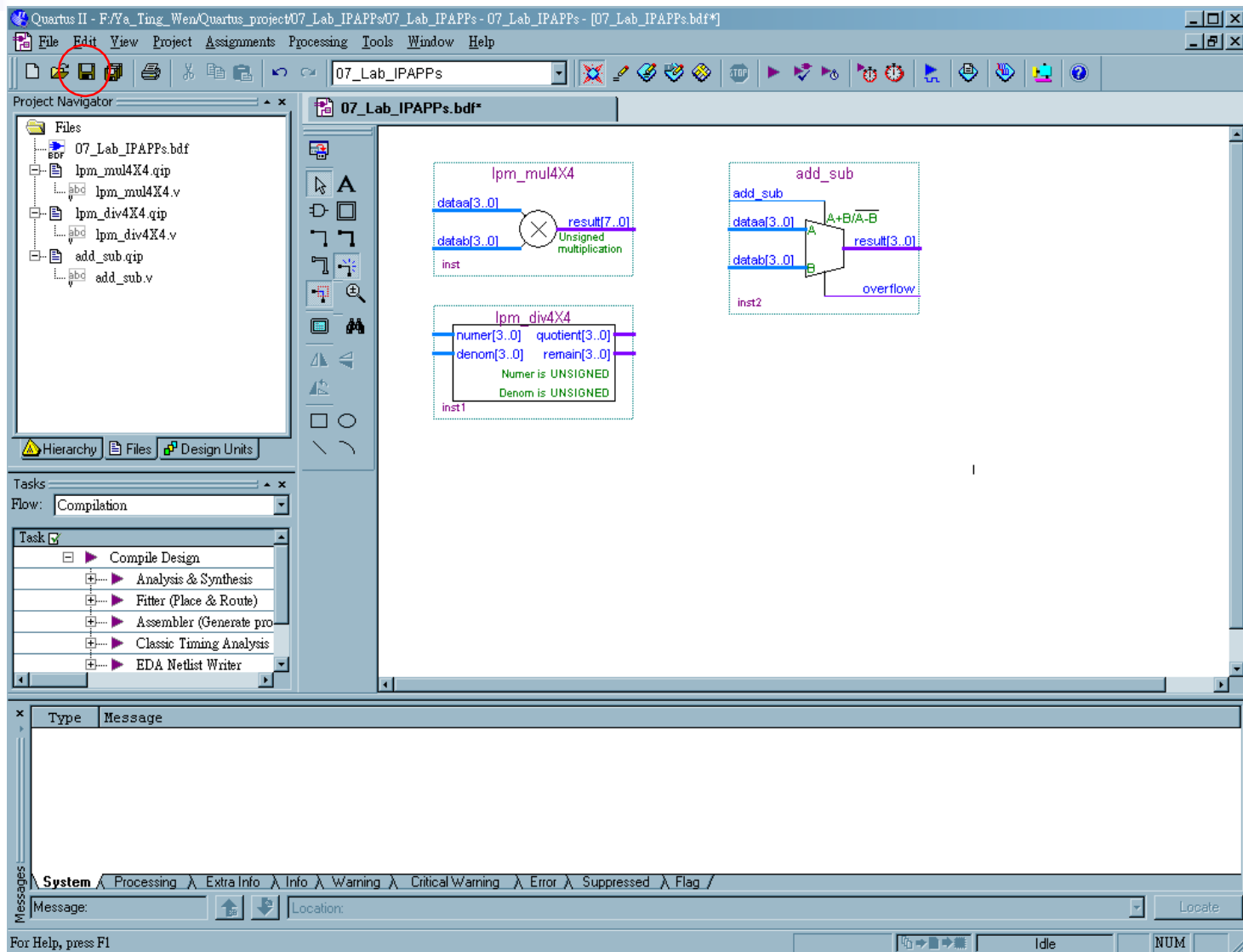
無號數之四則運算 Step 7

The image illustrates the process of inserting a 4x4 unsigned multiplication block into a Quartus II project, divided into three numbered steps:

- Step 1:** A dialog box titled "Quartus II IP Files" is displayed. It contains the text: "When you create an Altera IP variation, a Quartus II IP File is generated. Quartus II IP Files are used to represent the Altera IP in your design. Do you want to add the Quartus II IP File to the project?" Below the text is a checkbox labeled "Automatically add Quartus II IP Files to all projects" and a note: "(Note: Turning on this option permanently suppresses this dialog box. You can change this setting in the Options dialog box)". The "Yes" button is circled in red.
- Step 2:** The "Symbol" window is shown. The "Libraries" list on the left contains "e:/quartusii8.0/quartus/librari...". The main area displays a block symbol for "lpm_mul4X4" with inputs "dataa[3..0]" and "datab[3..0]", and output "result[7..0]". The block is labeled "Unsigned multiplication" and "inst". The "Name:" field at the bottom is set to "lpm_mul4X4". The "OK" button is circled in red.
- Step 3:** The "07_Lab_IPAPPs.bdf*" window is shown. It displays the same "lpm_mul4X4" block symbol inserted into the design. The block is labeled "Unsigned multiplication" and "inst". The "OK" button in the bottom right corner is circled in red.

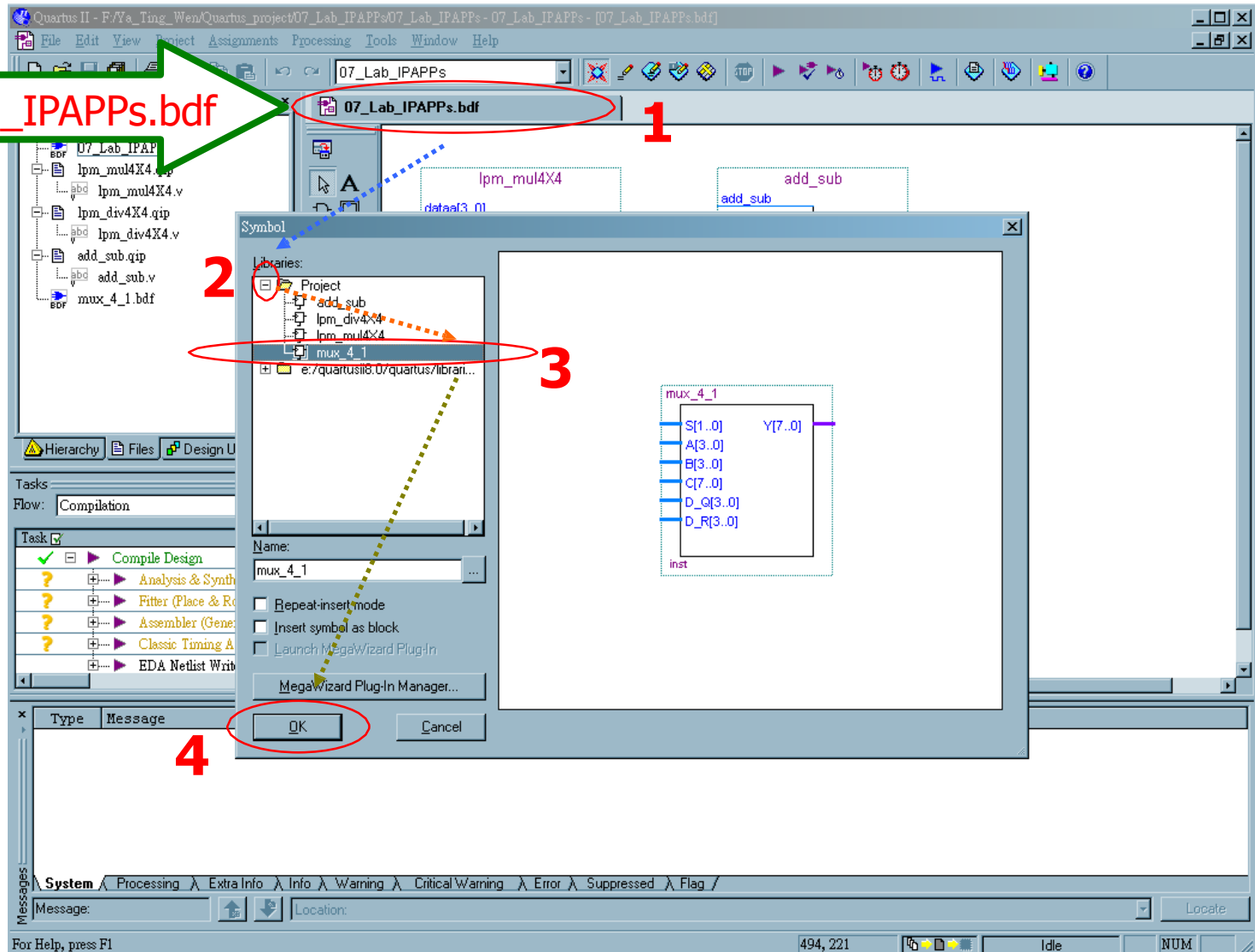
Arrows indicate the flow from Step 1 to Step 2, and from Step 2 to Step 3.

無號數之四則運算 Step 8



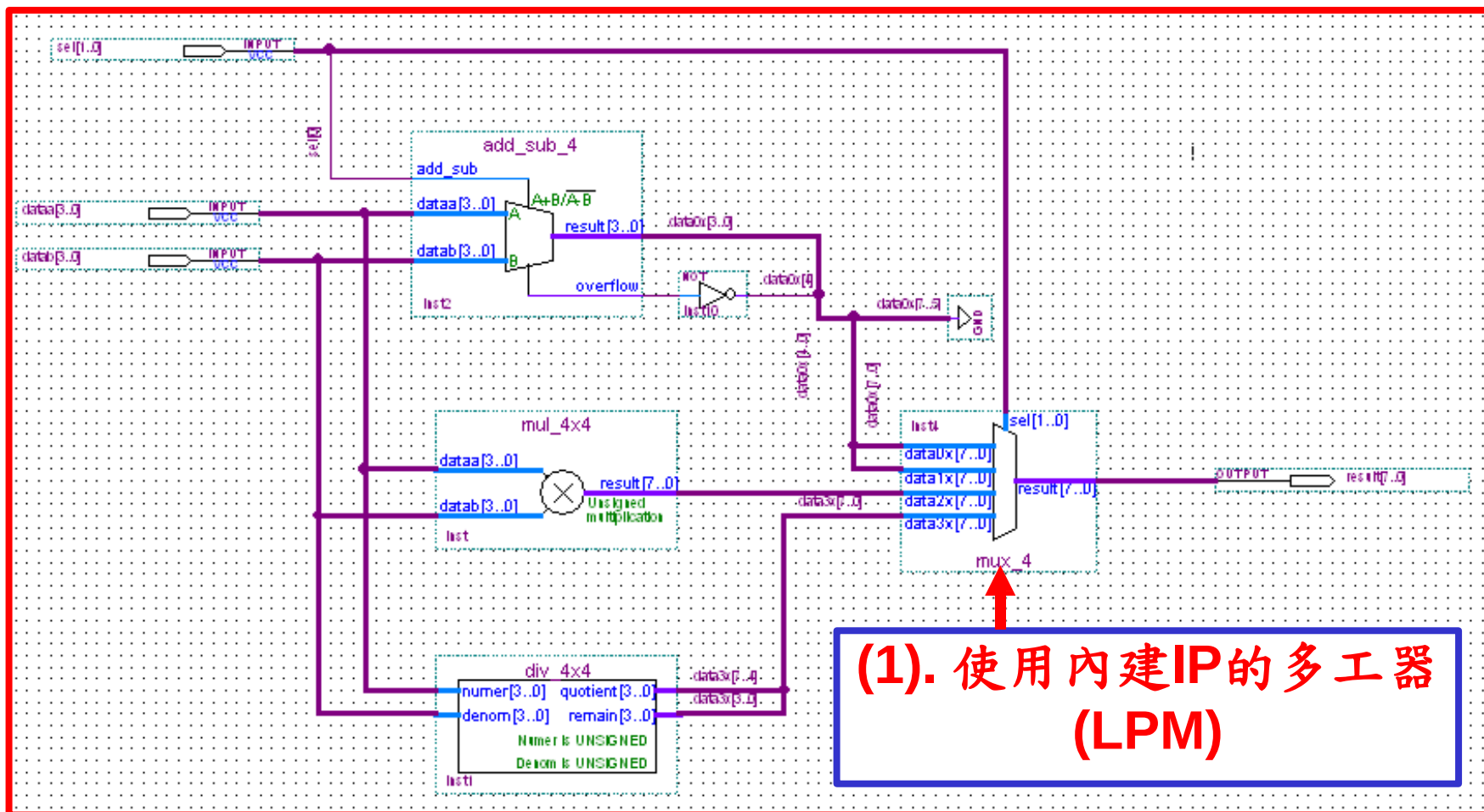
無號數之四則運算 Step 9

回到05_Lab_IPAPPs.bdf

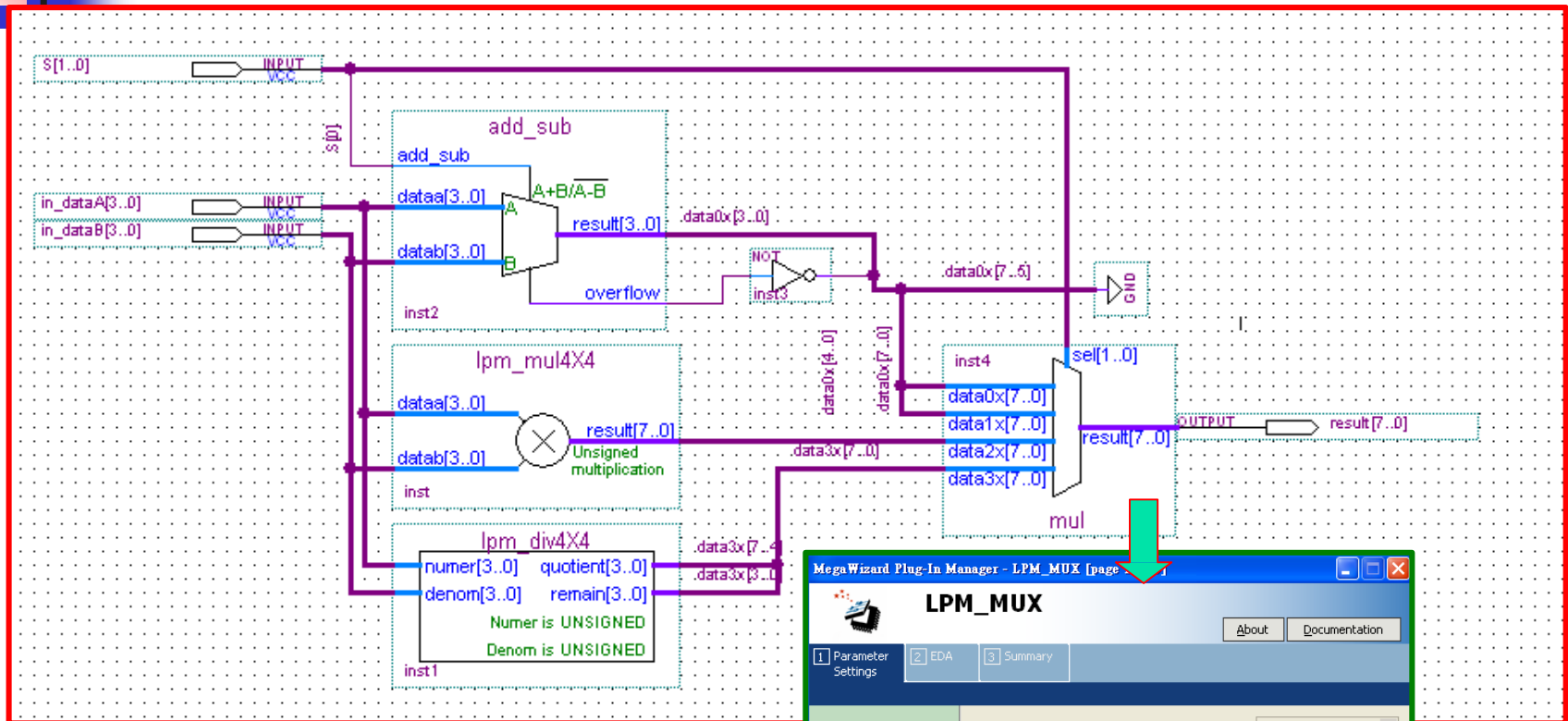


無號數之四則運算 Step 10

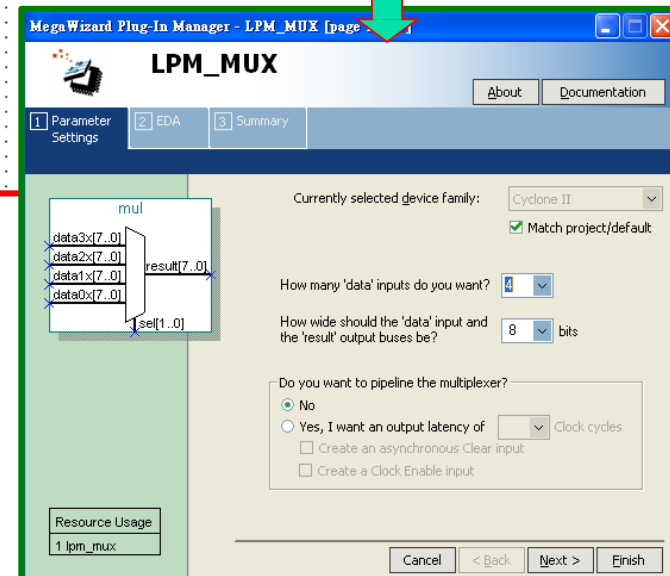
加入輸出、輸入腳位



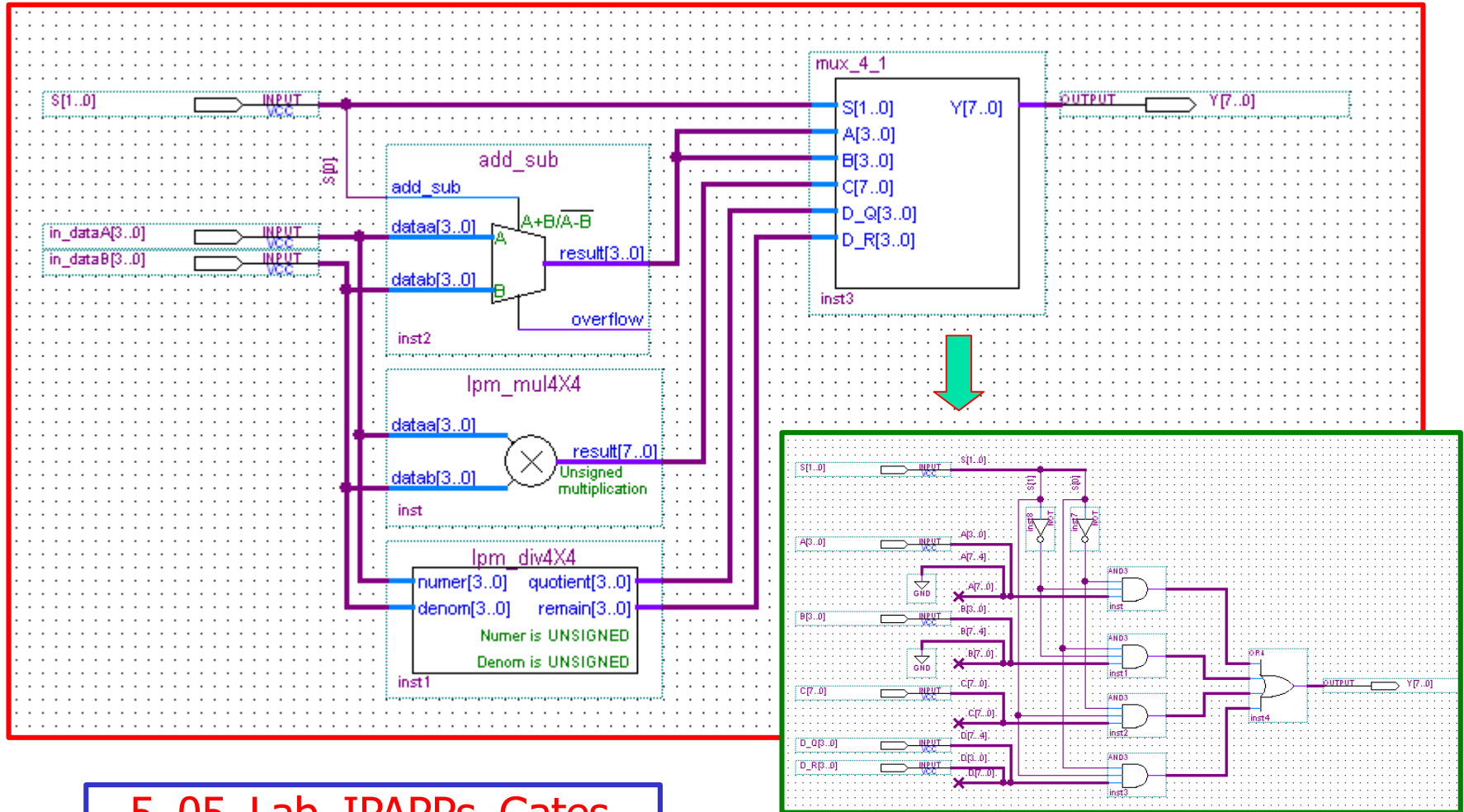
無號數之四則運算 Step 10 (Appendix-1)



5_04_Lab_IPAPPs_LPM

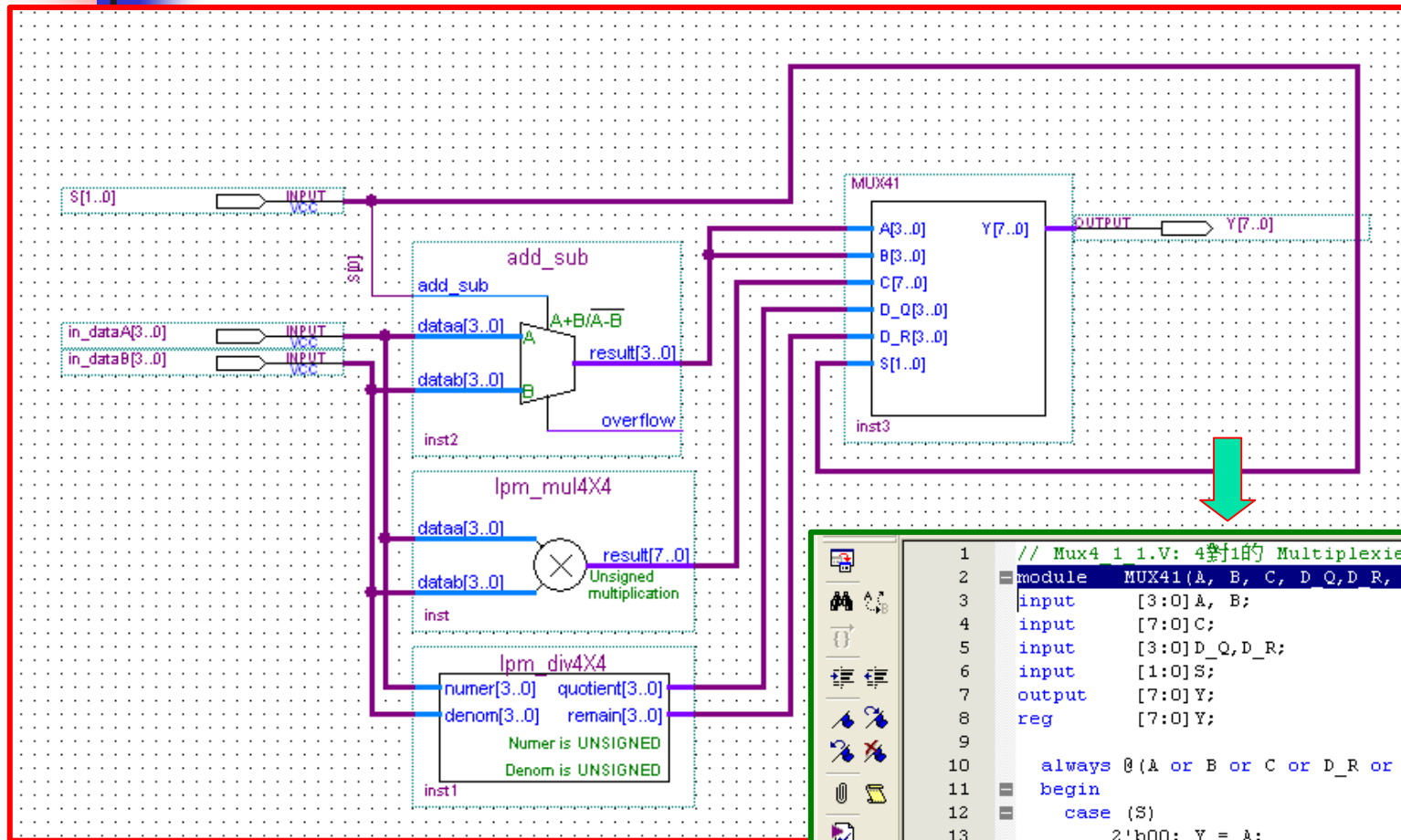


無號數之四則運算 Step 10 (Appendix-2)



5_05_Lab_IPAPPs_Gates

無號數之四則運算 Step 10 (Appendix-3)

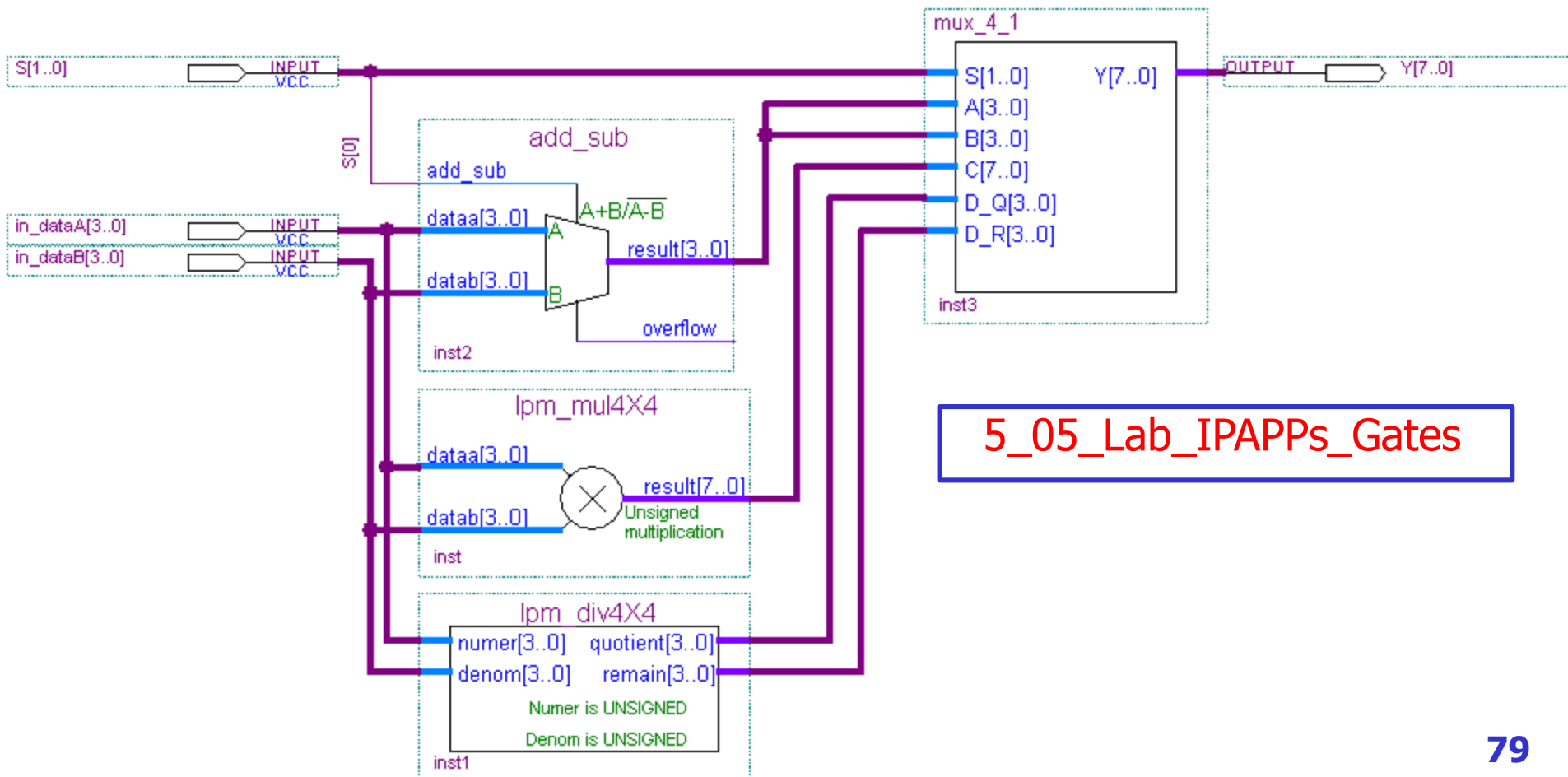


5_06_Lab_IPAPPs_Verilog

```
1 // Mux4 1 1.V: 4對1的 Multiplexier, 第1種寫法
2 module MUX41(A, B, C, D_Q,D_R, S, Y);
3     input    [3:0] A, B;
4     input    [7:0] C;
5     input    [3:0] D_Q,D_R;
6     input    [1:0] S;
7     output   [7:0] Y;
8     reg      [7:0] Y;
9
10    always @ (A or B or C or D_R or D_Q or S)
11    begin
12        case (S)
13            2'b00: Y = A;
14            2'b01: Y = B;
15            2'b10: Y = C;
16            2'b11: Y = {D_Q,D_R};
17        endcase
18    end
19 endmodule
20
```

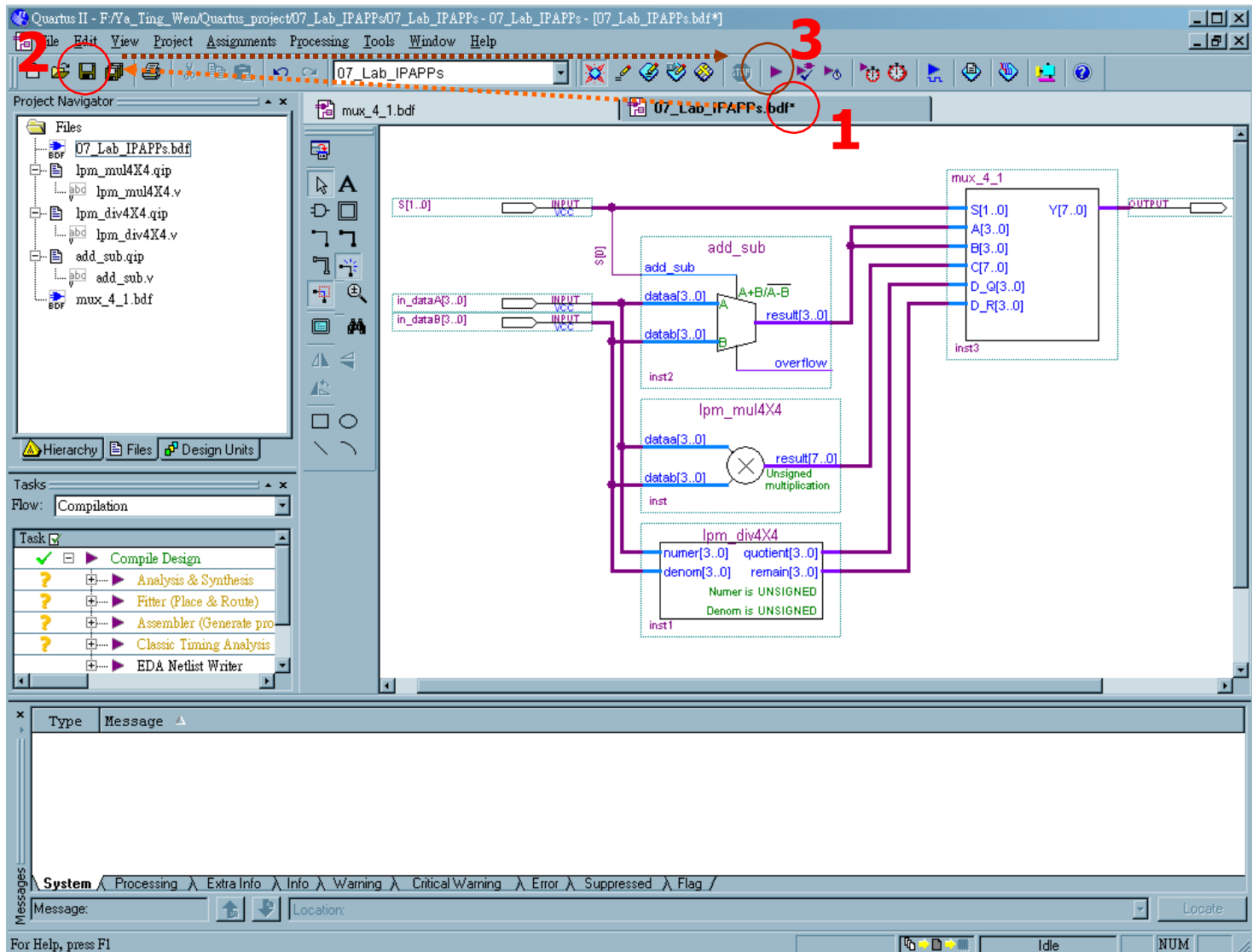
無號數之四則運算 Step 11

依圖示連線



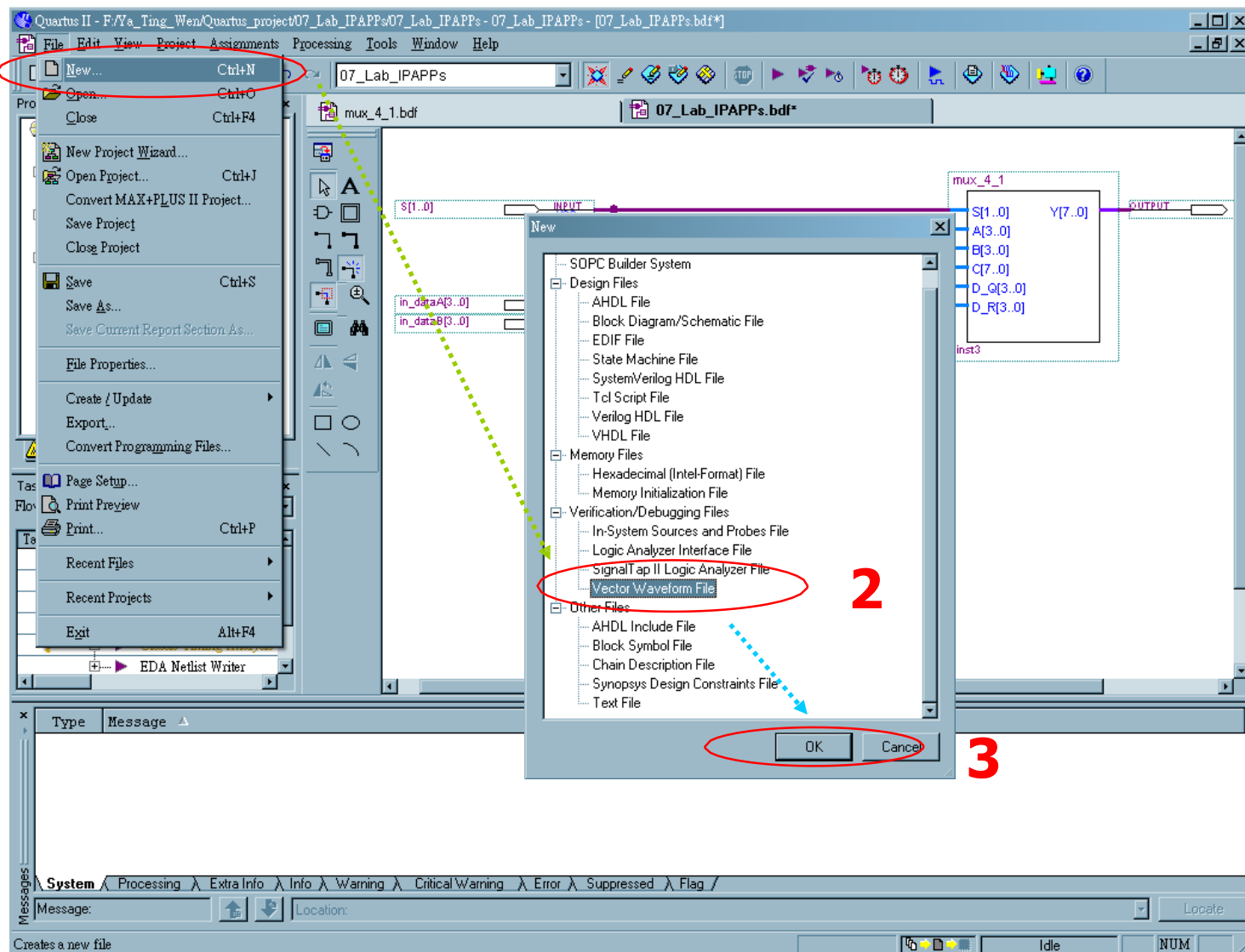
5_05_Lab_IPAPPs_Gates

無號數之四則運算 Step 12



無號數之四則運算 Step 13

1



2

3

無號數之四則運算 Step 14

點二下

選取所要觀看的腳位

Node Finder

Named: * Filter: Pins: all Customize... List OK

Look in: |07_Lab_IPAPPs| Include subentities

Name	Assignments
in_dataA	Unassigned
in_dataA[0]	Unassigned
in_dataA[1]	Unassigned
in_dataA[2]	Unassigned
in_dataA[3]	Unassigned
in_dataB	Unassigned
in_dataB[0]	Unassigned
in_dataB[1]	Unassigned
in_dataB[2]	Unassigned
in_dataB[3]	Unassigned
S	Unassigned
S[0]	Unassigned
S[1]	Unassigned
Y	Unassigned
Y[0]	Unassigned
Y[1]	Unassigned
Y[2]	Unassigned
Y[3]	Unassigned
Y[4]	Unassigned
Y[5]	Unassigned

Selected Nodes:

Name	Assignments	T
07_Lab_IPAPPs in_dataA	Unassigned	Ir
07_Lab_IPAPPs in_dataB	Unassigned	Ir
07_Lab_IPAPPs S	Unassigned	Ir
07_Lab_IPAPPs Y	Unassigned	O

Insert Node or Bus

Name: Type: INPUT Value type: 9-Level Radix: ASCII Bus width: 1 Start index: 0 Display gray code count as binary count

Node Finder...

無號數之四則運算 Step 15

Quartus II - F:\Ya_Ting_Wen\Quartus_project\07_Lab_IPAPPs\07_Lab_IPAPPs - 07_Lab_IPAPPs - [Waveform1.vwf]

File Edit View Project Assignments Processing Tools Window Help

Project Navigator

- Files
 - 07_Lab_IPAPPs.bdf
 - lpm_mul4X4.qip
 - lpm_div4X4.v
 - lpm_div4X4.qip
 - lpm_div4X4.v
 - add_sub.qip
 - add_sub.v
 - mux_4_1.bdf
- Tasks
 - Flow: Compilation
 - Task
 - Compile Design
 - Analysis & Synthesis
 - Fitter (Place & Route)
 - Assembler (Generate pro
 - Classic Timing Analysis
 - EDA Netlist Writer

Master Time Bar: 9.575 ns | Pointer: 350 ps | Interval: -9.23 ns | Start: | End: |

Waveform1.vwf

Insert Node or Bus

Name: **Multiple Items** OK Cancel

Type: **Multiple Items**

Value type: 9-Level

Radix: ASCII

Bus width: **Multiple Items**

Start index: **Multiple Items**

☐ Display gray code count as binary count

Messages

Type	Message
Info	*****
Info	Info: Running Quartus II Classic Timing Analyzer
Info	Info: Command: quartus_tan --read_settings_files=off --write_settings_files=off 07_Lab_IPAPPs -c 07_Lab_IPAPPs --timing_analysis_o
Info	Info: Longest tpd from source pin "in_dataB[0]" to destination pin "Y[1]" is 25.225 ns
Info	Info: Quartus II Classic Timing Analyzer was successful. 0 errors, 0 warnings
Info	Info: Quartus II Full Compilation was successful. 0 errors, 18 warnings

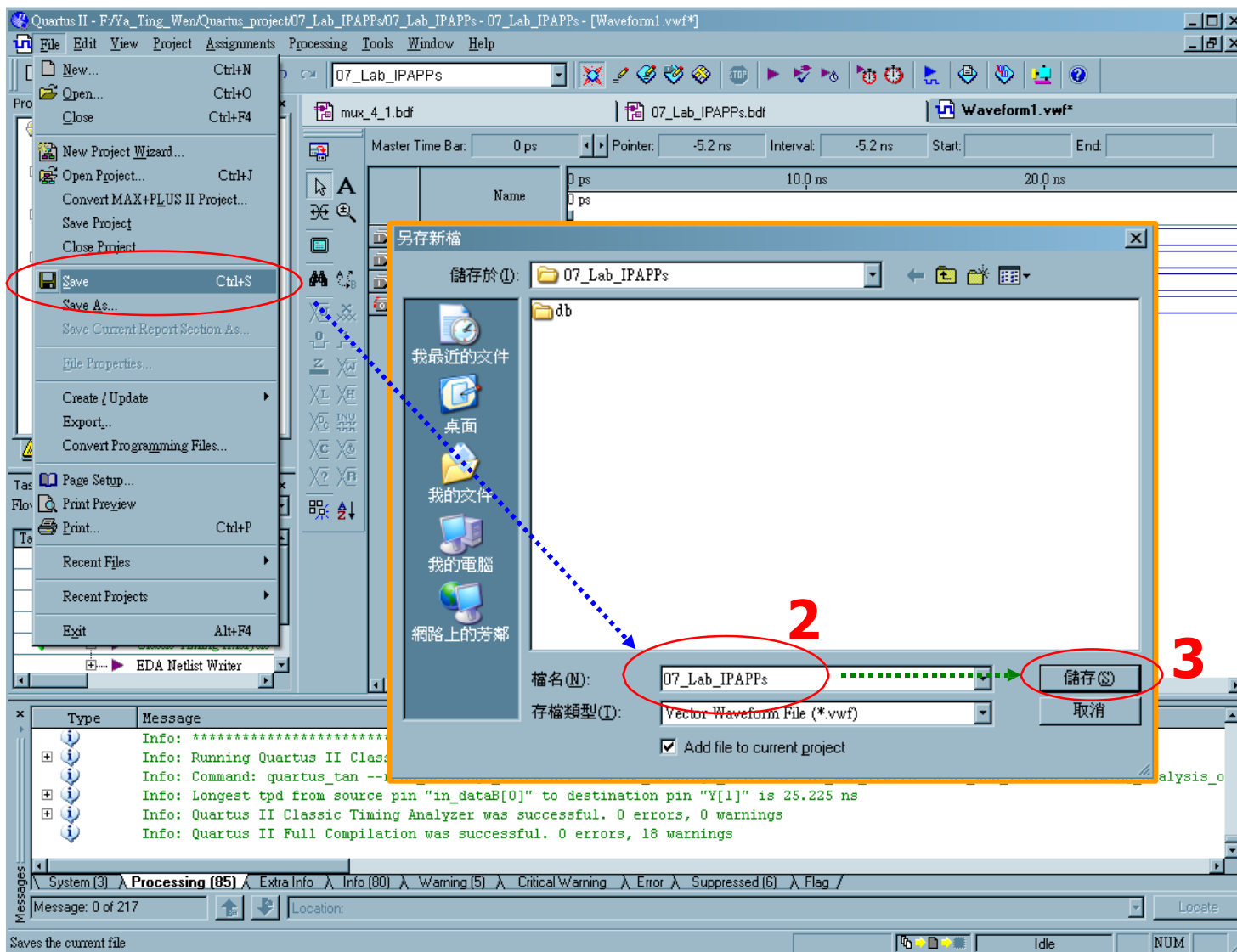
System [3] | Processing [85] | Extra Info | Info [80] | Warning [5] | Critical Warning | Error | Suppressed [6] | Flag /

Message: 0 of 217 | Location: | Locate

For Help, press F1

Idle NUM

無號數之四則運算 Step 16



無號數之四則運算 Step 17

按右鍵

The screenshot shows the Quartus II interface with the waveform editor open. The 'in_dataA' signal is selected, and a right-click context menu is displayed. The 'Value' option is highlighted, and a sub-menu is shown with 'Arbitrary Value...' selected. A green arrow points from the text '按右鍵' to the right-click action.

Project Navigator:

- Files
 - 07_Lab_IPAPPs.bdf
 - lpm_mul4X4.qip
 - lpm_div4X4.v
 - add_sub.qip
 - add_sub.v
 - mux_4_1.bdf
 - 07_Lab_IPAPPs.vwf

Tasks:

- Flow: Compilation
- Task
 - Compile Design
 - Analysis & Synthesis
 - Fitter (Place & Route)
 - Assembler (Generate pro
 - Classic Timing Analysis
 - EDA Netlist Writer

Messages:

- Info: Started Full Compilation at Wed May 06 13:45:11 2009 台北標準時間
- Info: Ended Full Compilation at Wed May 06 13:45:52 2009 台北標準時間
- Info: Arguments associated with the following assignment have been removed because the assignment was changed
- Info: Arguments associated with the following assignment have been removed because the assignment was changed
- Info: Arguments associated with the following assignment have been removed because the assignment was changed
- Info: Vector file 07_Lab_IPAPPs.vwf is saved in text format. You can compress it into Compressed Vector Waveform File format in order.

System [7] Processing [85] Extra Info Info [80] Warning [5] Critical Warning Error Suppressed [6] Flag

Message: 0 of 10

Overwrite Arbitrary Value

無號數之四則運算 Step 18

in_dataA

Arbitrary Value

Time range
Start time: 0 ps
End time: 1.0 us

Arbitrary value
Radix: Unsigned Decimal
Numeric or named value: 5

OK Cancel

in_dataB

Arbitrary Value

Time range
Start time: 0 ps
End time: 1.0 us

Arbitrary value
Radix: Unsigned Decimal
Numeric or named value: 3

OK Cancel

Quartus II

Nodes have different radices. Do you want to change the radix(es) of the selected node(s) to Unsigned Decimal?

是(Y) 否(N)

無號數之四則運算 Step 19

Quartus II - F:\Ya_Ting_Wen\Quartus_project\07_Lab_IPAPPs\07_Lab_IPAPPs - [07_Lab_IPAPPs.vwf*]

File Edit View Project Assignments Processing Tools Window Help

07_Lab_IPAPPs

Project Navigator

Files

- 07_Lab_IPAPPs.bdf
- lpm_mul4X4.qip
- lpm_mul4X4.v
- lpm_div4X4.qip
- lpm_div4X4.v
- add_sub.qip
- add_sub.v
- mux_4_1.bdf
- 07_Lab_IPAPPs.vwf

Hierarchy Files Design Units

Tasks

Flow: Compilation

Task

- Compile Design
- Analysis & Synthesis
- Fitter (Place & Route)
- Assembler (Generate pro
- Classic Timing Analysis
- EDA Netlist Writer

Master Time Bar: 0 ps Pointer: 500 ps Interval: 500 ps Start: 0 ps End: 1.0 us

Name

Signal	Value
in_dataA	5
in_dataB	3
S	0
Y	0

Context Menu:

- Cut (Ctrl+X)
- Copy (Ctrl+C)
- Paste
- Delete (Del)
- Select Entire Waveform Interval
- Insert
 - Value
 - Stretch or Compress Waveform Interval... (Ctrl+Alt+S)
 - Offset Waveform Interval... (Ctrl+Alt+O)
 - Uninitialized (U) (Ctrl+Alt+U)
 - Forcing Unknown (X) (Ctrl+Alt+X)
 - Forcing Low (0) (Ctrl+Alt+0)
 - Forcing High (1) (Ctrl+Alt+1)
 - High Impedance (Z) (Ctrl+Alt+Z)
 - Weak Unknown (W) (Ctrl+Alt+W)
 - Weak Low (L) (Ctrl+Alt+L)
 - Weak High (H) (Ctrl+Alt+H)
 - Don't Care (DC) (Ctrl+Alt+D)
 - Invert (Ctrl+Alt+I)
 - Count Value... (Ctrl+Alt+V)
 - Clock... (Ctrl+Alt+K)
 - Arbitrary Value... (Ctrl+Alt+B)
 - Random Values... (Ctrl+Alt+R)
 - Grouping
 - Display Format
 - Nodes
 - Group and Bus Bit Order
 - Locate
 - Zoom
 - Properties

Messages

Type Message

- Info: Started Full Compilation at Wed May 06 13:45:11 2009 台北標準時間
- Info: Ended Full Compilation at Wed May 06 13:45:52 2009 台北標準時間
- Info: Arguments associated with the following assignment have been removed because the assignment was changed
- Info: Arguments associated with the following assignment have been removed because the assignment was changed
- Info: Arguments associated with the following assignment have been removed because the assignment was changed
- Info: Vector file 07_Lab_IPAPPs.vwf is saved in text format. You can compress it into Compressed Vector Waveform File format in order to save space.

System [7] Processing [85] Extra Info [80] Info [80] Warning [5] Critical Warning Error Suppressed [6] Flag

Message: 0 of 10 Location: Overwrite Count Value

無號數之四則運算 Step 20

1

Count Value

Counting | Timing

Radix: ASCII

Start value: ASCII

End value: Fractional

Increment by: Octal

Signed Decimal

Unsigned Decimal

Count type

☒ Binary

☐ Gray code

確定 取消

2

Count Value

Counting | Timing

Start time: 0 ps

End time: 1.0 us

Transitions occur

☐ Relative to clock settings:

☐ Positive edge

☐ Negative edge

☒ At absolute times:

Count every: 100 ns

Multiplied by: 1

確定 取消

3

Quartus II

Nodes have different radices. Do you want to change the radix(es) of the selected node(s) to Unsigned Decimal?

是(Y) 否(N)

無號數之四則運算 Step 21

The screenshot displays the Quartus II software interface for a project named "07_Lab_IPAPPs". The top menu bar includes File, Edit, View, Project, Assignments, Processing, Tools, Window, and Help. The toolbar contains various icons, with the "Run" icon (a green play button) circled in red. The Project Navigator on the left shows the project hierarchy, including files like "mux_4_1.bdf" and "07_Lab_IPAPPs.vwf". The central pane shows the "mux_4_1.bdf" file, which is a multiplexer circuit. The right pane displays a timing diagram for the circuit, showing signals "in_dataA", "in_dataB", "S", and "Y" over time. The timing diagram has a scale of 10.0 ns and shows a sequence of values for each signal. The bottom pane shows the "Messages" window, which contains information about the compilation process, including the start and end times of the full compilation and a note about the vector file format.

Quartus II - F:\Ya_Ting_Wen\Quartus_project\07_Lab_IPAPPs\07_Lab_IPAPPs - [07_Lab_IPAPPs.vwf*]

File Edit View Project Assignments Processing Tools Window Help

Project Navigator

Files

- 07_Lab_IPAPPs.bdf
- lpm_mul4X4.qip
- lpm_mul4X4.v
- lpm_div4X4.qip
- lpm_div4X4.v
- add_sub.qip
- add_sub.v
- mux_4_1.bdf
- 07_Lab_IPAPPs.vwf

Hierarchy Files Design Units

Tasks

Flow: Compilation

Task

- Compile Design
- Analysis & Synthesis
- Fitter (Place & Route)
- Assembler (Generate pro
- Classic Timing Analysis
- EDA Netlist Writer

Master Time Bar: 0 ps Pointer: 18.5 ns Interval: 18.5 ns Start: End:

Name

Signal	Value
in_dataA	5
in_dataB	3
S	0
Y	[0]

Messages

Type Message

- Info: Started Full Compilation at Wed May 06 13:45:11 2009 台北標準時間
- Info: Ended Full Compilation at Wed May 06 13:45:52 2009 台北標準時間
- Info: Arguments associated with the following assignment have been removed because the assignment was changed
- Info: Arguments associated with the following assignment have been removed because the assignment was changed
- Info: Arguments associated with the following assignment have been removed because the assignment was changed
- Info: Vector file 07_Lab_IPAPPs.vwf is saved in text format. You can compress it into Compressed Vector Waveform File format in order

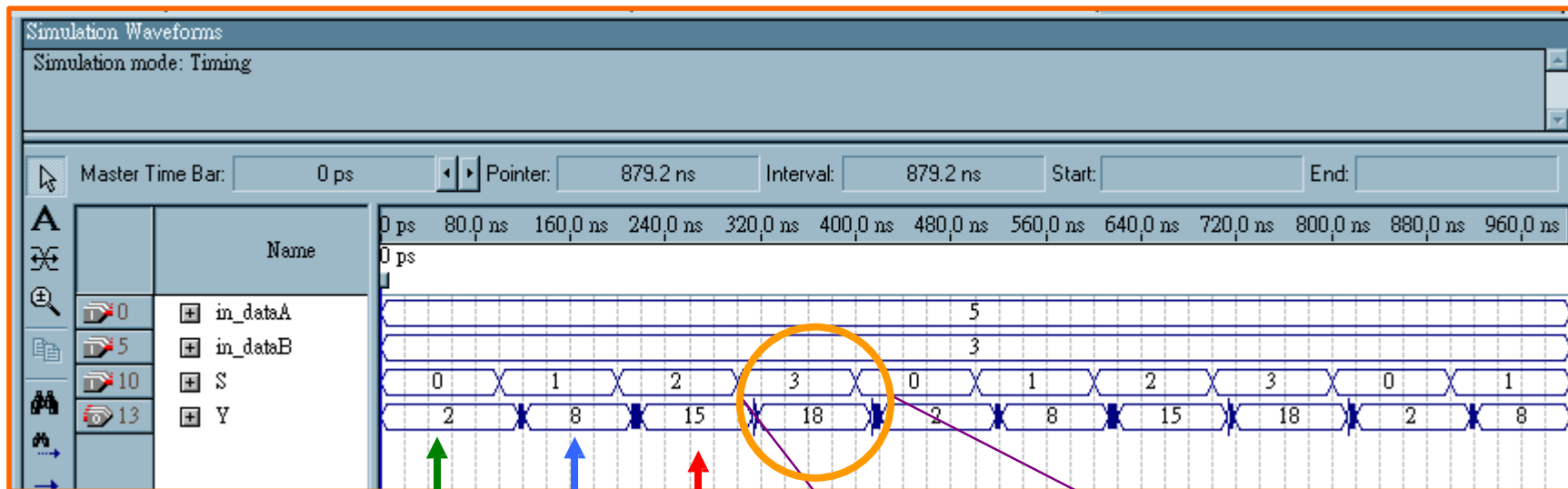
System (7) Processing (85) Extra Info Info (80) Warning (5) Critical Warning Error Suppressed (6) Flag

Message: 0 of 10 Location: Locate

For Help, press F1

Idle NUM

無號數之四則運算 Step 22



$$5-3=2$$

$$5+3=8$$

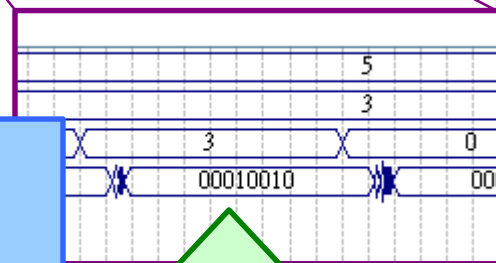
$$5 \times 3 = 15$$

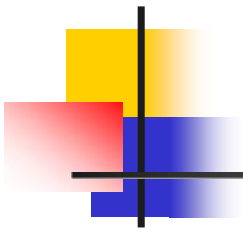
$$5 \div 3 = 1 \dots 2$$

商 : 0001

餘 : 0010

OUTPUT : 00010010





Appendix: IP Applications (Gates)

除法器(IP應用) Step 1

The image displays three sequential screenshots of the MegaWizard Plug-In Manager (MegaWizard) interface, illustrating the steps to create a custom megafunction for a divider.

Step 1: The first screenshot shows the initial MegaWizard Plug-In Manager dialog. The "Which action do you want to perform?" section has "Create a new custom megafunction variation" selected. The "Next >" button is circled in red.

Step 2: The second screenshot shows the "MegaWizard Plug-In Manager [page 2a]" dialog. The "Which device family will you be using?" dropdown is set to "Cyclone II". The "Which type of output file do you want to create?" section has "Verilog HDL" selected. The "Browse..." button next to the output file name is circled in red. The "Next >" button is also circled in red.

Step 3: The third screenshot shows the "MegaWizard Plug-In Manager - LPM_DIVIDE [page 3 of 6]" dialog. The "General" tab is selected. The "LPM_DIVIDE" megafunction is highlighted in the list. The "How wide should the 'numerator' input bus be?" and "How wide should the 'denominator' input bus be?" dropdowns are both set to "4" bits. The "Next >" button is circled in red.

Arrows indicate the flow from Step 1 to Step 2, and from Step 2 to Step 3.

除法器(IP應用) Step 2

The image displays three overlapping screenshots of the MegaWizard Plug-In Manager for the LPM_DIVIDE IP core, illustrating the configuration steps:

- Step 1:** The "General" tab is selected. The "Do you want to pipeline the function?" option is set to "No". The "Which do you wish to optimize?" section shows "Default Optimization" selected. The "Always return a positive" section shows "Yes" selected. The "Resource Usage" section indicates "25 lut". The "Next >" button is circled in red.
- Step 2:** The "EDA" tab is selected. The "Simulation Libraries" section is visible. The "Next >" button is circled in red.
- Step 3:** The "Summary" tab is selected. The "Turn on the files you wish to generate" section is visible, showing a list of files to be generated. The "Finish" button is circled in red.

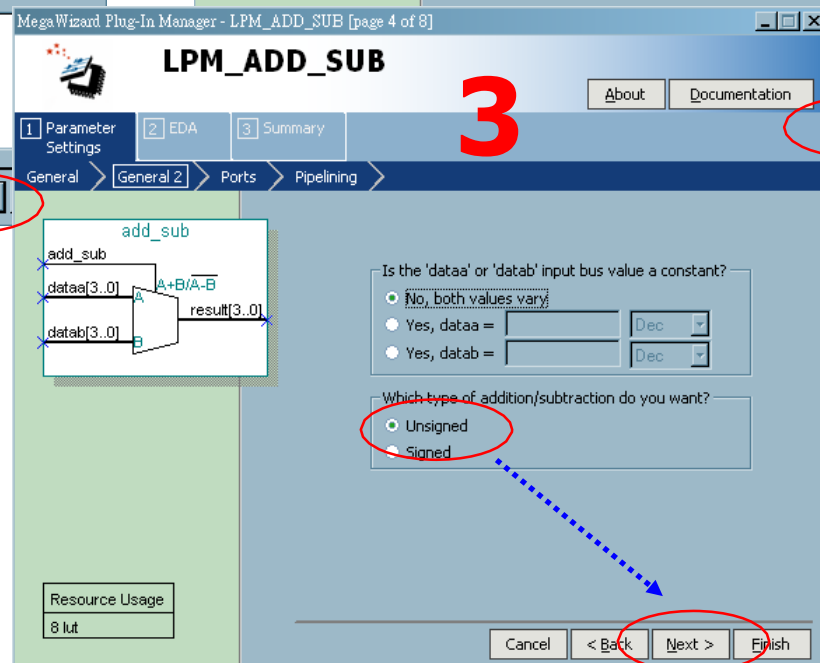
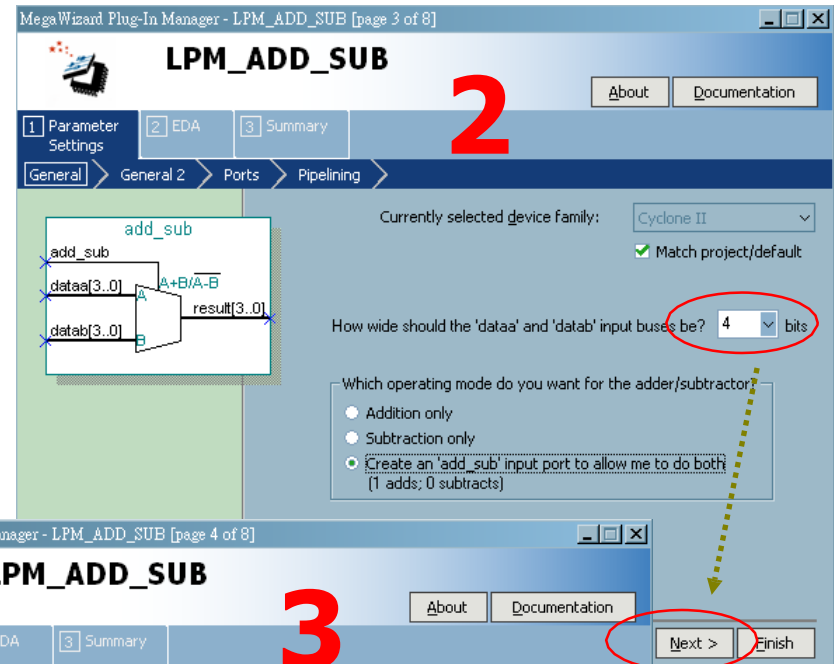
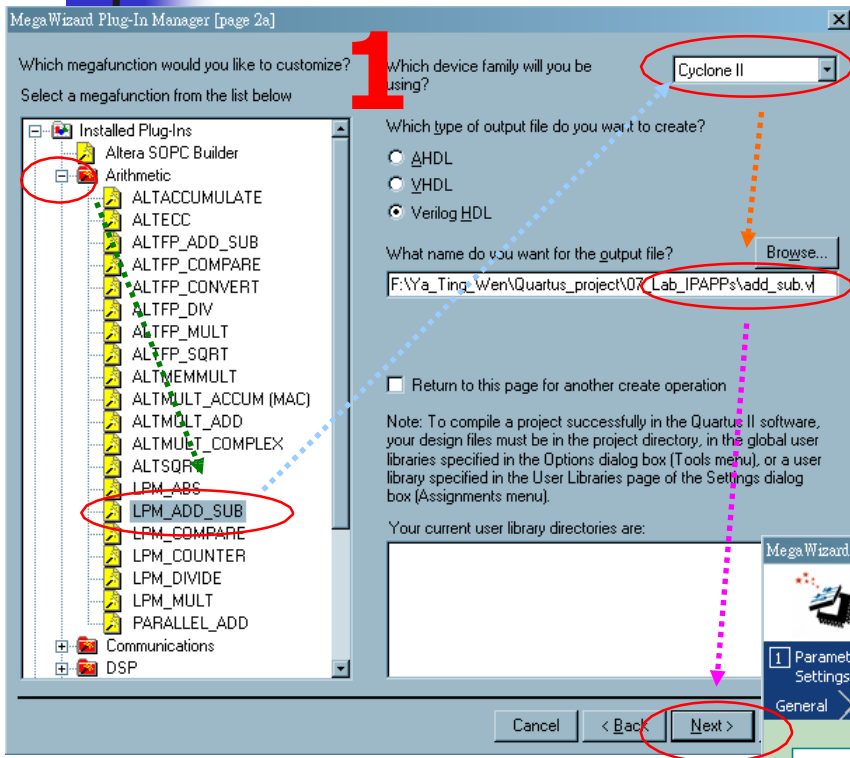
File List:

File	Description
☒ lpm_div4x4.v	Variation file
☐ lpm_div4x4.inc	AHDL Include file
☐ lpm_div4x4.cmp	VHDL component declaration file
☒ lpm_div4x4.bsf	Quartus II symbol file
☐ lpm_div4x4_inst.v	Instantiation template file
☒ lpm_div4x4_bb.v	Verilog HDL black-box file

除法器(IP應用) Step 3

The screenshot displays the Quartus II software interface for a project named "07_Lab_IPAPPs". The "Project Navigator" on the left shows the project files, including "07_Lab_IPAPPs.bdf", "lpm_mul4X4.qip", "lpm_div4X4.qip", and their corresponding ".v" files. The "Tasks" pane on the left lists the compilation steps: Compile Design, Analysis & Synthesis, Fitter (Place & Route), Assembler (Generate pro), Classic Timing Analysis, and EDA Netlist Writer. The main workspace shows the schematic design for "07_Lab_IPAPPs.bdf". It contains two IP blocks: "lpm_mul4X4" and "lpm_div4X4". The "lpm_mul4X4" block has inputs "dataa[3..0]" and "datab[3..0]", and an output "result[7..0]". The "lpm_div4X4" block has inputs "numer[3..0]" and "denom[3..0]", and outputs "quotient[3..0]" and "remain[3..0]". The "lpm_div4X4" block also displays the text "Numer is UNSIGNED" and "Denom is UNSIGNED". The "Messages" pane at the bottom shows the "System" message, indicating the project is in the "Processing" state. The status bar at the bottom indicates the project is in the "Idle" state.

加、減法器(IP應用) Step 1



加、減法器(IP應用) Step 2

Step 1: Ports Configuration

Do you want any optional inputs or outputs?

Input:

- ☐ Create a carry/borrow-out input

Outputs:

- ☐ Create a carry/borrow-in output
- ☒ Create an overflow output

Step 2: Pipelining Configuration

Do you want to pipeline the function?

- ☒ No
- ☐ Yes, I want an output latency of Clock cycles
- ☐ Create an asynchronous Clear input
- ☐ Create a Clock Enable input

Step 3: Summary Configuration

Turn on the files you wish to generate. A gray checkmark indicates a file that is automatically generated, and a red checkmark indicates an optional file. Click Finish to generate the selected files. The state of each checkbox is maintained in subsequent MegaWizard Plug-In Manager sessions.

The MegaWizard Plug-In Manager creates the selected files in the following directory:

F:\Ya_Ting_Wen\Quartus_project\07_Lab_IPAPPs\

File	Description
☒ add_sub.v	Variation file
☐ add_sub.inc	AHDL Include file
☐ add_sub.cmp	VHDL component declaration file
☒ add_sub.bsf	Quartus II symbol file
☐ add_sub_inst.v	Instantiation template file
☒ add_sub_bb.v	Verilog HDL black-box file
☐ add_sub_waveforms.html	Sample waveforms in summary
☐ ...add_sub_wave*.jpg	Sample waveform file(s)

Step 4: EDA Configuration

Simulation Libraries

To properly simulate the generated design files, the following simulation mode file(s) are needed

File	Description
lpm	LPM megafunction simulation library

Timing and resource estimation

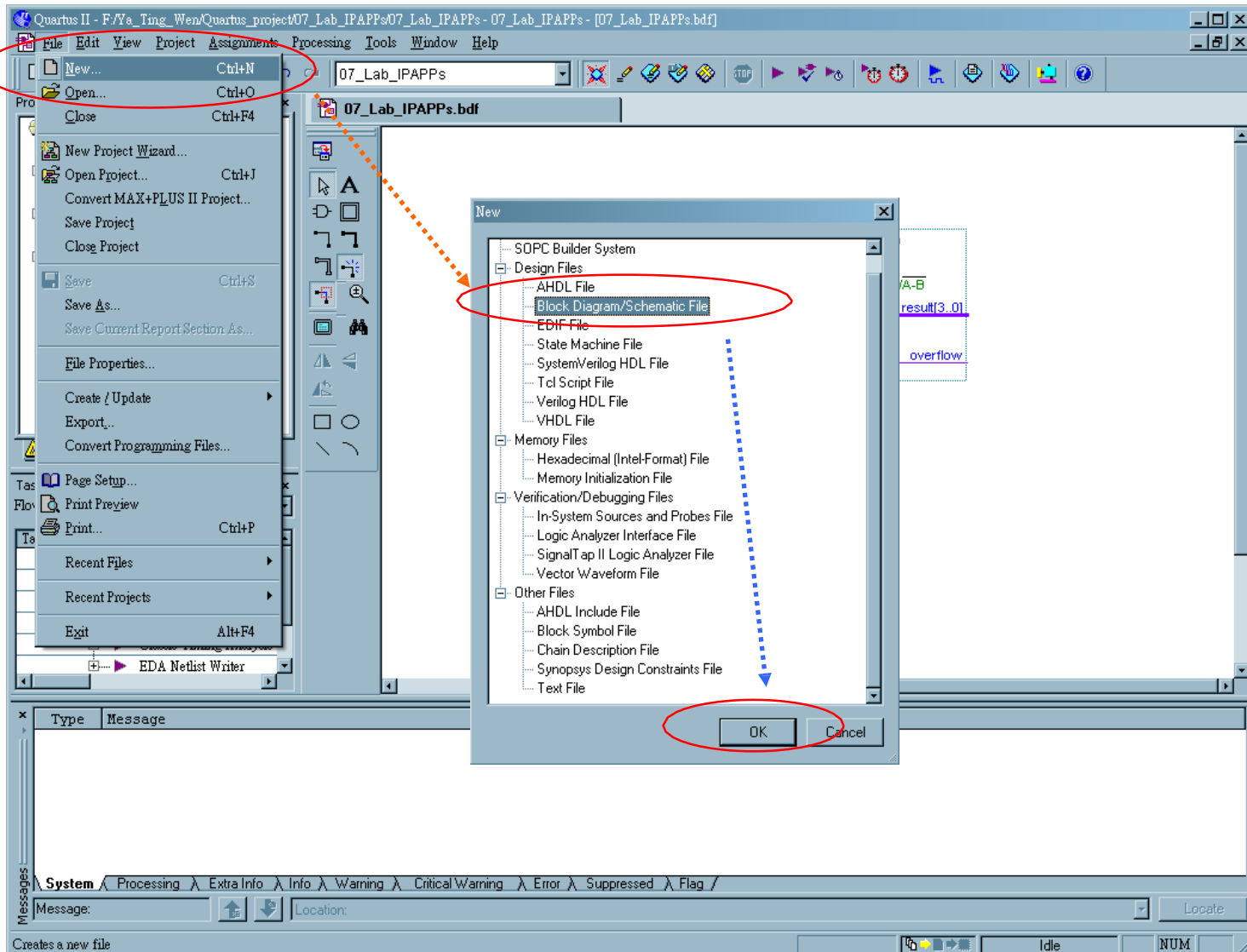
Generates a netlist for timing and resource estimation for this megafunction. If you are synthesizing your design with a third-party synthesis tool, using a timing and resource estimation netlist can allow for better design optimization.

Not all third-party synthesis tools support this feature - check with the tool vendor for complete support information.

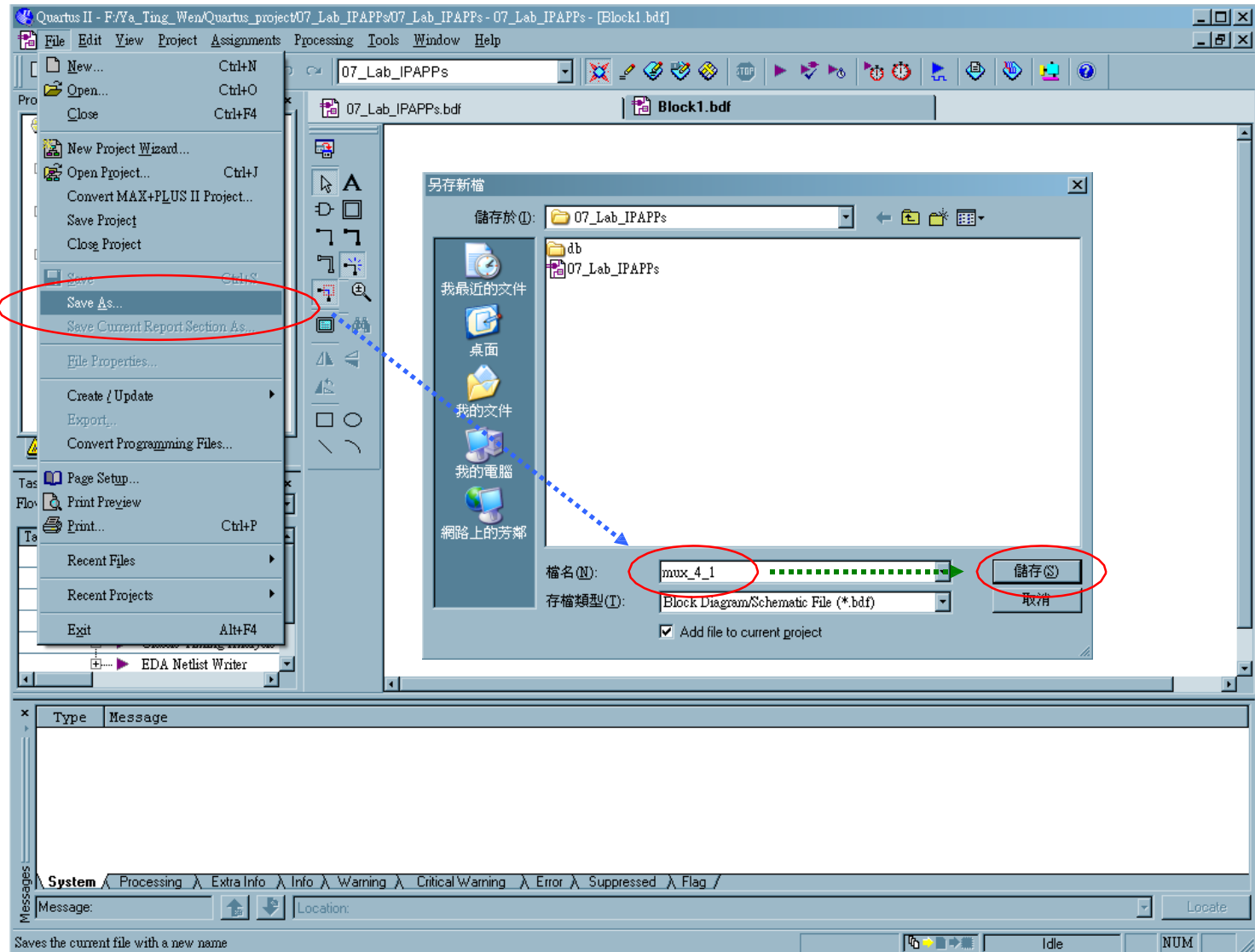
Note: Netlist generation can be a time-intensive process. The size of the design and the speed of your system affect the time it takes for netlist generation to complete.

☐ Generate netlist

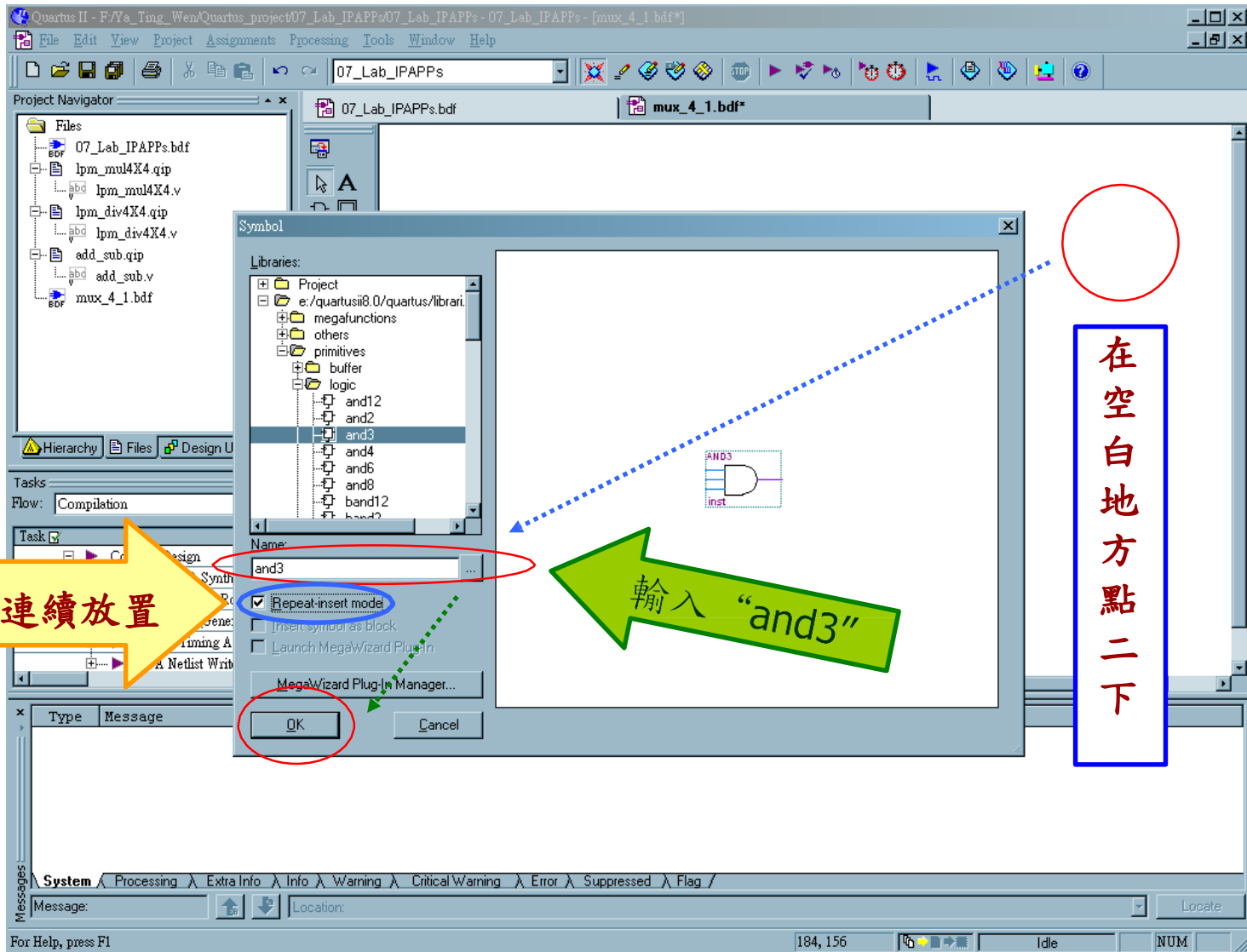
多工器(Gates) Step 1



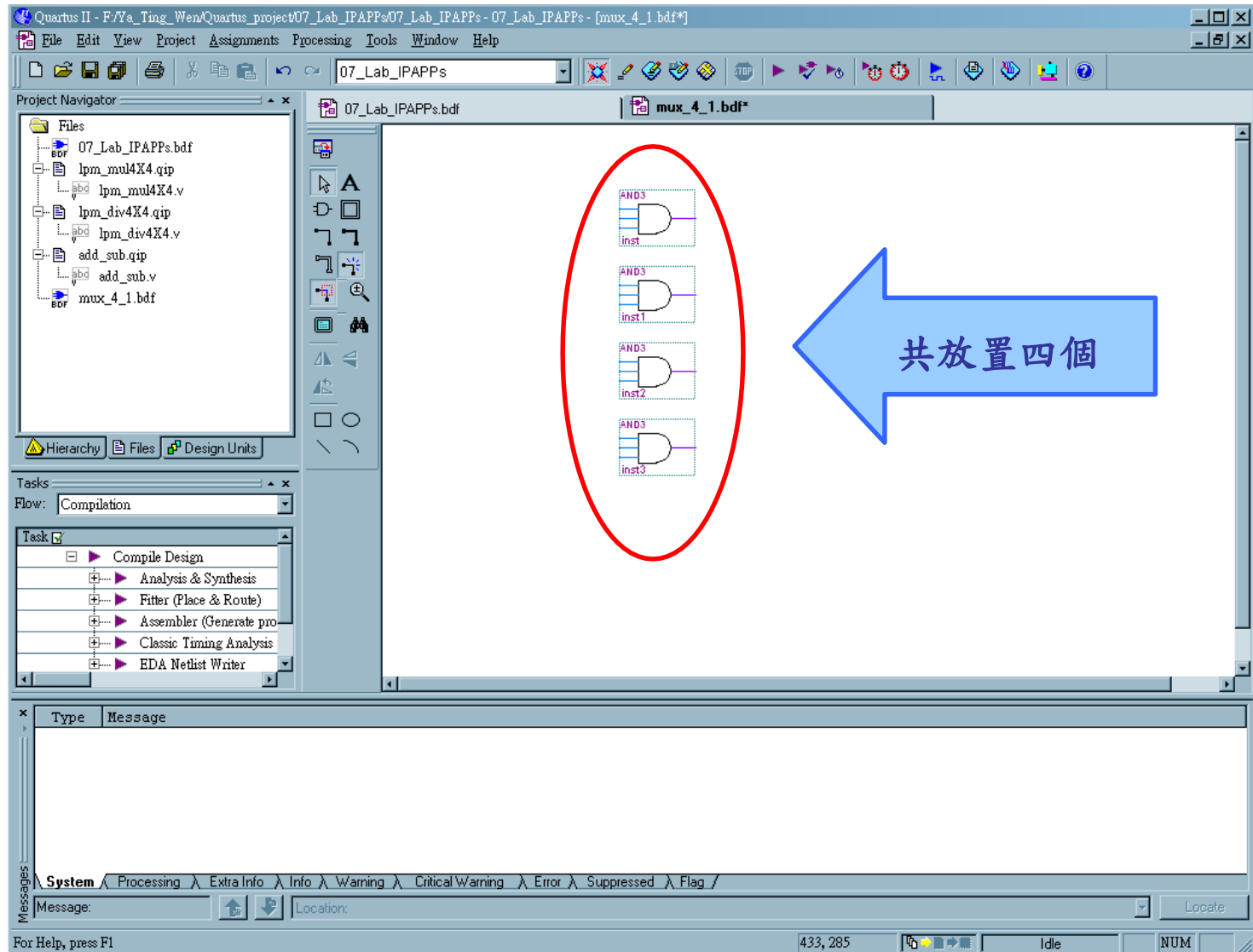
多工器(Gates) Step 2



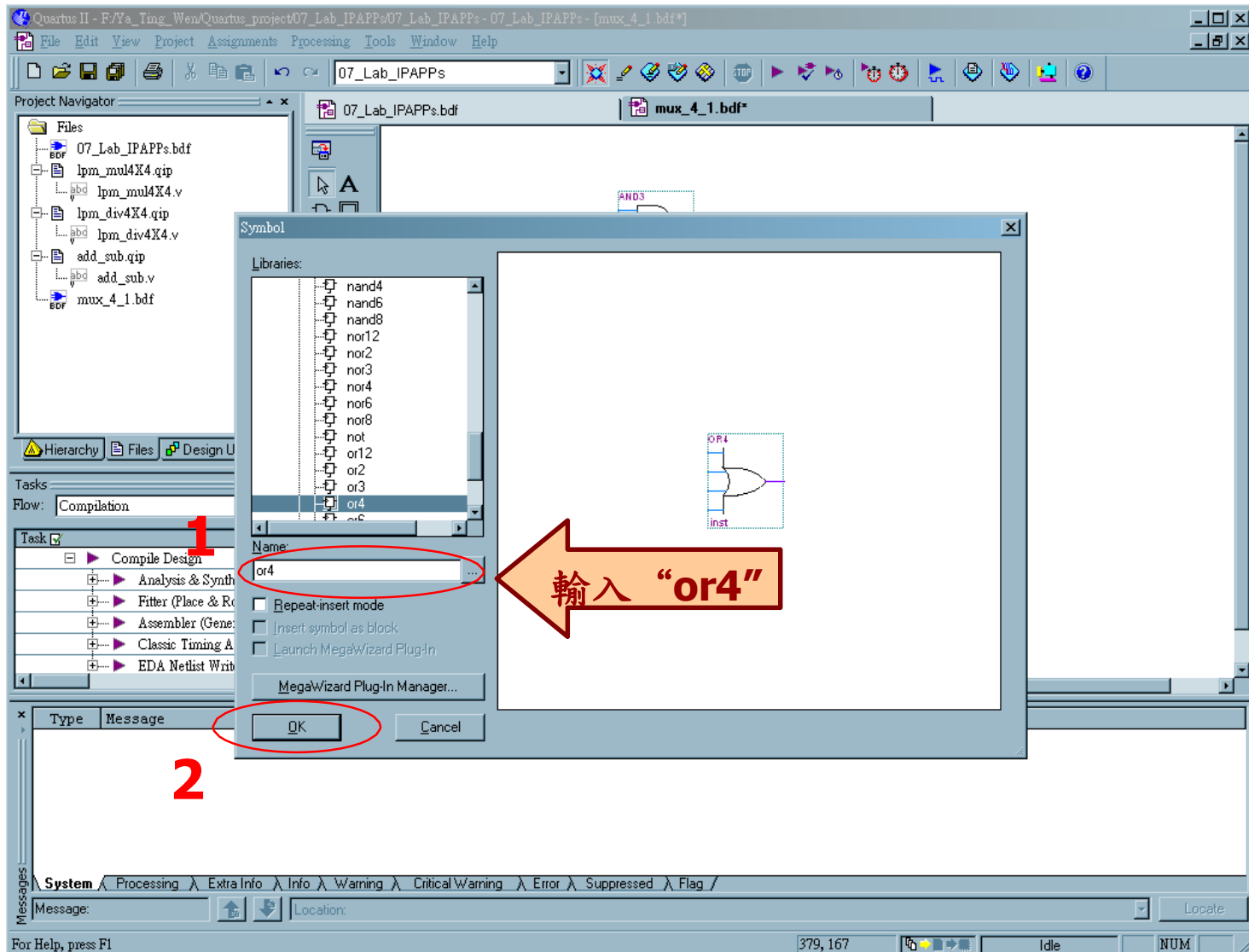
多工器(Gates) Step 3



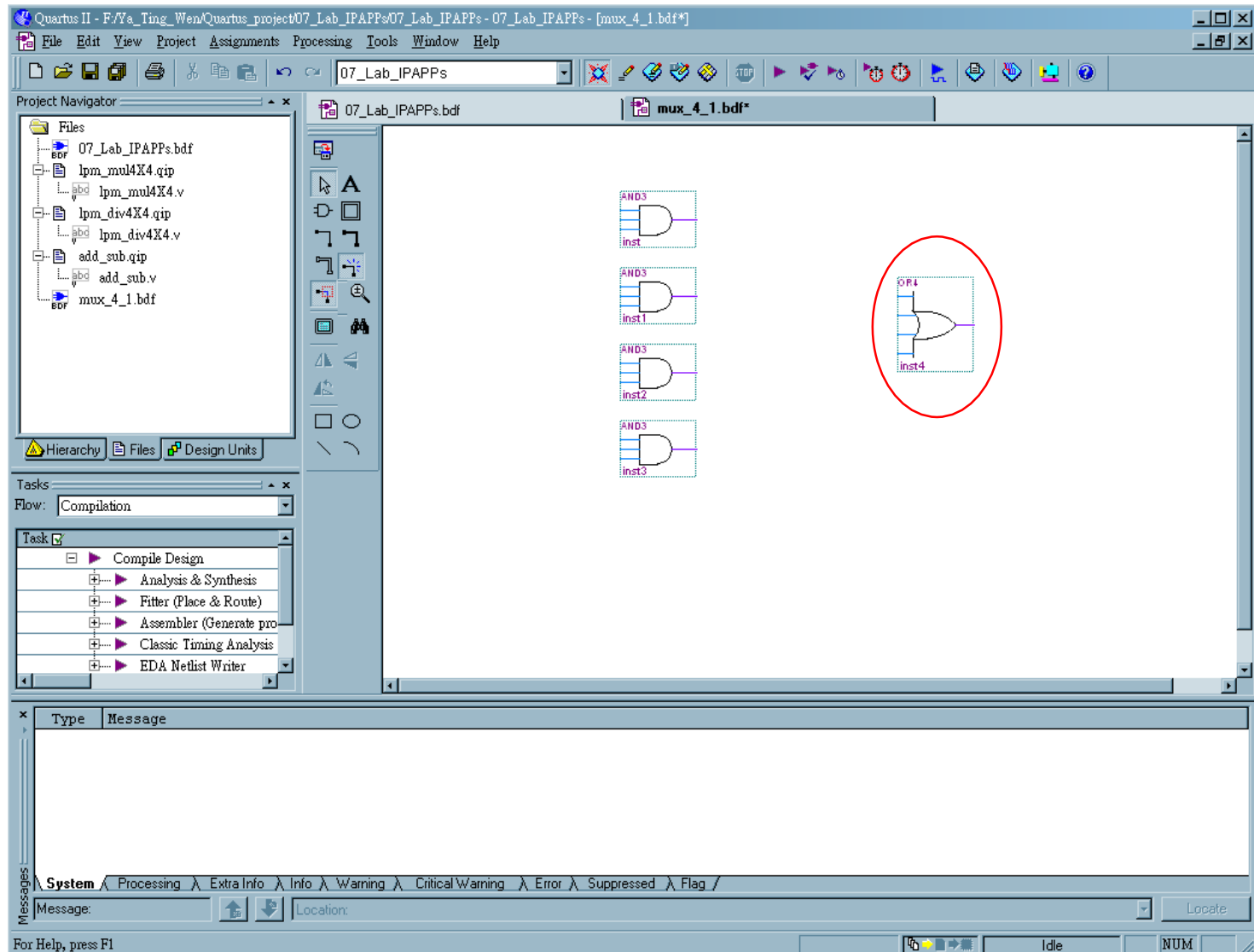
多工器(Gates) Step 4



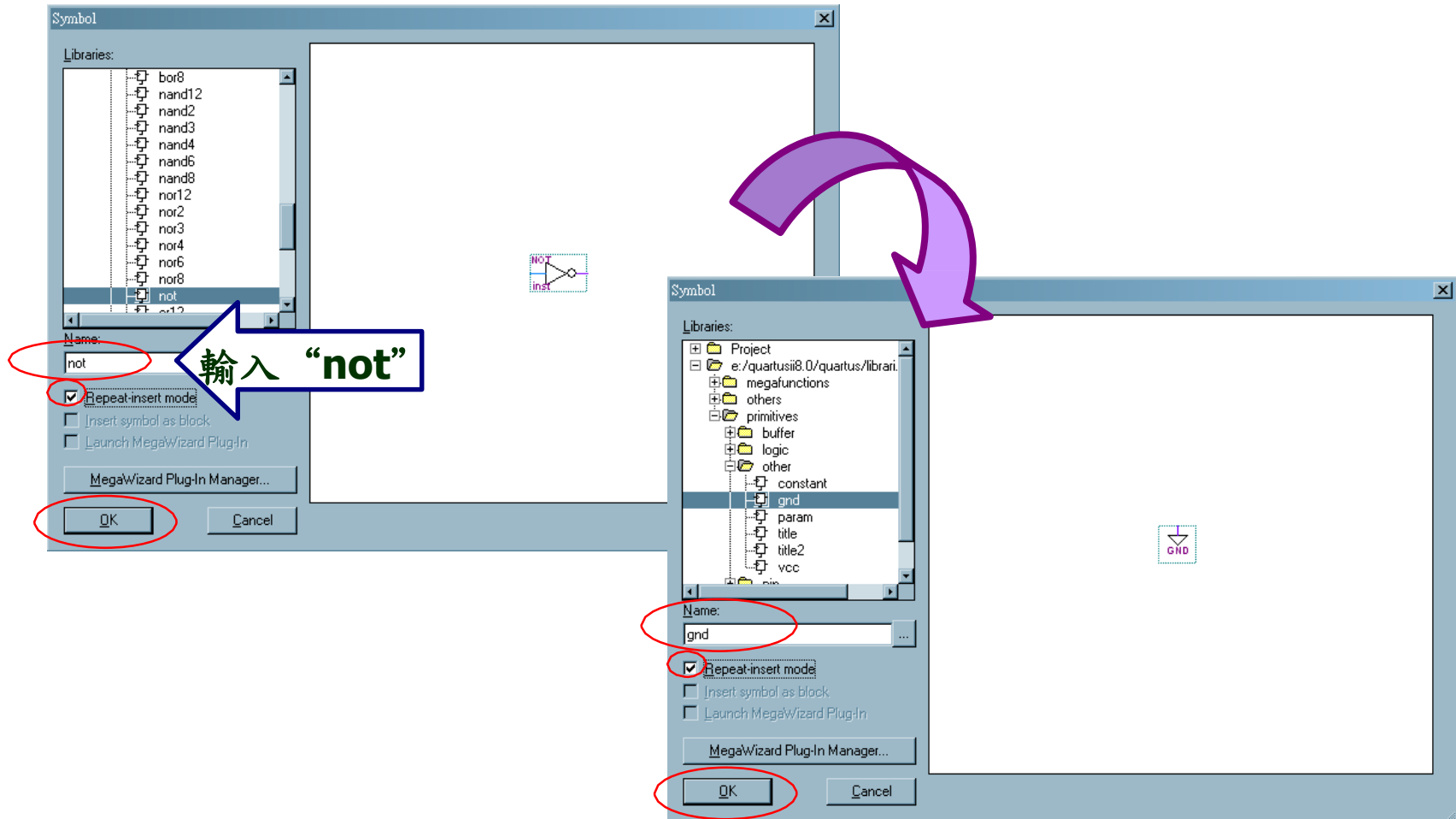
多工器(Gates) Step 5



多工器(Gates) Step 6



多工器(Gates) Step 7

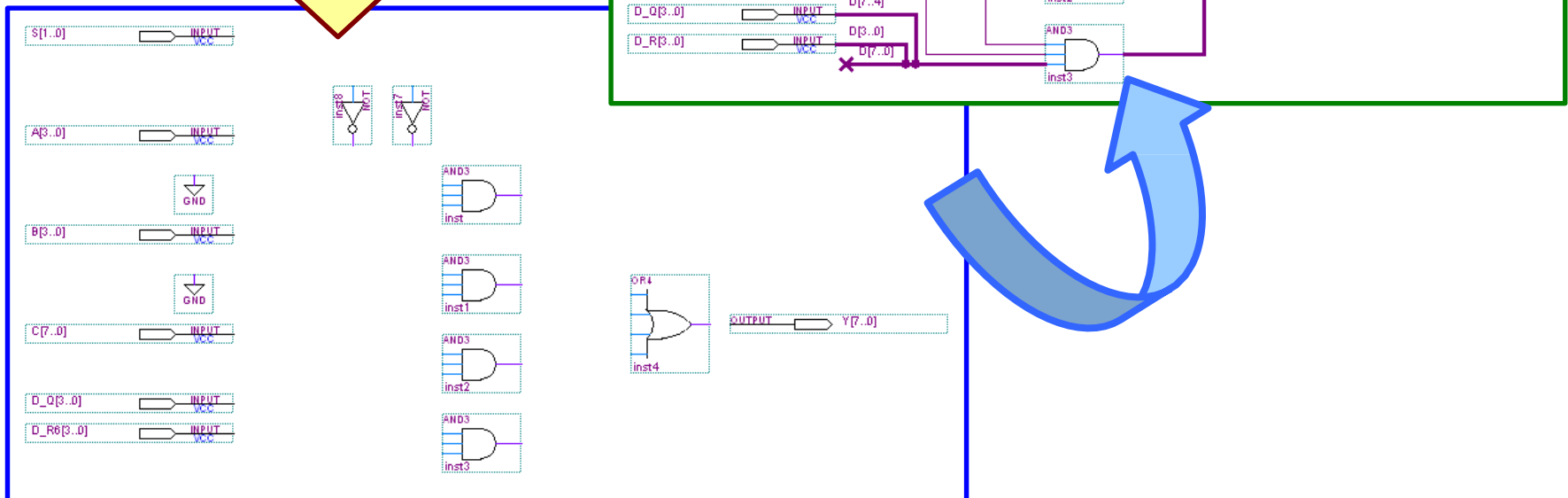


在 .bdf 檔，各加上兩個

多工器(Gates) Step 8

加入輸入與輸出腳位

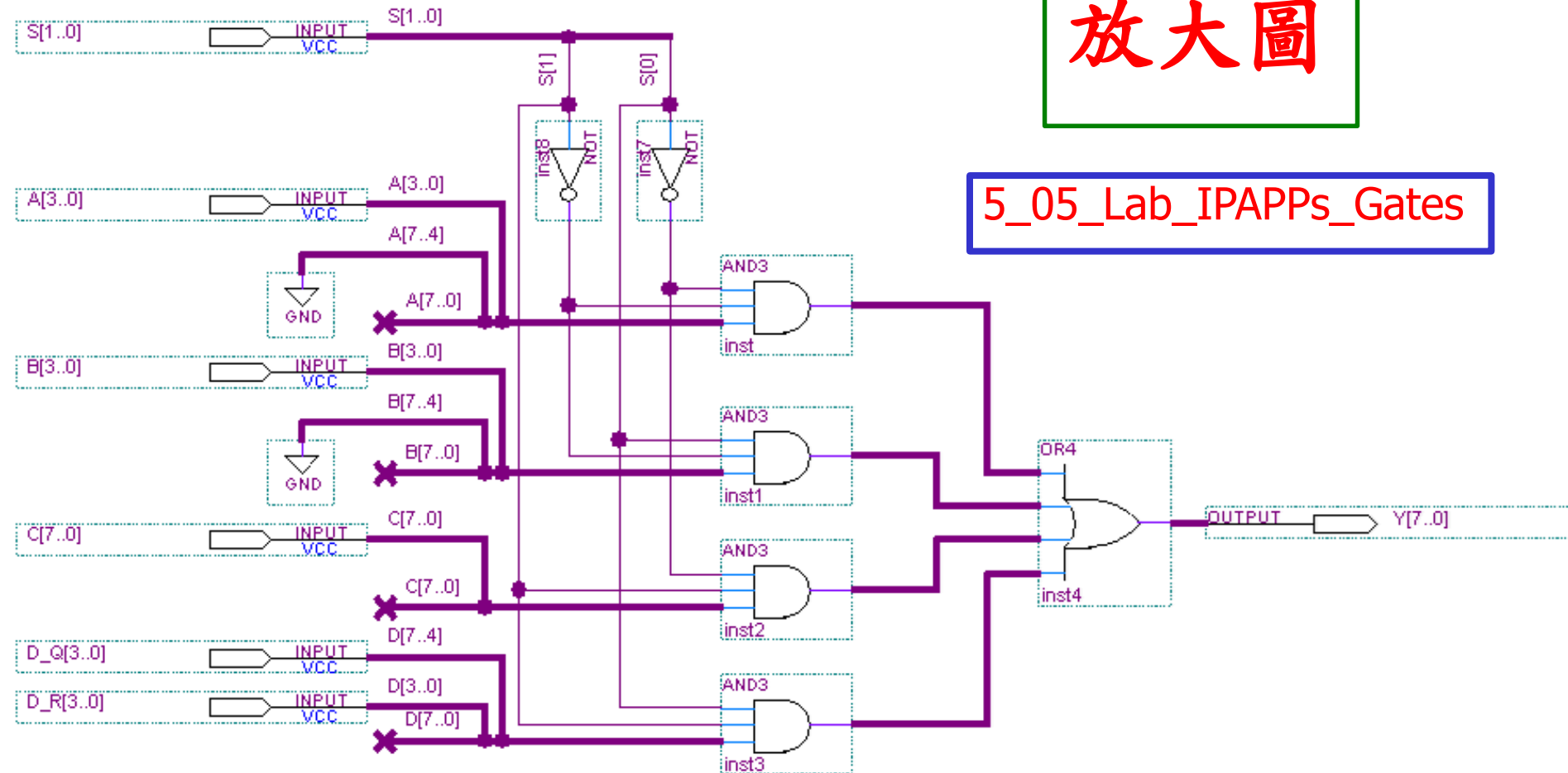
依照圖示接線
(後有放大圖)



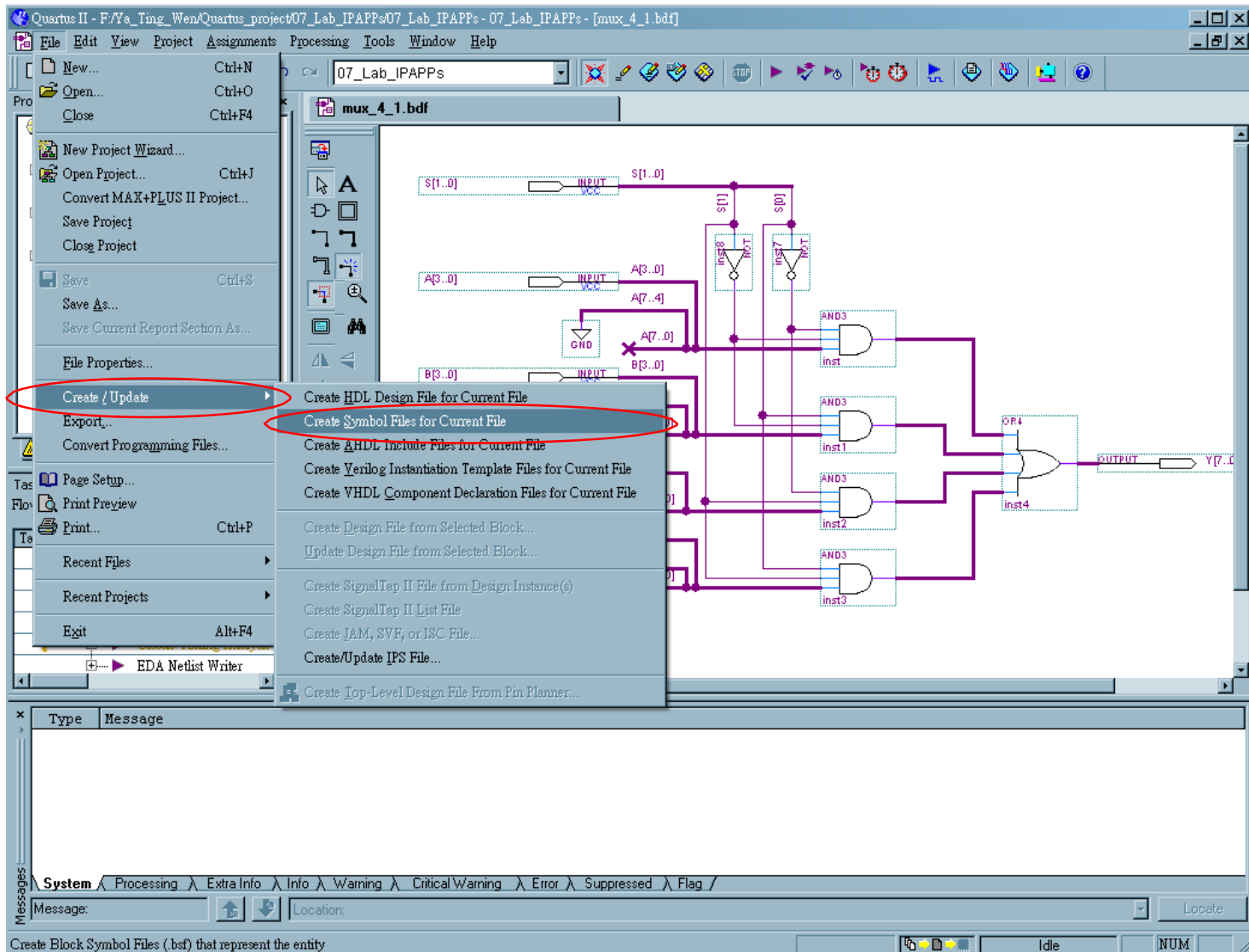
多工器(Gates) Step 9

放大圖

5_05_Lab_IPAPPs_Gates



多工器(Gates) Step 10



多工器(Gates) Step 11

The screenshot shows the Quartus II software interface. The 'Project Navigator' on the left lists files for '07_Lab_IPAPPs.bdf', including 'lpm_mul4X4.qip', 'lpm_div4X4.v', 'add_sub.qip', and 'mux_4_1.bdf'. The main workspace displays a circuit diagram with a 4-to-1 multiplexer (MUX4) and an output labeled 'OUTPUT' connected to 'V[P.0]'. A 'Create Symbol File' dialog box is open, showing the file 'mux_4_1.bdf' in the '07_Lab_IPAPPs' directory. The dialog box has three red annotations: '1' points to the 'File name' field containing 'mux_4_1', '2' points to the 'Save' button, and '3' points to the 'OK' button in the message window below. The message window displays the text: 'Created Block Symbol File F:/Ya_Ting_Wen/Quartus_project07_Lab_IPAPPs/mux_4_1.bsf'. The status bar at the bottom indicates 'Idle' and 'NUM'.

Quartus II - F:/Ya_Ting_Wen/Quartus_project07_Lab_IPAPPs/07_Lab_IPAPPs - [mux_4_1.bdf]

File Edit View Project Assignments Processing Tools Window Help

07_Lab_IPAPPs

Project Navigator

Files

- 07_Lab_IPAPPs.bdf
- lpm_mul4X4.qip
- lpm_mul4X4.v
- lpm_div4X4.qip
- lpm_div4X4.v
- add_sub.qip
- add_sub.v
- mux_4_1.bdf

Hierarchy Files Design Units

Tasks

Flow: Compilation

Task

- Compile Design
- Analysis & Synthesis
- Fitter (Place & Route)
- Assembler (Generate pro
- Classic Timing Analysis
- EDA Netlist Writer

Create Symbol File

儲存於 (I): 07_Lab_IPAPPs

我最近的文件

桌面

我的文件

我的電腦

網路上的芳鄰

檔名 (N): mux_4_1

儲存類型 (T): Symbol File (*.bsf)

儲存 (S)

取消

Quartus II

Created Block Symbol File F:/Ya_Ting_Wen/Quartus_project07_Lab_IPAPPs/mux_4_1.bsf

確定

System Processing Extra Info Info Warning Critical Warning Error Suppressed Flag

Message: Location: Locate

For Help, press F1

Idle NUM